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Quality Assurance (IM353IA): Unit IV

Department of Industrial Engineering and Management

Go, change the world

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Unit IV

Experimental Design for Process Improvement

Introduction to Design of Experiments

Introduction to Design of Experiments

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Design of Experiments

Module Objectives

- Review types of experiments
- Define Design of Experiments (“DOE”)
- Understand how to use a DOE
- Relate a DOE to other **6s** tools
- Analyze a 2^2 full-factorial experiment

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How do we learn?

- In order to learn, two things must occur simultaneously:



+



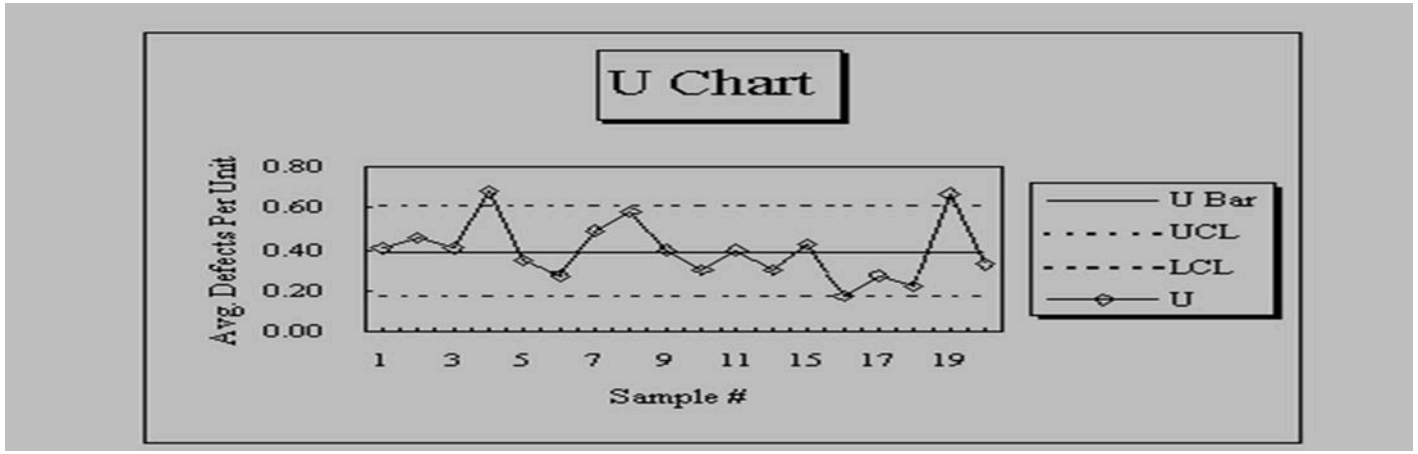
A perceptive observer

A significant event

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SPC is passive data generator

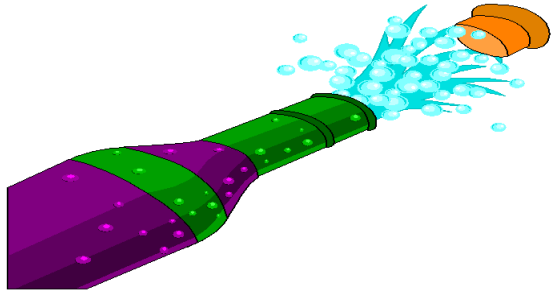
- Using SPC we constantly strive to reduce variation by detecting and reducing unexpected variation (significant events)
- Most basic SPC methods are passive--after the fact. SPC increases the opportunity to see a significant event and learn from it.
- SPC waits for an event to happen and helps you to observe the results
- This passive method of obtaining knowledge will not be enough to allow us to compete quickly in our changing markets



Example

- Wine has been in existence since the beginning of recorded history
- Champagne is made after the second fermentation of wine
- First discovered by a Benedictine monk in the 1600's
- Why did it take so long to discover champagne?





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Methods of Experimentation

TRIAL & ERROR

- Treats symptoms
- Looks for “quick fix”
- Makes random changes to process variables

ONE-FACTOR-AT-A-TIME (“OFAT”)

- Varies one factor while holding others constant
- Doesn’t considering combined effect of variables

FULL-FACTORIAL DOE

- Examines all combinations of variables
- Looks at combined effect of variables

FRACTIONAL-FACTORIAL DOE

- Looks at a fraction of the possible combinations
- Used for screening & getting preliminary information
- Looks at combined effect of variables
- Yields less information than a full factorial

Experiment Example

OBJECTIVE

Find out the fastest route to work?

POSSIBLE ROUTES

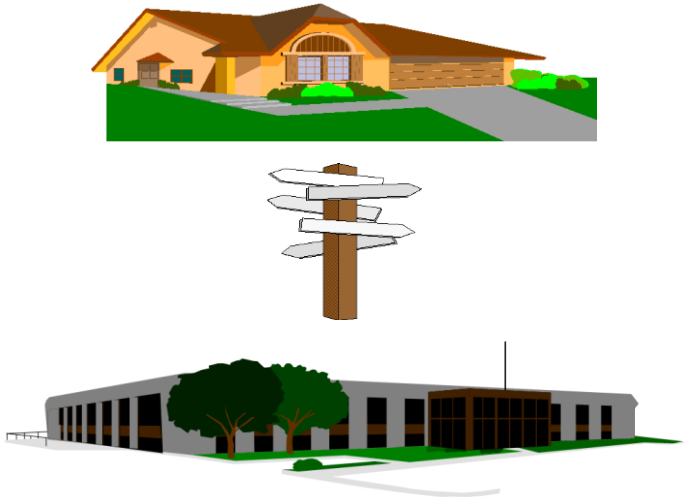
Normal route

Shortcut route

EXPERIMENT

Monday normal route

Tuesday shortcut route



Experiment Example

Results:

TRIAL	ROUTE	TIME
Monday	Normal	18.50 min
Tuesday	Shortcut	16.25 min

QUESTION

NOW THAT YOU HAVE THE RESULTS, WHICH IS THE FASTER ROUTE?

- Normal route?
- Shortcut route?

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Steps to Experimentation

ESTABLISH OBJECTIVE

- What to minimize?
- What to maximize?
- What to make more consistent?

IDENTIFY FACTORS

- What factors are known to affect the ability to minimize, maximize, or make more consistent?
- Brainstorm what other factors could possibly affect the ability to minimize, maximize, or make more consistent

CONDUCT EXPERIMENT

- Design, conduct, and analyze experiment
- Gain process knowledge

IMPLEMENT

- Implement process changes to optimize process

What is a DOE?

DEFINITION

- A structured experimental strategy …
- That allows for the *simultaneous* evaluation of processing variables (X's) …
- On their ability to influence a product or process characteristic (Y's).

Structured Experimental Approach

Benefits of a DOE

A DOE IS USEFUL TO

- Identify important factors
- Establish process stability
- Find best operating conditions

DOE Terminology

RESPONSE

- A process characteristic that is measurable
- A product that is measurable.
- A response is generally a “Y”
- Example: Release, Gloss, Opacity, Adhesive Coat Weight

FACTOR

- A process variables being investigated
- A factor is generally an “X”
- Example: Liner Lot, Oven Temperature, Cross-Linker Level

LEVEL

- The values or settings at which a factors is evaluated
- Example: Oven Temperature (i.e.,100°F & 220°F)

INTERACTION

- The failure of one factor to produce the same effect on the response at different levels of another factor
- Example: Time & Temperature

Full Factorials - Standard Order 2n Designs

2^n

- 2= # of levels
- n= # of factors

EXAMPLE

- 2^2full factorial design consists of 2 factors
- Each evaluated at two levels
- A total of four experiments are run (2 x 2 = 4)

CODING

- "-" = low level of a factor
- "+" = high level of a factor

TRIAL	FACTOR 1	FACTOR 2
1	-	-
2	+	-
3	-	+
4	+	+

How to Analyze a DOE

SIX SIGMA RULES OF ANALYSIS

Practical

- Does the difference in the response make sense?
- Are any initial patterns or trends apparent?

Graphical

- Can a graphical method be used to display the results?
- Are there any patterns in the data?

Analytical

- Can an analytical method be used to quantify the results?
- Which variables appear to have the most impact?

More DOE Terminology

MAIN EFFECT

- The effect or impact of a factor on a response variable.

INTERACTION EFFECT

- The combined effect or impact of factors on a response variable.

NOTE

- The effect of a given factor (or interaction) is calculated numerically as the average response when a factor is high less the average response when the factor is low.

AVERAGE AVERAGE

$$\text{EFFECT} = \text{OF HIGH} - \text{OF LOW}$$

RESPONSES RESPONSES

EFFECT	**=**	**AVERAGE OF HIGH RESPONSES**	**-**	**AVERAGE OF LOW RESPONSES**

Square Geo-Gram

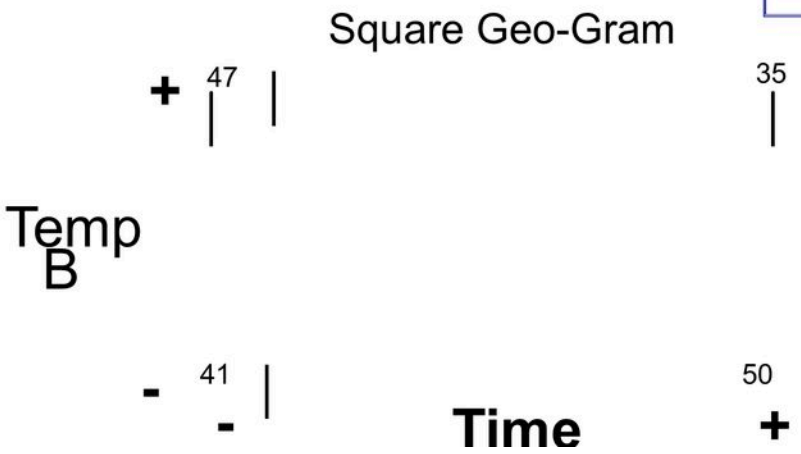
Geo-Gram:

The next step is to look for patterns in the data. This is greatly facilitated with the use of graphical techniques.

The geo-gram is a geometrical representation of the data.

The shape is determined by the number of factors (i.e. 2 factors is a square, 3 factors is a cube), the number of levels and the distance between levels.

TRIAL	TEMP FACTOR 1	TIME FACTOR 2	RESPONSE
1	325	12 min	41
2	350	12 min	47
3	325	18 min	50
4	350	18 min	35



THIS DEFINES THE INFERENCE SPACE OR THE EXPERIMENTAL BOUNDARIES OF YOUR EXPERIMENT WITHIN YOUR PROCESS.

Calculating the Effects

In order for us to answer the question:

“ Which factor (X) has the biggest impact on the Response (Y)?”

we have the following:

MAIN EFFECT

The effect or impact of a factor on a response variable.

INTERACTION EFFECT

The combined effect or impact of factors on a response variable.

Before we begin analyzing the importance and or significance of each factor, we want to look at coding the levels for each factor as (+) and (-).

	TEMP	TIME	
TRIAL	FACTOR 1	FACTOR 2	RESPONSE
1	325 (-)	12 min (-)	41
2	350 (+)	12 min (-)	47
3	325 (-)	18 min (+)	50
4	350 (+)	18 min (+)	35

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Calculating Interaction Effects

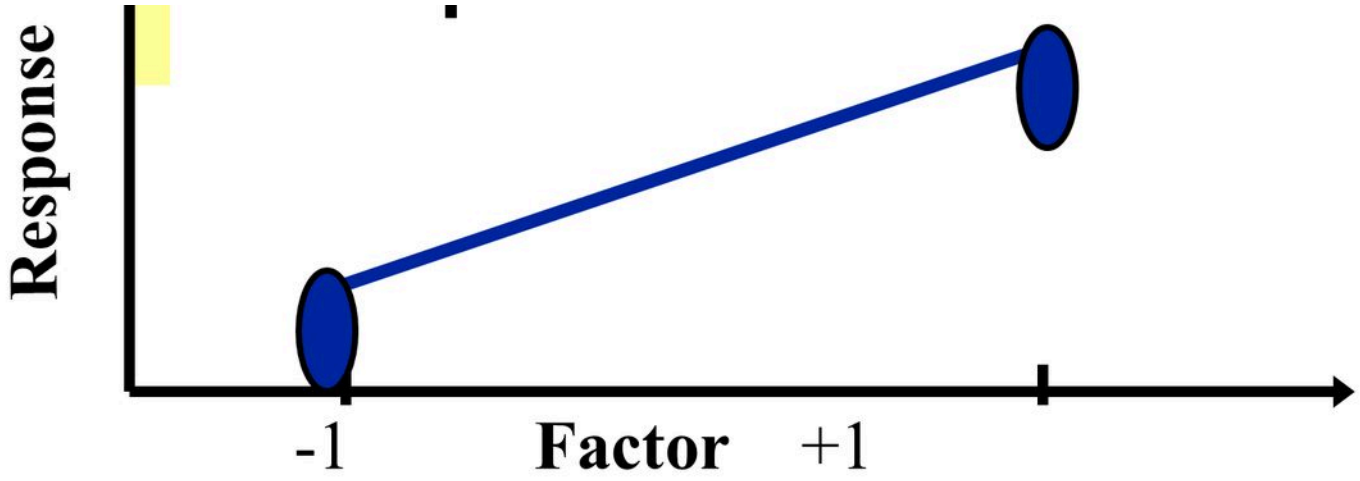
INTERACTION EFFECT

- Convert the low and high values of the individual factors to "-" and "+" signs, respectively
- For each trial, algebraically multiply the signs of the two factors together to determine the sign of the interaction effect

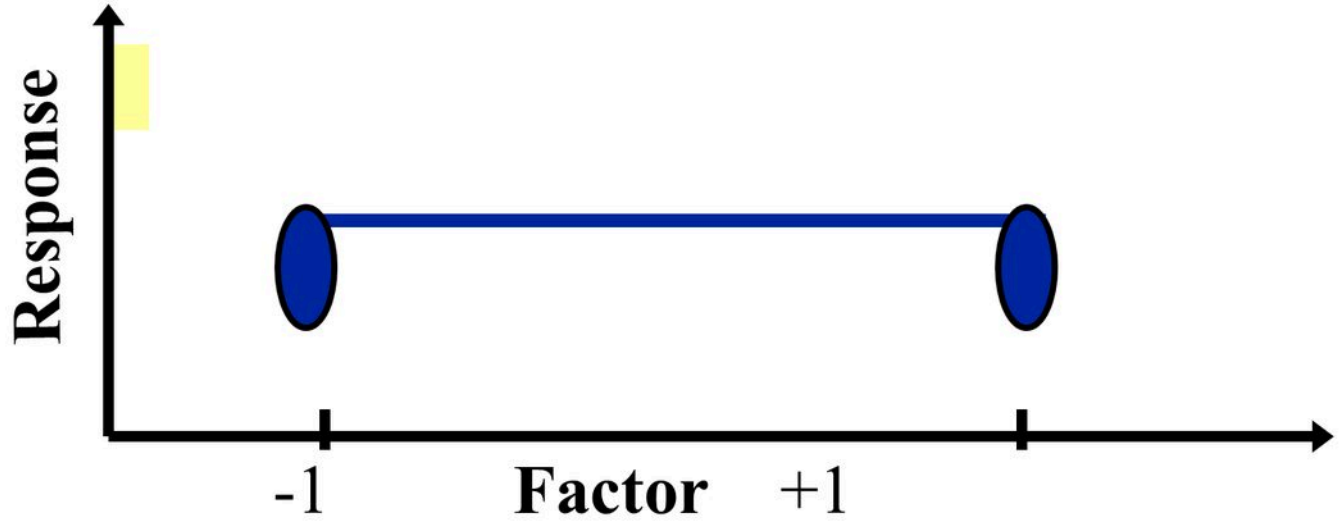
	TEMP	TIME	TEMP x TIME	
TRIAL	FACTOR 1	FACTOR 2	INTERACTION	RESPONSE
1	-	-	+	41
2	+	-	-	47
3	-	+	-	50
4	+	+	+	35

Interpreting Main Effects Graphs

Strong Effect

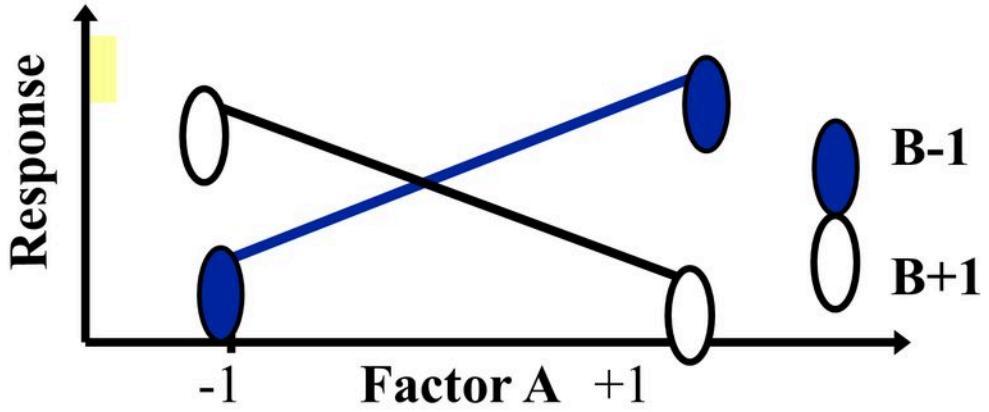


No Effect

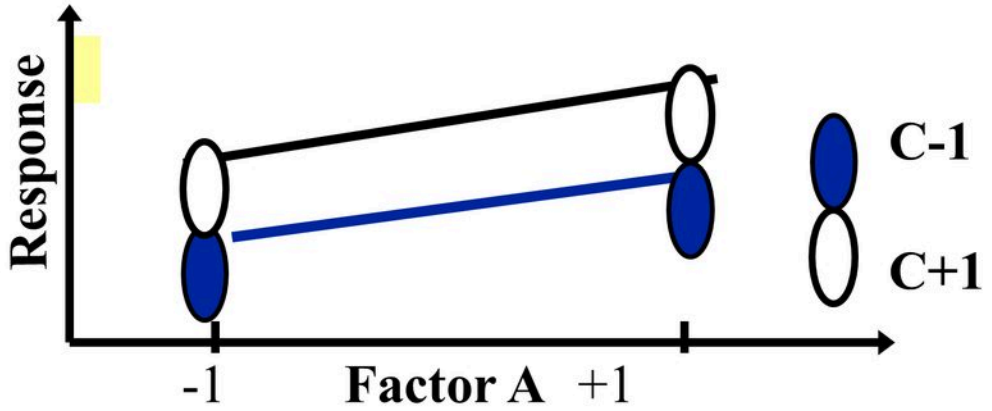


Interpreting Interaction Effect Graphs

Strong Interaction



No Interaction



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Interpreting the Graphs

- Now that we have calculated the effect for time, temperature and the time/temperature interaction we can compare the relative importance of each:

Factor Effect
Time -1.5
Temperature -4.5
Time x Temperature -10.5

MAIN EFFECT

- Oven temperature appears to impact taste more than time

INTERACTION EFFECT

- In this experiment, the Time x Temp Interaction has the largest effect and is therefore the most important.

How to Analyze a DOE

ANALYTICAL ANALYSIS

There are number of methods that are available to statistically quantify DOE results. They are beyond the scope of this course.

ANOVA

DOE and the Role of Questions

Questions play an important role in ensuring DOE success

KEY QUESTIONS

- What is the objective of the DOE?
- Is the objective of the DOE something we already know?
- What are the responses?
- Are the responses qualitative or quantitative?
- Can the responses be measured?
- Are the measurement system adequate?
- How many measurements should be taken?
- Is the response affected by time?
- Are there any important noise variables?
- Do the levels chosen make sense?

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Ingredients for a Good Experimental Strategy

CLEARLY DEFINED OBJECTIVES

- What that we don't know now, will we know when the experiment is completed?
- Who is the customer of the experiment?

KNOWLEDGE OF THE PROCESS

- Involve the process experts
- Involve a new set of "eyes"
- Use process maps and FMEAs
- Determine number of measurements needed for each response

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Ingredients for a Good Experimental Strategy

SELECTION OF FACTORS AND LEVELS

- Use of process knowledge
- Setting factor levels:
  - set factor levels wide enough to detect active effect
  - further apart then normally feel comfortable with
  - not so far that response has no value
  - examine all factor level combinations for hazards or useless results

KNOWLEDGE OF ENVIRONMENT

- Where are the noise variables?
- Do you need to randomize the trials?
- What is the intended range of validity of the results?

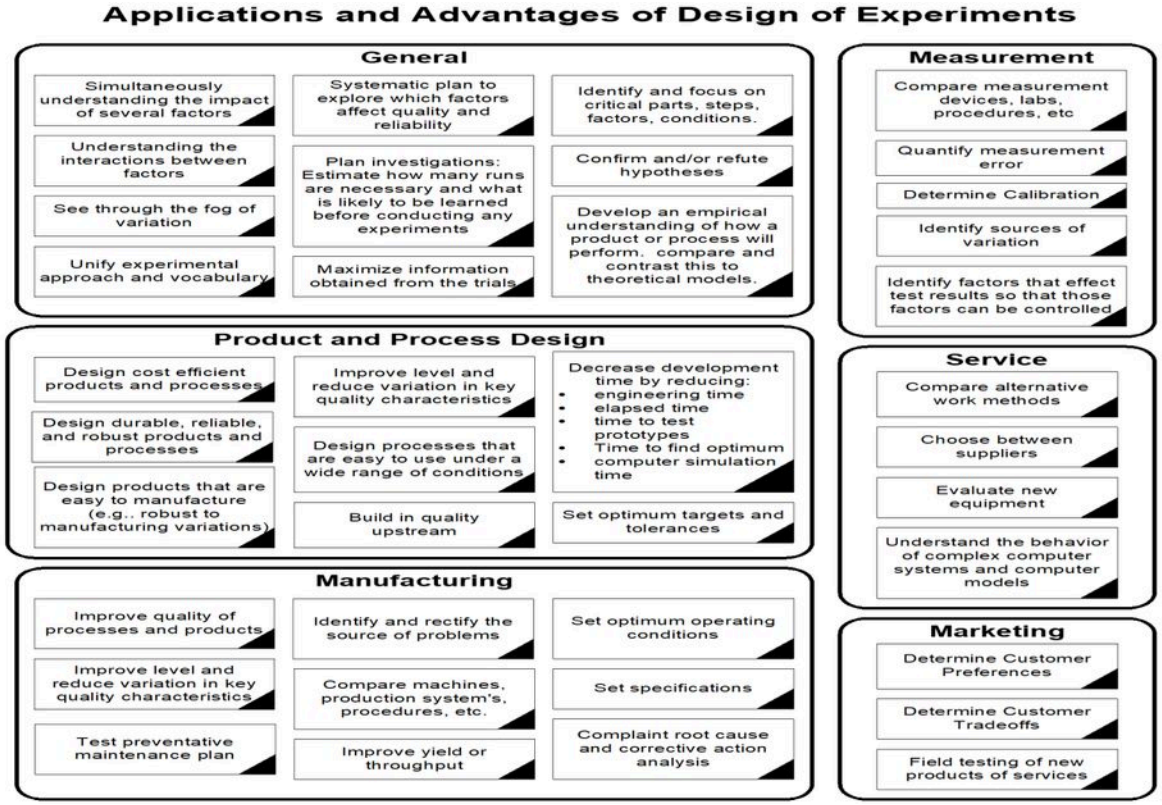
PREPARE A DOE PLAN

- Review with team before running DOE

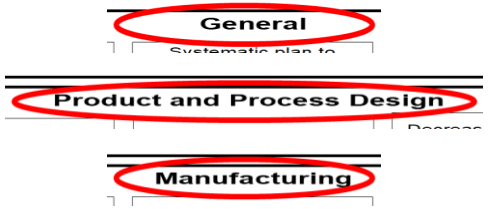
Appendix

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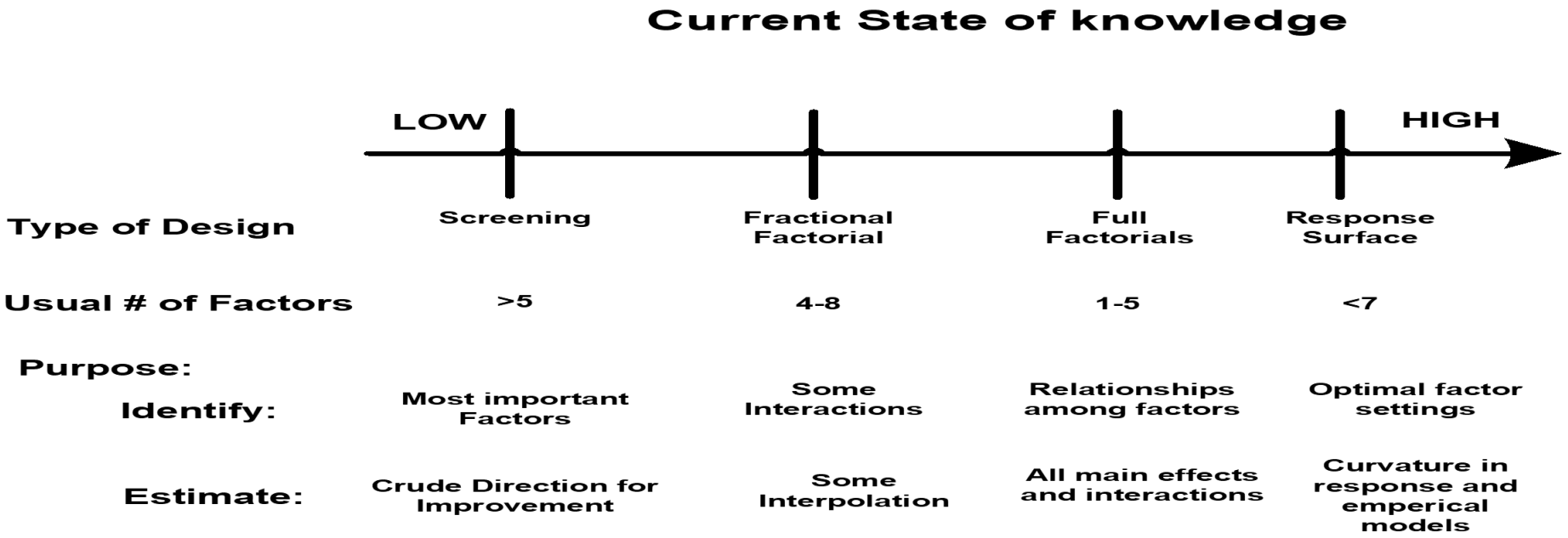
DOE Applications







The DOE Knowledge Line



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More Facts about Full Factorial Experiments

- Full factorials can be used when investigating a small number of factors (2-4), but are not recommended when there is a large number of factors (5 or more).
- The number of runs needed increases exponentially with the number of factors. The cost and time involved in running the experiment become prohibitive.
- The limitation of a full factorial is the time it takes to run the experiments. The resources and time needed can be significant.
- There are other experimentation techniques to deal with this issue - Fractional Factorials

Factors	Levels	Symbol	Experiments
2	2	2^2	4
3	2	2^3	8
7	2	2^7	128
15	2	2^{15}	32,768
2	3	3^2	9
3	3	3^3	27
7	3	3^7	2,187

DOE Recipes - Standard Order Designs

2^2 DESIGN

Trial	A	B
1	–	–
2	+	–
3	–	+
4	+	+

2^3 DESIGN

Trial	A	B	C
1	–	–	–
2	+	–	–
3	–	+	–
4	+	+	–
5	–	–	+
6	+	–	+
7	–	+	+
8	+	+	+

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DOE Recipes - Interactions

2^2 DESIGN

Trial	A	B	AB
1	-	-	+
2	+	-	-
3	-	+	-
4	+	+	+

2^3 DESIGN

Trial	A	B	C	AB	AC	BC	ABC
1	-	-	-	+	+	+	-
2	+	-	-	-	-	+	+
3	-	+	-	-	+	-	+
4	+	+	-	+	-	-	-
5	-	-	+	+	-	-	+
6	+	-	+	-	+	-	-
7	-	+	+	-	-	+	-
8	+	+	+	+	+	+	+

Design of Experiments-Learning Objectives

- Define the purpose & benefits of conducting a design of experiment.
- Describe screening of DOE and optimization DOE and how they are used in the improve phase.
- Review full factorial design.

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Design of Experiments-Learning Objectives

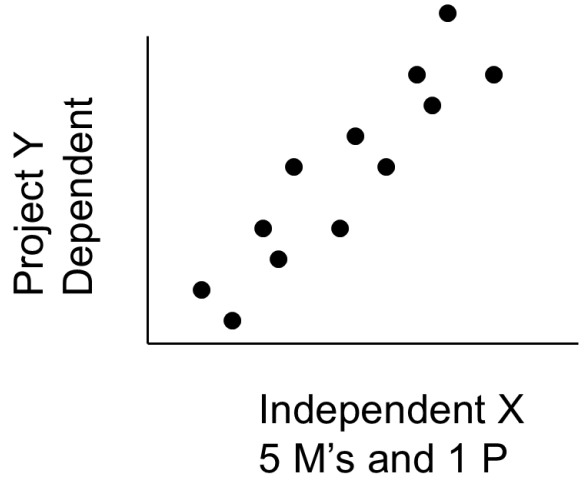
- Describe the 5 steps in analyzing DOE data, the key components of each step and the use of Minitab to analyze the data
  - Plot the raw data
  - Plot the residual
  - Examine the factor effects
  - Confirm impression with statistical procedures
  - Summarize conclusions
- Describe the use of Fractional Factorial Design and explain the benefits & the trade of over Full Factorial Design

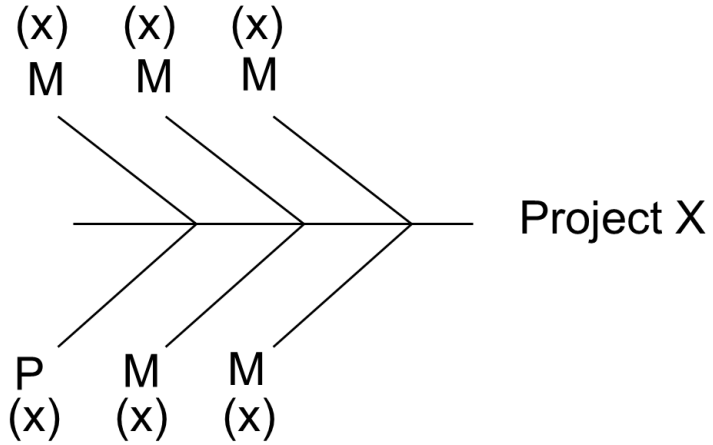
Design of Experiments

- **With the use of Design of Experiments (DOE), we will be able to:**
  - **Verify the "Vital Few" X's that impact the quality of the Project Y**
  - **Identify the best combination X-values to optimize process performance and meet customer CTQ's**

Design of Experiments - Terminology

- **Independent Variables-X's**
  - **Also called factors**
  - **Factors or variables we select in advance**
  - **The causes**
- **Dependent Variables-Y**
  - **Also called responses**
  - **The quantity (Y) that we measure to determine the impact of the X's**
  - **The effect.**





Design of Experiments - Terminology

- **Levels**
  - **The test settings for X**
- **Main effects**
  - **Differences between each factor level**
- **Interactions**
  - **Differences between two or more factor level combinations**

For Example: Is the on-time delivery different for:

1. 62ft trucks vs 66ft trucks?
2. Brand X nor Brand Y tires?

For Example: Is on-time delivery different when:

1. 62ft truck, Brand X tire
2. 62ft truck, Brand Y tire.
3. 66ft truck, Brand X tire
4. 66ft truck, Brand Y tire.

For Example: In our Capital Logistics GB Case Study:

Y=on-time delivery of shipments

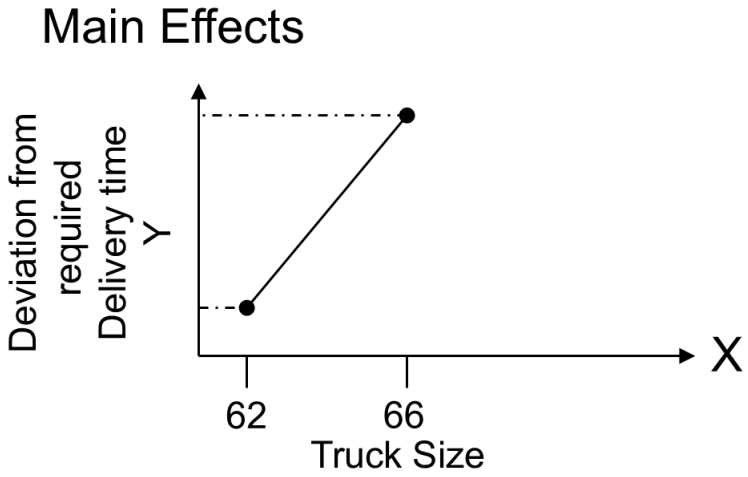
X_1 = Truck size

Levels = 62ft 66ft truck.

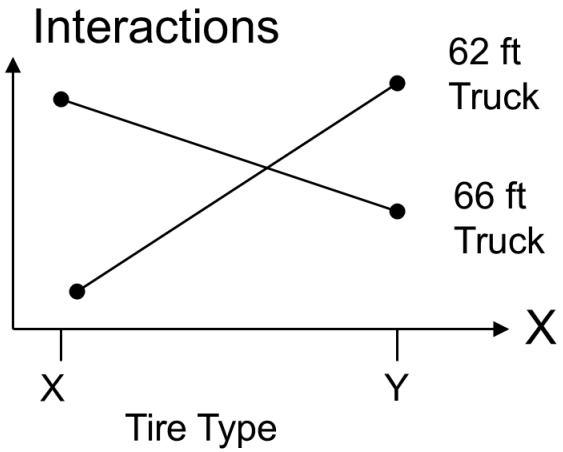
X_2 = The Type

Levels = Brand X Brand Y

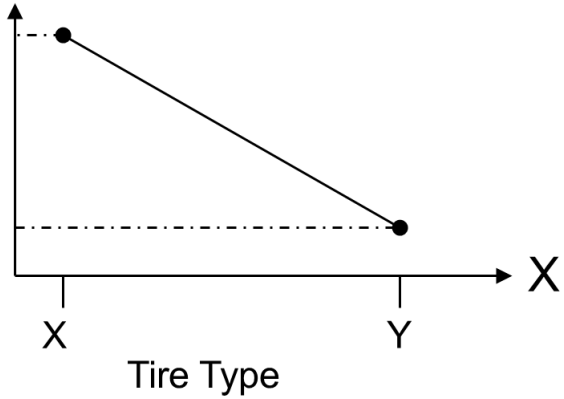
Design of Experiments - Terminology



Deviation from required Delivery time Y



Deviation from required Delivery time Y



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Strategy for Choosing Appropriate Design

- The Knowledge line:
  - Current state of process knowledge

	LOW			HIGH
Type of Design	Main effect screening	Fractional Factorials	Full Factorials	Response Surface
Usual # of factors	>5	4-10	1-5	2-3
Purpose Identify	Main effect critical factors Vital few	Some interactions	Relationship among factors	Optimal factor setting
Estimate	Crude direction for improvement	Some interactions	All main effects & interactions	Curvature in response, empirical models
STEP 7	**Screening DOE →**			
STEP 8			**Optimization DOE →**	

DOE Benefits

- **Can be used to identify "Vital Few" sources of variation**
- **Define the relationship between the inputs & outputs.**
- **Allows you to measure the influence of the "Vital Few" variables on a response variable.**
  - **Also allows you to measure the interactions between the "Vital Few" variables.**
- **Is more effective & efficient than testing one factor at a time**
- **Minimizes the number of test runs you have to make to draw valid conclusions about X & Y linkages.**
- **Provides knowledge of best set-up conditions of X's for improved Y performance.**

Statistical Design of Experiments Raises
Your Batting Average

The Process of Experimentation

- Define Project
  - Identify responses
- Establish current situation
- Perform analysis
  - Identify factors
  - Choose factor levels
  - Select design
  - Randomize runs
  - Collect data
  - Analyze data
  - Draw conclusions
  - Verify results
- Determine Solutions

The Process of Experimentation

- Record results
- Standardize
- Determine Future Plans

The Process of Experimentation

- **Define Project (Problem)**
  - **The mileage on my new car is not up to advertised standards. I want to improve my car's mileage.**
- **Establish Current situation (& state hypothesis)**
  - **Currently do not get the manufacturer's advertised gas mileage on my car.**
  - **Some combination of speed, gas octane, and tire pressure will provide me with the optimum gas mileage.**

The Process of Experimentation

- **Perform Analysis**
  - **Dependent Variable – Gas mileage (Y)**
  - **Identify factors (independent variables)**
    - **Tire Pressure (X)**
    - **Octane (X)**
    - **Speed (X)**
  - **Choose Factor Levels**

The Process of Experimentation

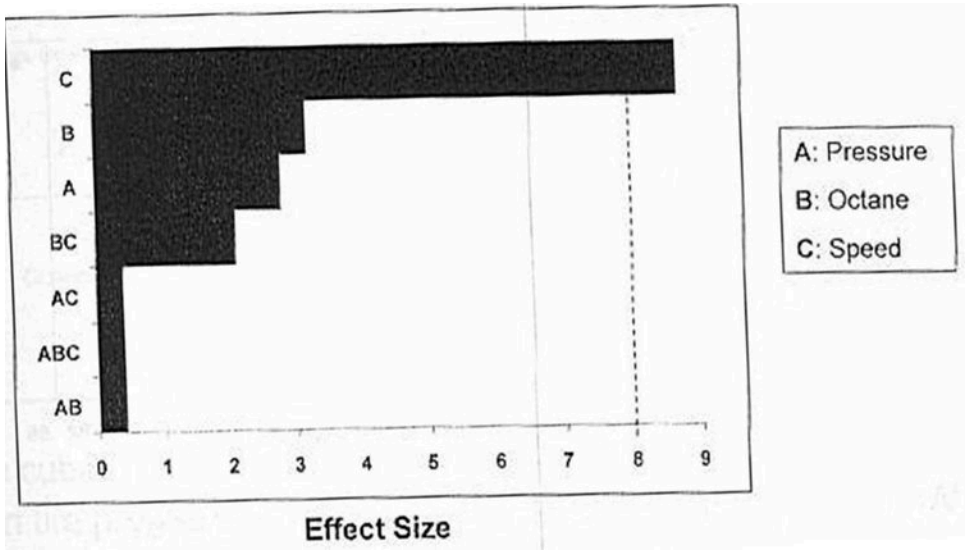
- **Perform Analysis**
  - **Select Design**
    - **Number of test levels (number of Factors) = 2^3 = trials**

The Process of Experimentation

- **Perform Analysis**
  - **Randomize runs**

The Process of Experimentation

- **Perform Analysis**
  - **What effects (main and interaction) seem to be significant?**
    - **Main effect is the effect of one X on the Y**
    - **Interaction effects are the combined effects of two or more X's on the Y.**
  - **Collect & Analyze Data**



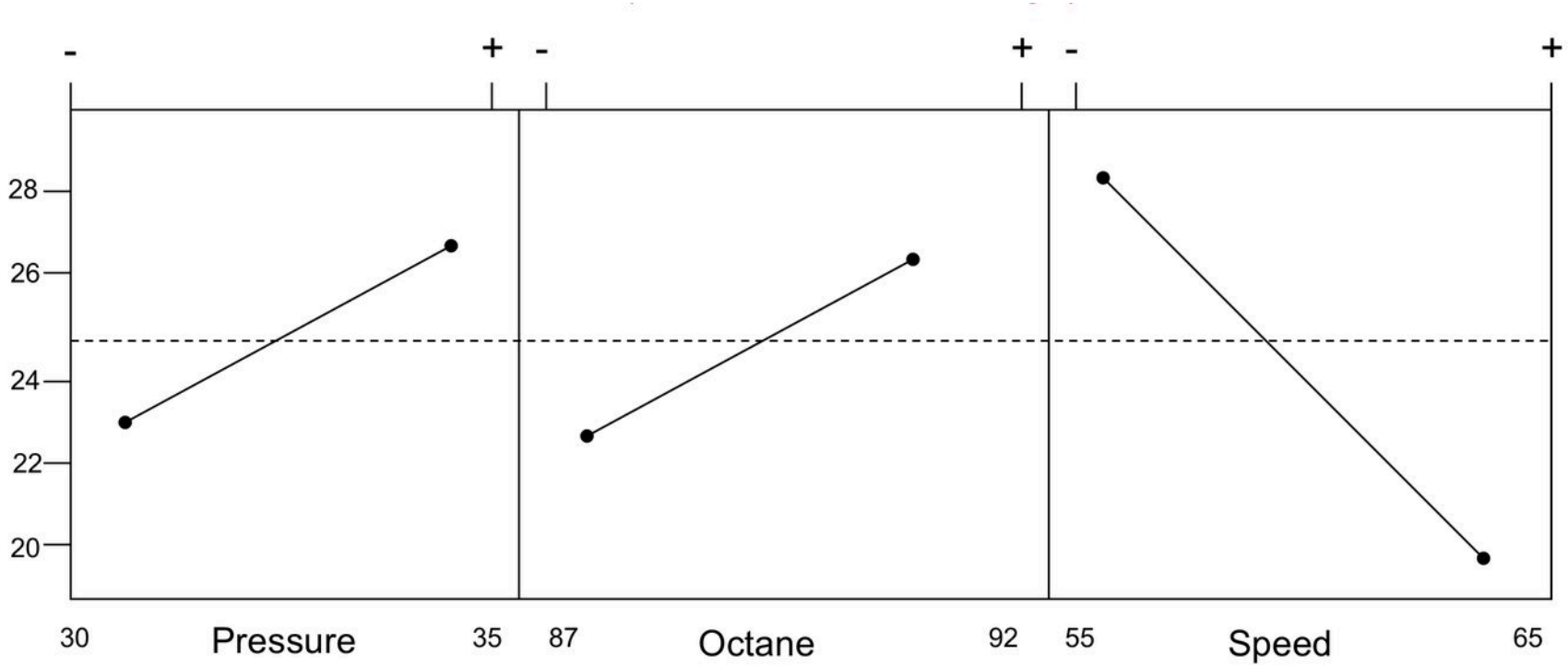
- What factor has the biggest effect on gas mileage?

The Process of Experimentation

– **Perform Analysis**

- **Collect & Analyze Data**

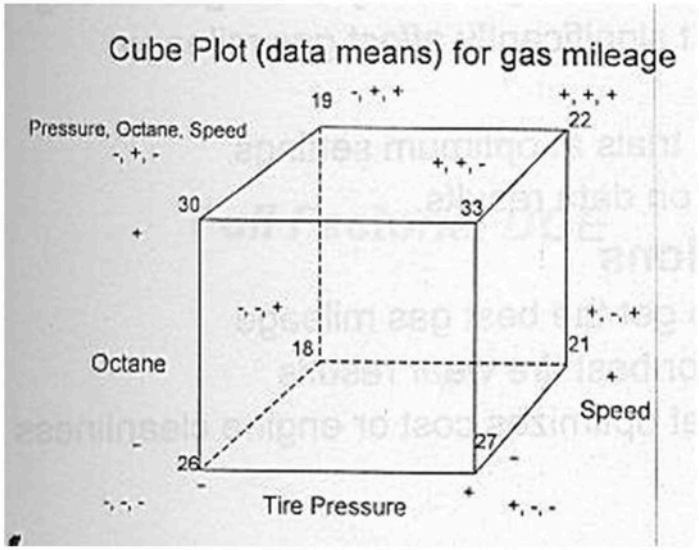
Main Effects Plot (Data Means for Mileage)



– What is the best speed to get the best gas mileage?

The Process of Experimentation

– **Perform Analysis**
  - **Collect & Analyze Data**



– Each corner of the cube represents an experimental trial from the trial table
– For example, when tire pressure is -, octane is –and speed is -, the gas mileage is 26 (x,y,z is the convention)
– Can you tell which combination of the tire pressure, octane and speed is the best by looking at the cube?

The Process of Experimentation

- Perform Analysis
  - Draw Conclusions
    - Speed has an important effect on gas mileage
    - Tire pressure does not significantly affect gas mileage.
    - Octane does not significantly affect gas mileage
  - Verify Results
    - Run additional trials at optimum settings
    - Conduct MSA on data results.
- Determine Solutions
  - Drive at 55 mph to get the best gas mileage
  - Set the tire pressure for best tire wear results
  - Buy the octane that optimizes cost or engine cleanliness
- Record Results
- Standardize
- Determine Future Plans

Full Factorial DOE

Full Factorial DOE

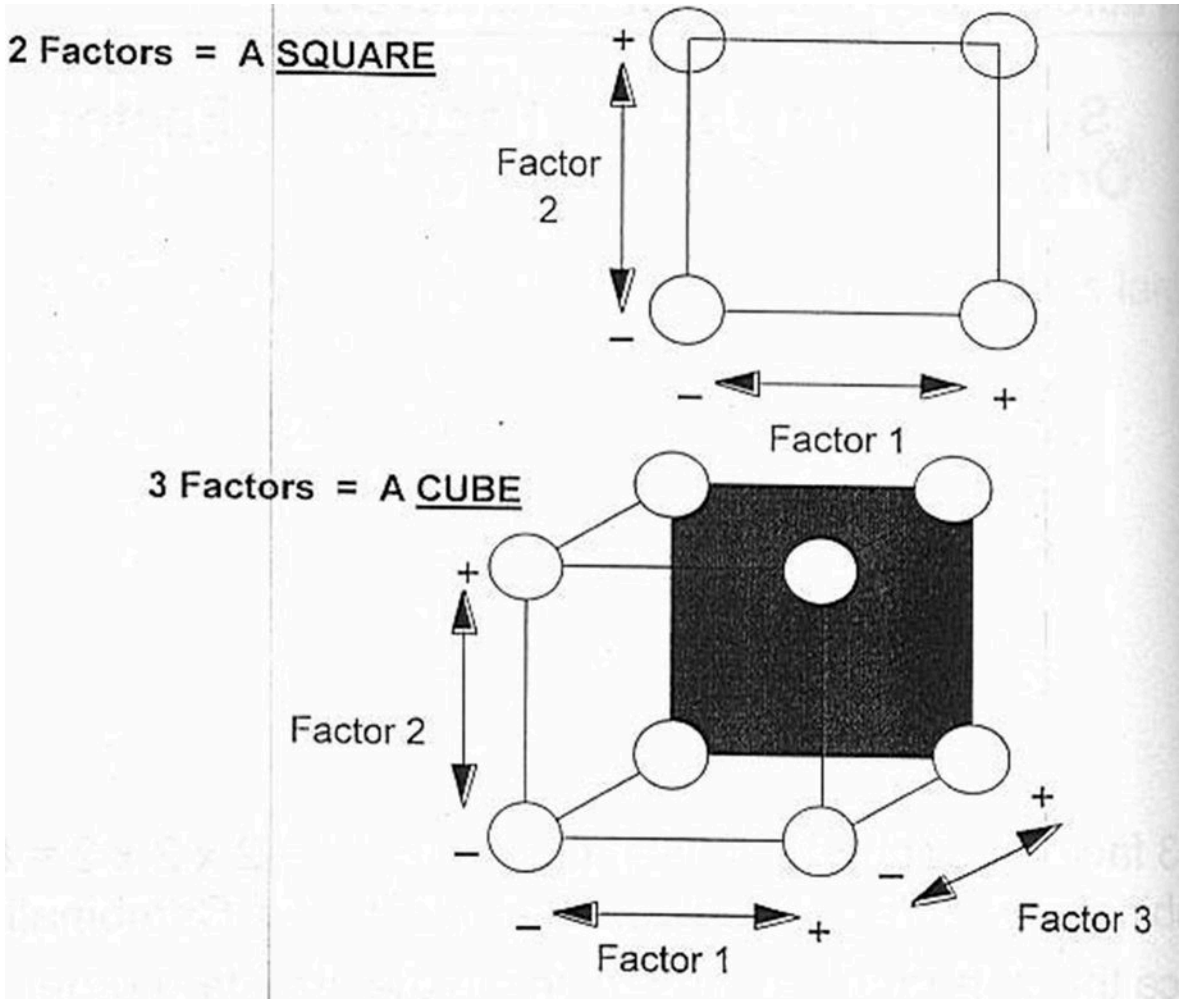
- **Full Factorial Design of Experiment**
  - **Is used to study the Vital Few X variables and all possible interactions**
  - **Can be quite costly when the number of Vital Few X's is large**
  - **Can study Main effects**
  - **Can study all interactions**

Full Factorial Layout: 3 Factors, 2 Levels

– **For 3 Factors, each at 2 levels, there are 2^3 = 2 x 2 x 2 = 8 combinations of factor settings. (Levels ^{Factors} = Combinations). Notice the pattern of factor settings in the standard order.**

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Visualizing the Experimental Space



The Factorial Pattern of Experimentation

k Factors	Order	X_1	X_2	X_3	X_4	X_5	⋯	X_k
	1	−	−	−	−	−		
k = 1	2	+	−	−	−	−		
	3	−	+	−	−	−		
k = 2	4	+	+	−	−	−		
	5	−	−	+	−	−		
	6	+	−	+	−	−		
	7	−	+	+	−	−		
k = 3	8	+	+	+	−	−		
	9	−	−	−	+	−		
	10	+	−	−	+	−		
	11	−	+	−	+	−		
	12	+	+	−	+	−		
	13	−	−	+	+	−		
	14	+	−	+	+	−		
	15	−	+	+	+	−		
k = 4	16	+	+	+	+	−		
	17	−	−	−	−	+		
	18	+	−	−	−	+		
	19	−	+	−	−	+		
	20	+	+	−	−	+		
	21	−	−	+	−	+		
	22	+	−	+	−	+		
	23	−	+	+	−	+		
	24	+	+	+	−	+		
	25	−	−	−	+	+		
	26	+	−	−	+	+		
	27	−	+	−	+	+		
	28	+	+	−	+	+		
	29	−	−	+	+	+		
	30	+	−	+	+	+		
	31	−	+	+	+	+		
k = 5	32	+	+	+	+	+		

– **The number of trials = (2 levels)** $^{(k\ \text{factors})s}$ **= 2^K**

Replication Vs Repetition

- Replication: Multiple execution of all or part of the experimental process with the same factor settings
  - Build different experimental units
  - In our car mileage example, replicate different experiments at a different time.
- Repeat tests: 2 or more observations that have the same levels for all factors
  - Performed on the same experimental unit

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Design Layout Example

2^2 design with 2 replication

Std. Over			Response
1	-	-	V_1
2	+	-	V_2
3	-	+	V_3
4	+	+	V_4
1	-	-	V_5
2	+	-	V_6
3	-	+	V_7
4	+	+	V_8

2^2 design with 2 replication

Std. Order			Obs1	Obs2	Avg
1	-	-	Y_{11}	Y_{21}	V_1
2	+	-	Y_{12}	Y_{22}	V_2
3	-	+	Y_{13}	Y_{23}	V_3
4	+	+	Y_{14}	Y_{24}	V_4

Randomization: The Experimenter's Insurance

- **Definition of Randomization: To assign the order in which the experimental trials will be run using random mechanism.**
  - **It is not the standard order**
  - **It is not running in an order that is convenient**
  - **Minitab will randomize for us**
- **Why randomization**
  - **Averages the effect of any lurking variables over all of the factors in the experiment**
- **What is a Lurking variable?**
  - **A variable that has an important effect and yet is not included among the factors under consideration**

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Analyzing DOE Data

The Process of Experimentation

- Define Project
  - Identify responses
- Establish current situation
- Perform analysis
  - Identify factors
  - Choose factor levels
  - Select design
  - Randomize runs
  - Collect data
  - Analyze data
  - Draw conclusions
  - Verify results
- Determine Solutions



The Process of Experimentation

- Record results
- Standardize
- Determine Future Plans

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Steps in Analysis: Full Factorial Replicated Designs

- Diagnostics: is data OK?
- Analysis: Make inferences

1. **Plot the raw data**
2. **Plot the residuals**
3. **Examine the factor effects**
4. **Confirm impressions with statistical procedures.**
5. **Summarize conclusions**

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Background for DOE: Capital Logistics GB Case Study

- In step 6 of the analyze Phase, the green Belt and the project team at capital logistics determined, based on the results of their Hypothesis testing, that there were 2 X's that an impact on the on-time delivery of shipments to customers. The 2 X's were :
  - Drivers
  - Distance
- The team determined that there wasn't anything that they could do to change the distance that each truck traveled to make their deliveries – the routes were automated by the system and were determined to be the shortest routes. They decided to focus on driver as a factor, as this was under their immediate control.

Background for DOE: Capital Logistics GB Case Study

- The team used the cause & Effect diagram to Brainstorm potential root causes as to why Driver / trucking Company influenced time. The resulting list included:
  - Size of the Vehicle
  - Dispatch mode
  - Type of engine
  - Type of tire
  - Fuel capacity

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Background for DOE: Capital Logistics GB Case Study

- After reviewing their data further , the team decided to conduct a Design of Experiment. They would use size of vehicle and tire type as possible vital X's in their DOE. Each of these factors has 2 levels:
  - Size of vehicle is either 66 or 62 feet.
  - Tire Type is either X or Y
- The DOE was set-up and conducted to determine the vital X that should have the greatest impact on the timeliness of shipment deliveries to their customers.

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Examine Factor Effects

– **The 3rd Step in "Analyze data"**

$$Y = f(X_1, X_2, X_3, \ldots\ldots X_n)$$

Y (response) = On-time Delivery of Shipment
Xi (Factors) = Size of Truck, Tire Type.
How do the factors affect the response?
How do the combinations (interactions) of factors affect the response?
We can write the equation that answer these questions.
This is your Y = f(X) Transfer function: On-Time Delivery = Constant + (-) Size of Truck + Tire Type + (-) Size of Truck ∗Tire Type Interaction Effect.

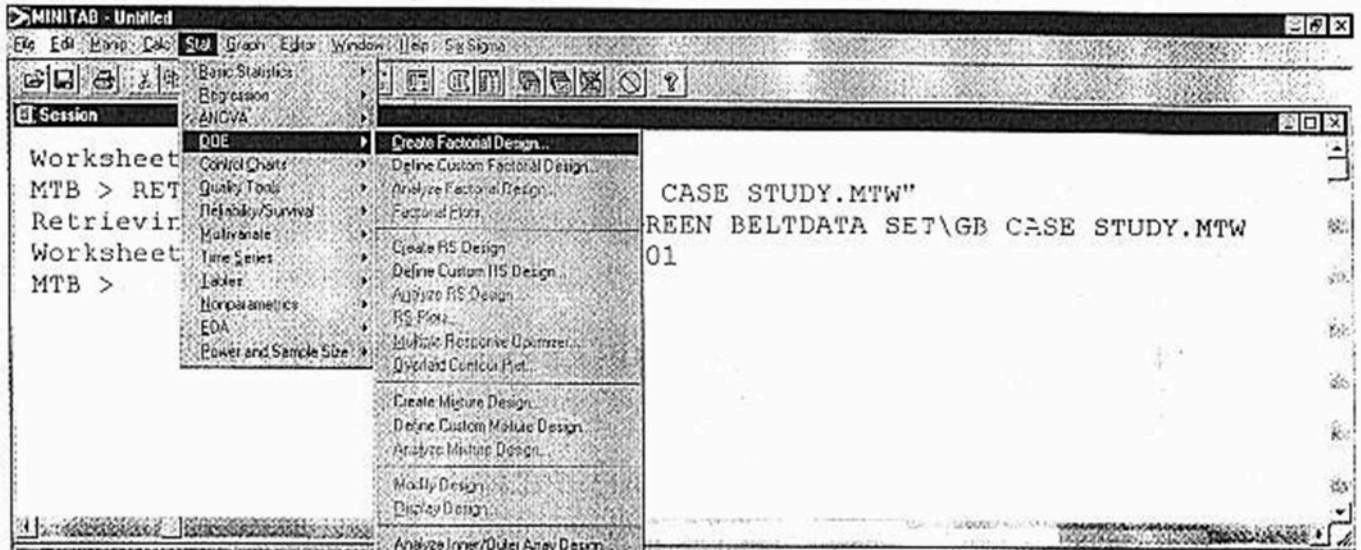
Factor Names & Interaction Symbols

- It is useful to abbreviate factor names. There are several ways to do this:
  - Choose the first letter or first two letters of each factor.
  - Assign factors a letter A,B,C,D,….
- Denote the interaction between Factor A and factor B as:
  - A x B or AB
- For two factors (A and B) there is one interaction: AB
- For Three factors there are
  - Three two-way interactions AB, AC, BC
  - One three-way interaction ABC
- How many interactions are there for four factors (A,B,C,D)?

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Examine Factor Effects – In Minitab

Create Factorial Design–in Minitab Data File: GB Case Study.mtw

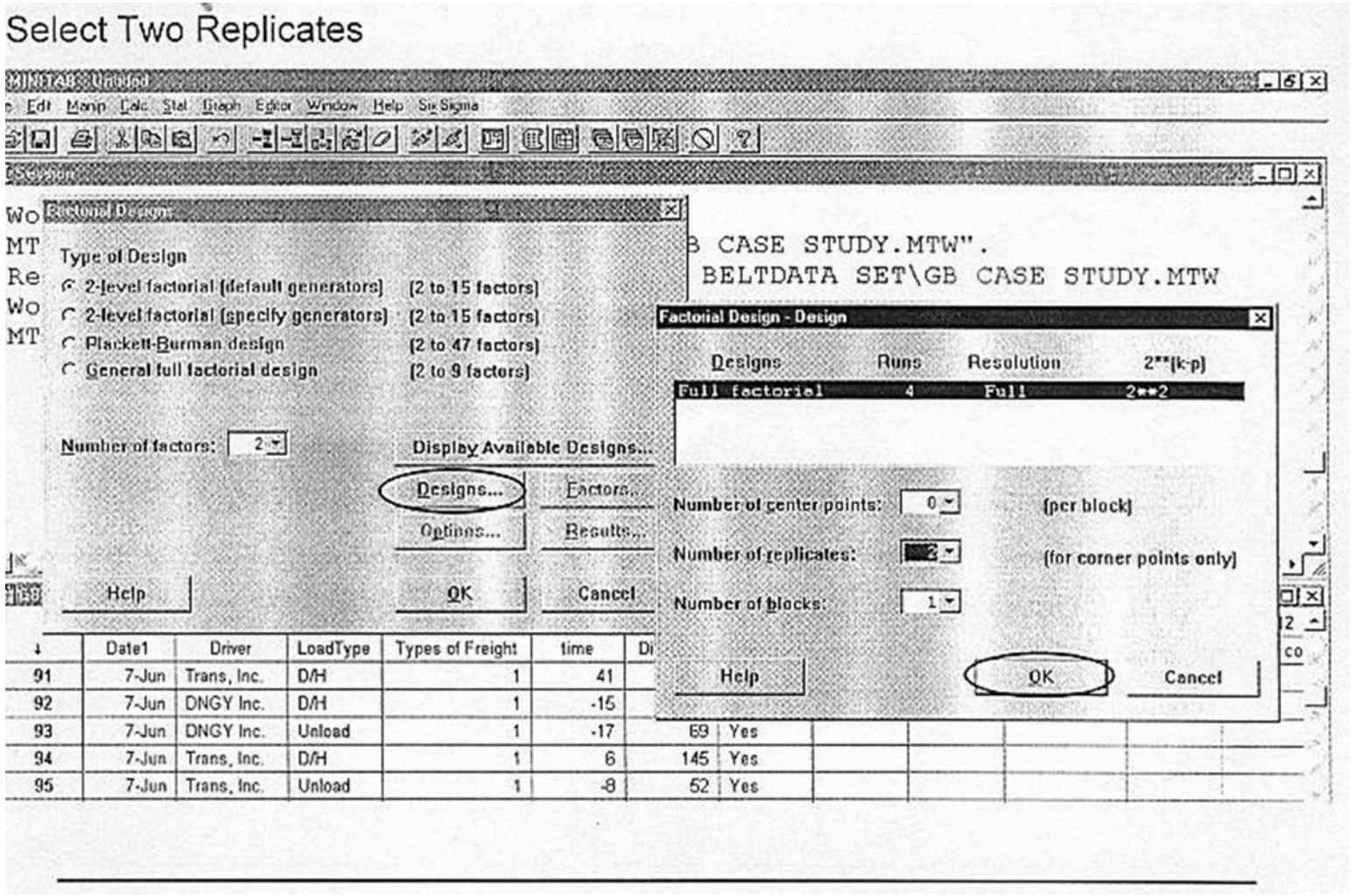


	C1-D	C2-T	C3-T	C4	C5	C6	C7-T	C8	C9	C10	C11-D	C12
	Date1	Driver	LoadType	Types of Freight	time	Distance	On Time	Units2	Defects	Opportunities	Date	defect co
1	3-Jun	Cap. Log	D/H	1	-60	144	Yes	50E	89	1	1/10/01	
2	3-Jun	Cap. Log.	Unload	1	-75	122	No				1/11/01	
3	3-Jun	Temp Labor	Unload	1	-0	80	Yes				1/12/01	
4	3-Jun	Temp Labor	D/H	1	27	53	Yes				1/13/01	
5	3-Jun	Temp Labor	D/H	1	24	40	Yes				1/14/01	
6	3-Jun	Trans, Inc.	Unload	1	-33	42	Yes				1/15/01	
7	3-Jun	Trans, Inc.	D/H	1	-33	72	Yes				1/16/01	
8	3-Jun	Trans, Inc.	D/H	1	-2	42	Yes				1/17/01	

Type of design = full factorial

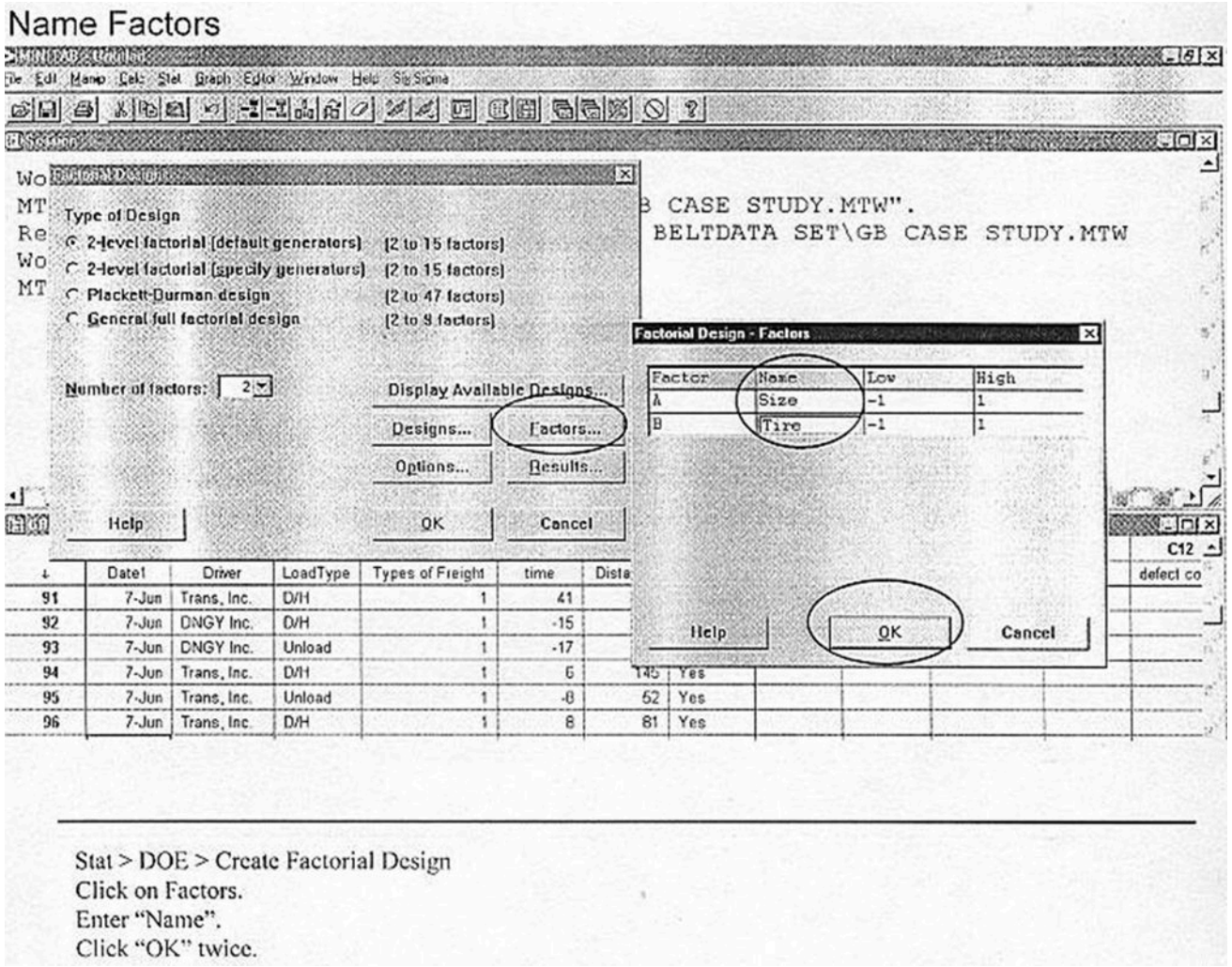
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Examine Factor Effects – In Minitab



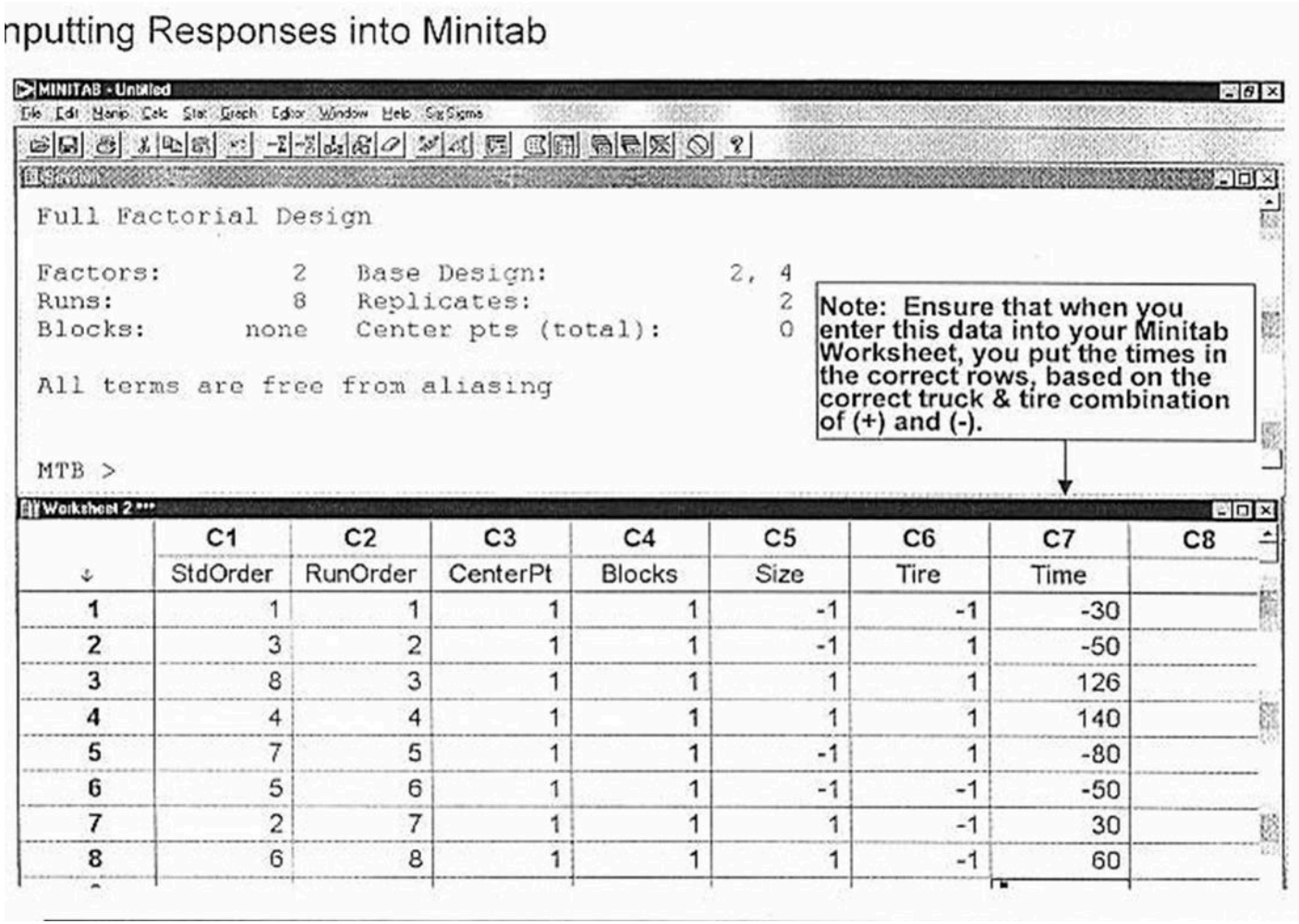
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Examine Factor Effects – In Minitab



Examine Factor Effects – In Minitab

nputting Responses into Minitab



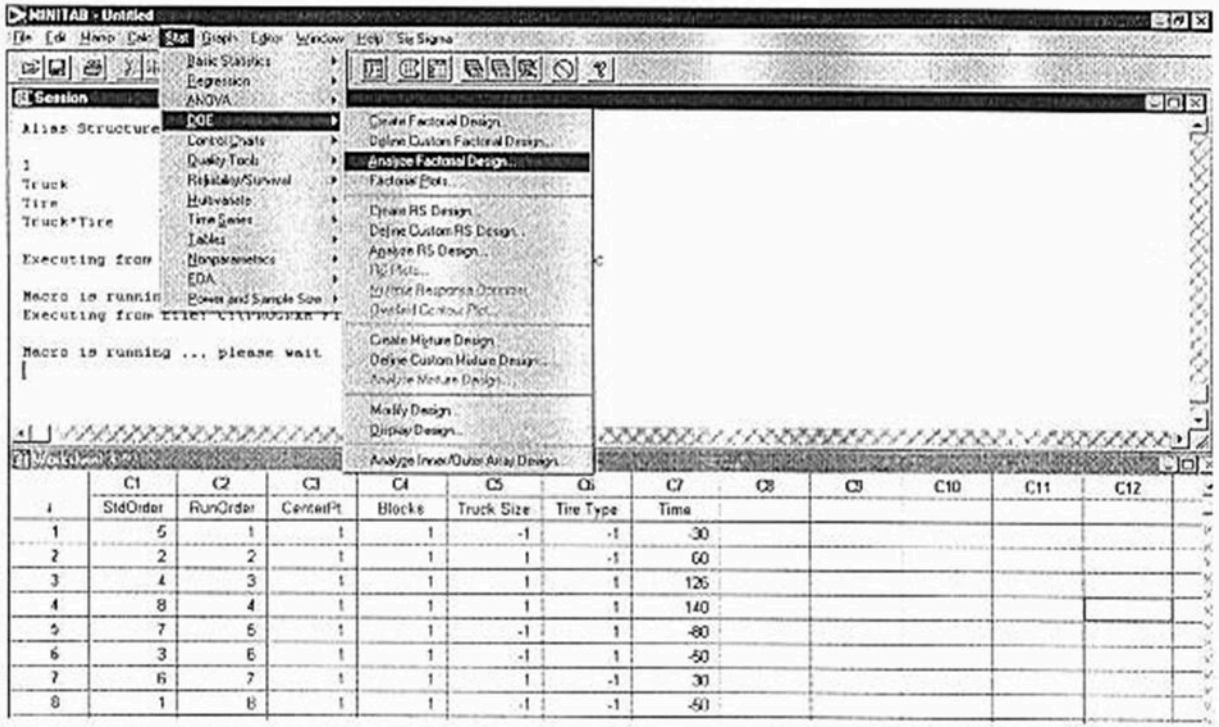
Worksheet 2 ***

	C1	C2	C3	C4	C5	C6	C7	C8
↓	StdOrder	RunOrder	CenterPt	Blocks	Size	Tire	Time	
1	1	1	1	1	-1	-1	-30	
2	3	2	1	1	-1	1	-50	
3	8	3	1	1	1	1	126	
4	4	4	1	1	1	1	140	
5	7	5	1	1	-1	1	-80	
6	5	6	1	1	-1	-1	-50	
7	2	7	1	1	1	-1	30	
8	6	8	1	1	1	-1	60	

Examine Factor Effects – In Minitab

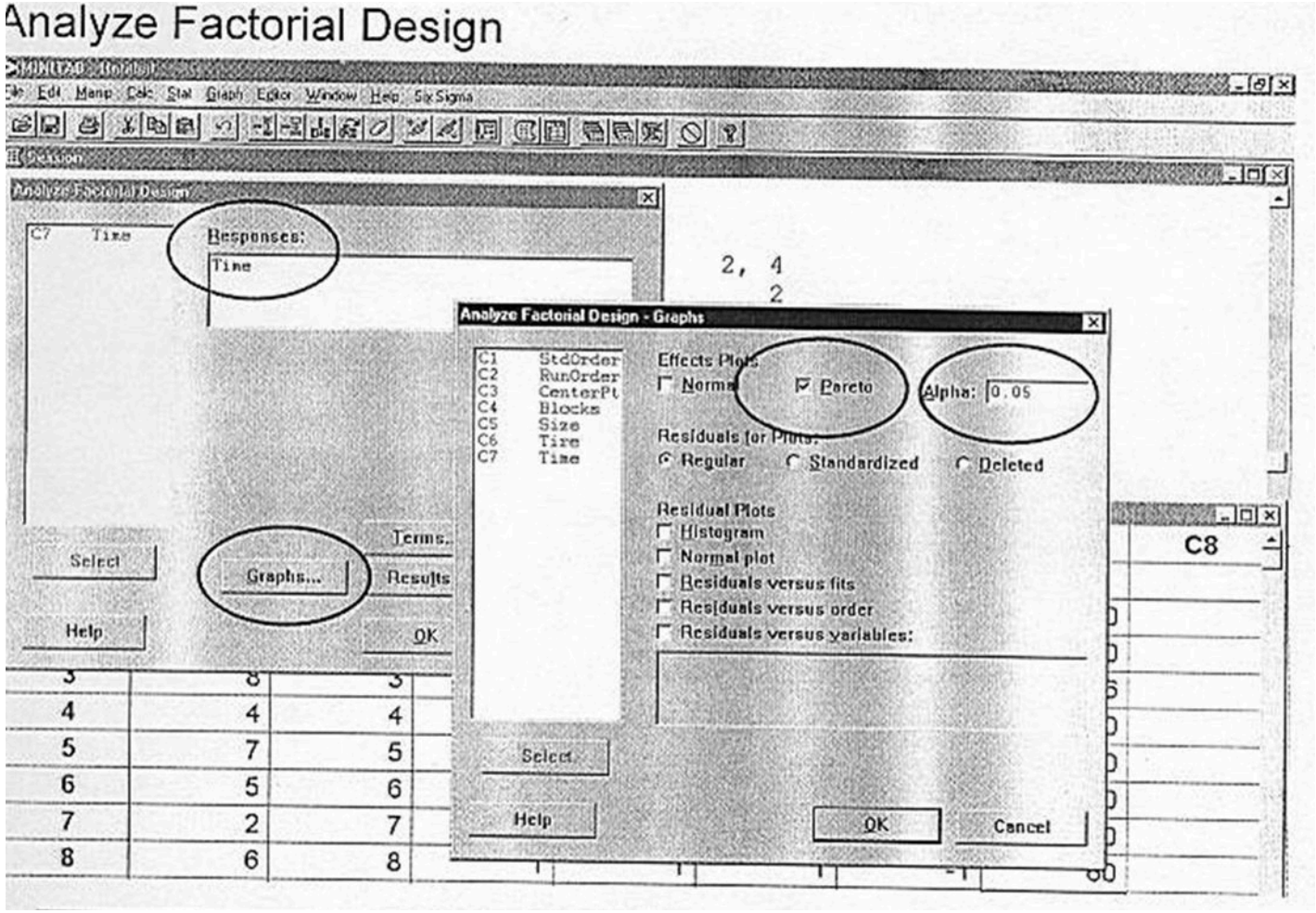
Analyze Factorial Design

In Minitab: Stat > DOE > Analyze Factorial Design



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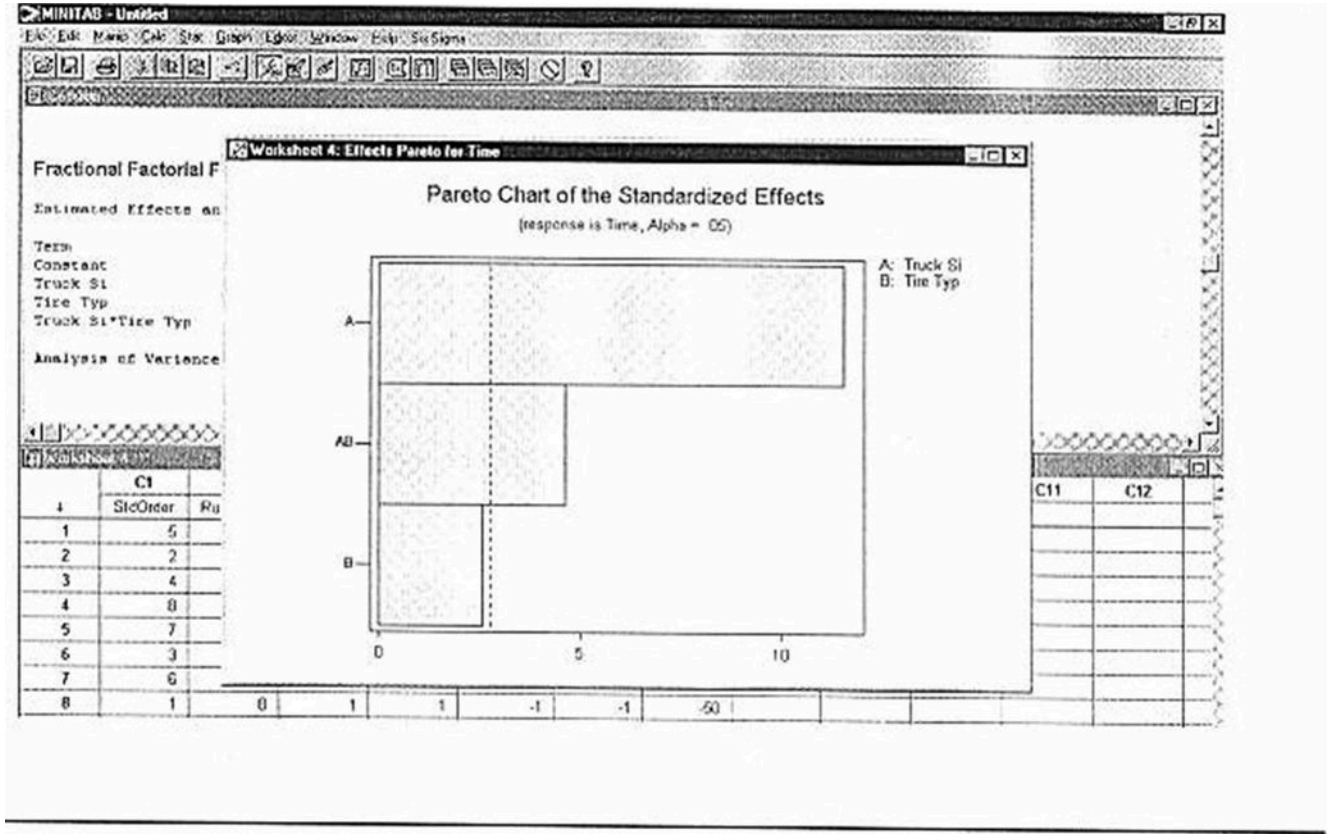
Examine Factor Effects – In Minitab



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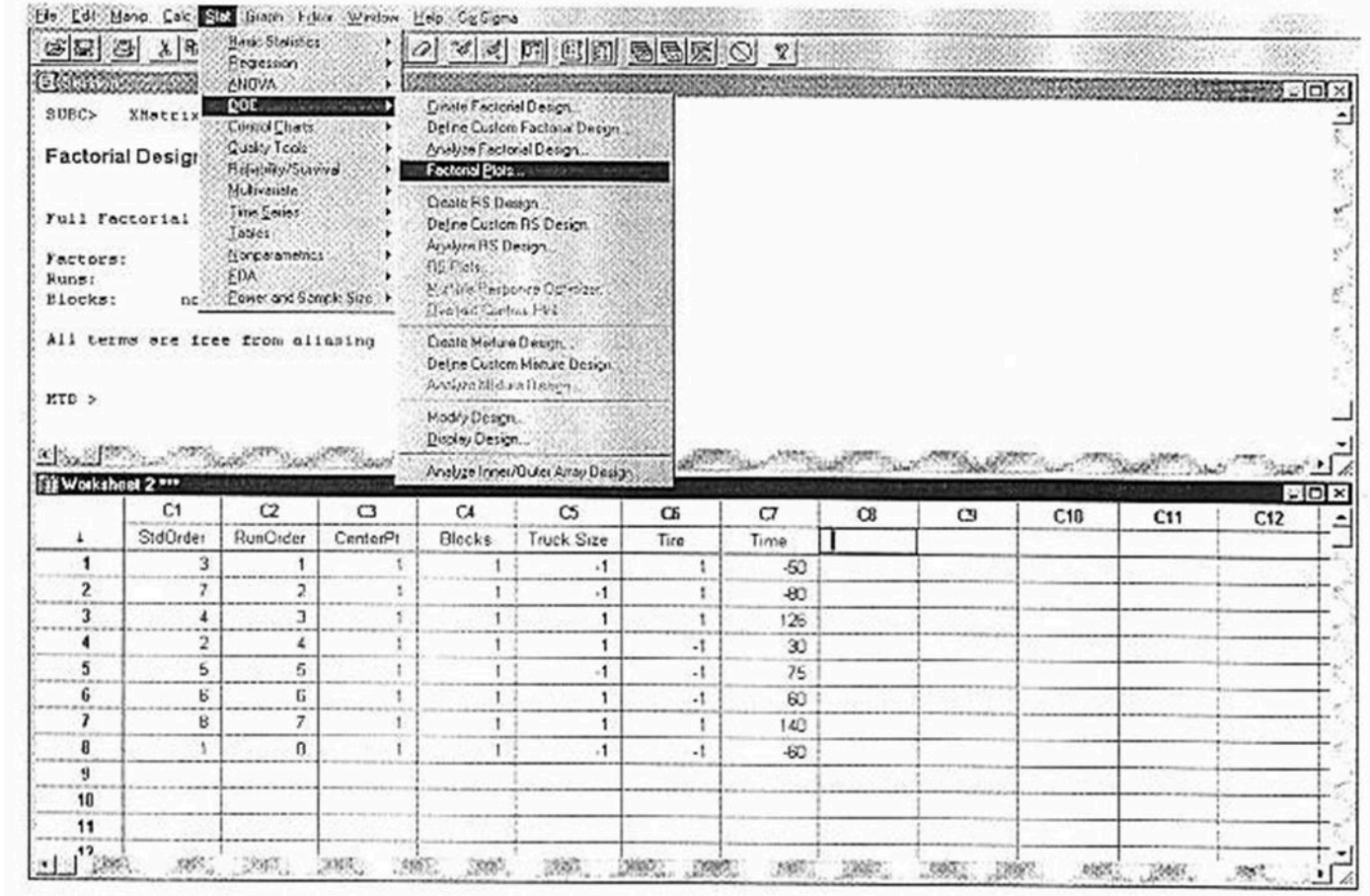
Examine Factor Effects – In Minitab

– **What does the Pareto Chart tell you about the factors & interactions?**



Examine Factor Effects – In Minitab

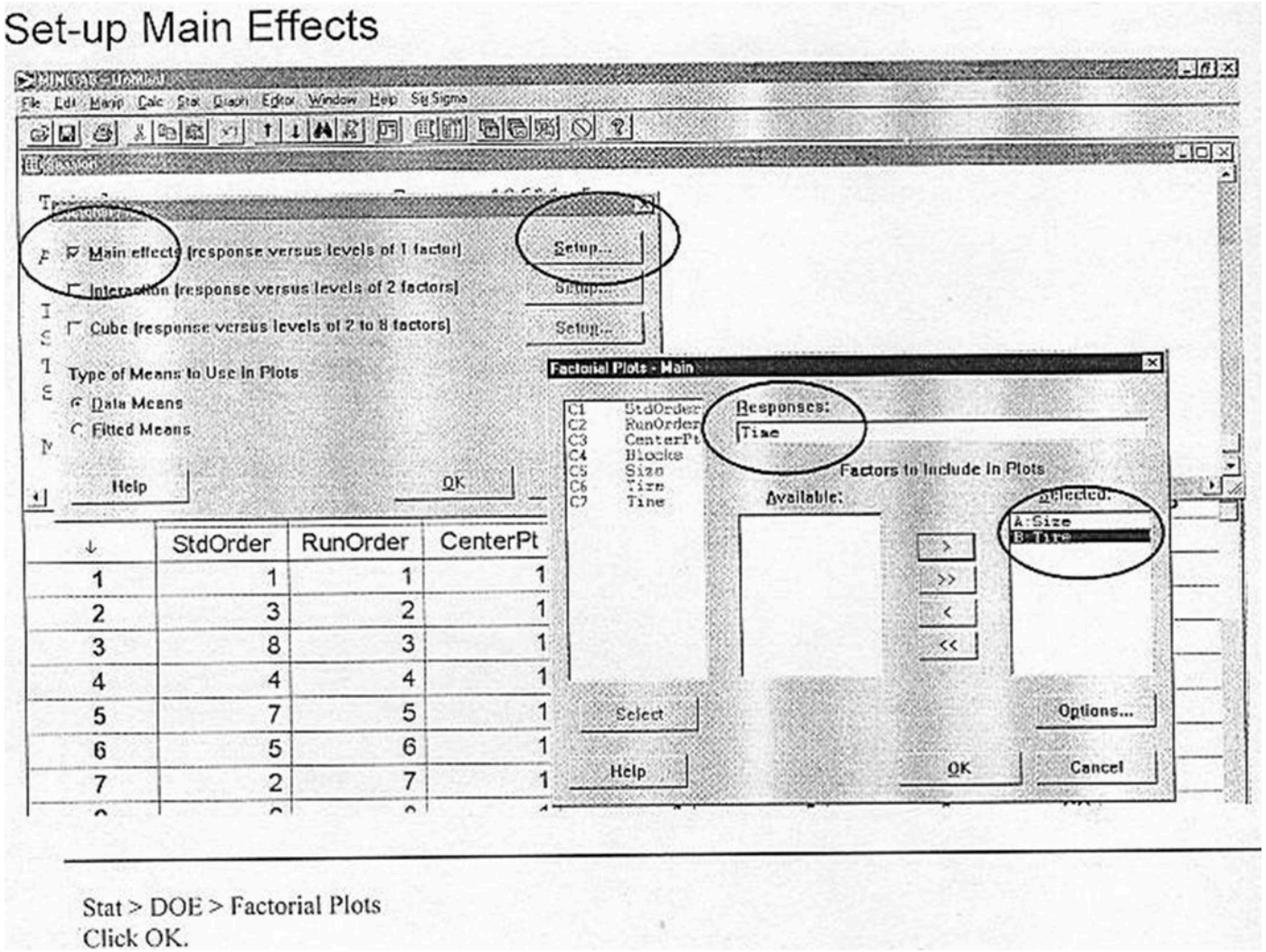
Analyze Factorial Plots



	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12
	StdOrder	RunOrder	CenterPt	Blocks	Truck Size	Tire	Time					
1	3	1	1	1	-1	1	-50					
2	7	2	1	1	-1	1	-80					
3	4	3	1	1	1	1	126					
4	2	4	1	1	1	-1	30					
5	5	5	1	1	-1	-1	75					
6	6	6	1	1	1	-1	60					
7	8	7	1	1	1	1	140					
8	1	8	1	1	-1	-1	-60					
9												
10												
11												
12												

Examine Factor Effects – In Minitab

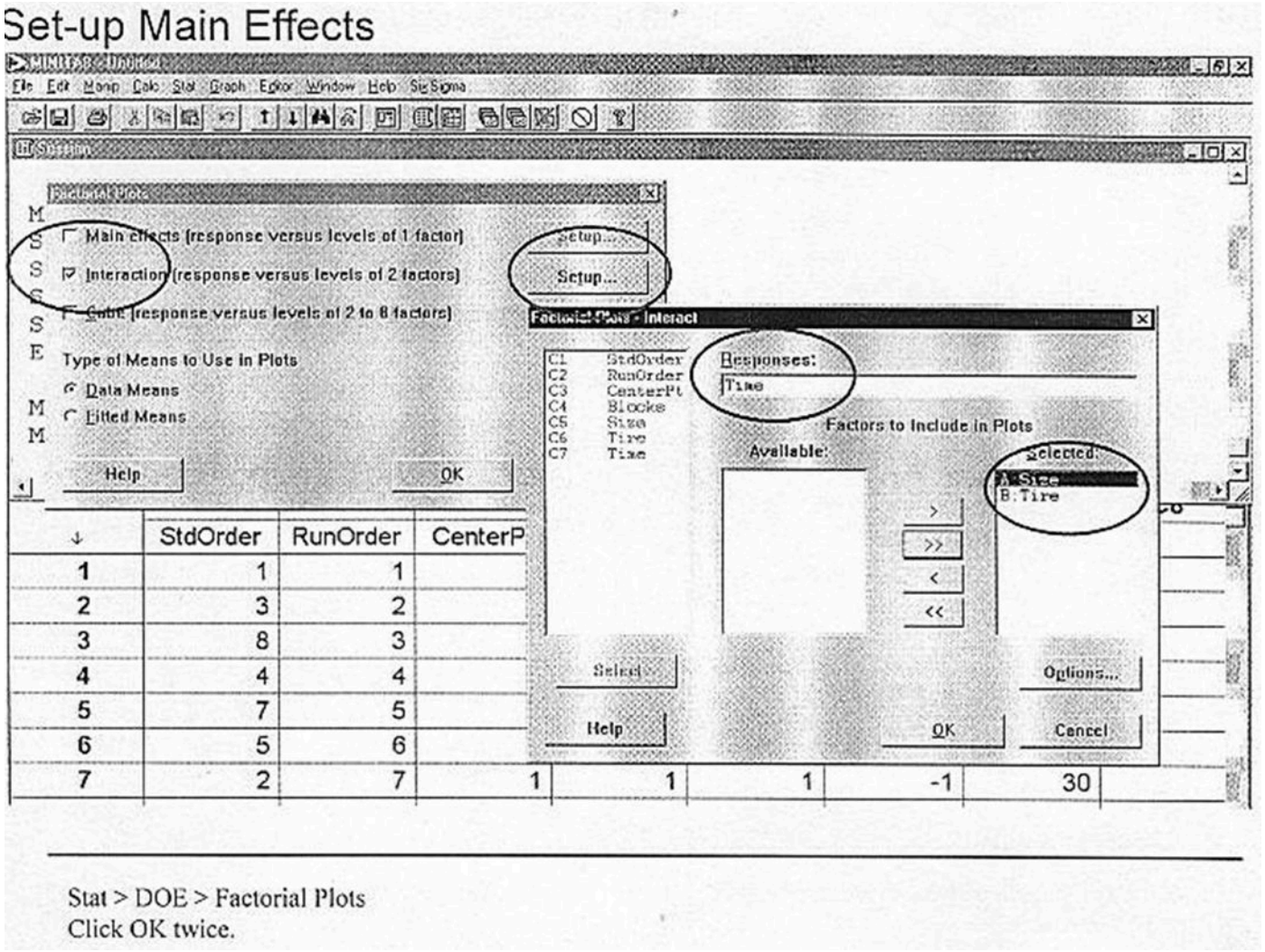
Set-up Main Effects



	StdOrder	RunOrder	CenterPt
1	1	1	1
2	3	2	1
3	8	3	1
4	4	4	1
5	7	5	1
6	5	6	1
7	2	7	1

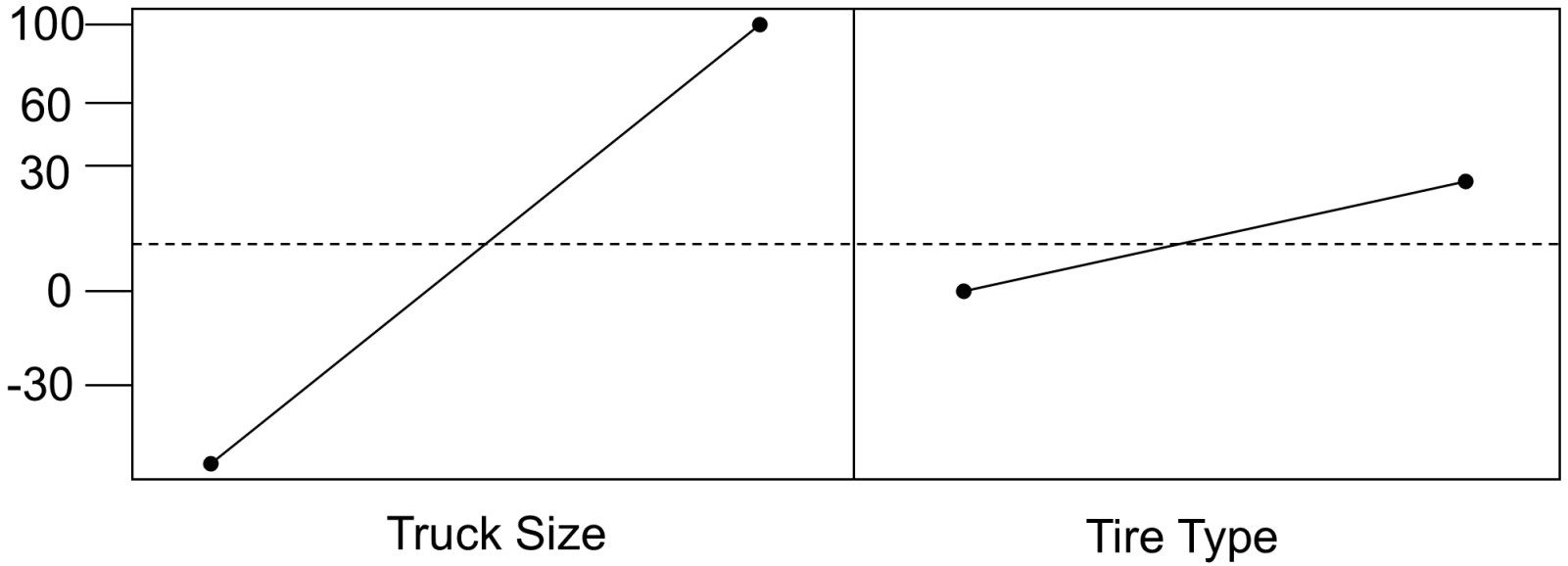
Stat > DOE > Factorial Plots
Click OK.

Examine Factor Effects – In Minitab



Main Effect Plot

– **In GB Capital Logistics on-time delivery Case Study**



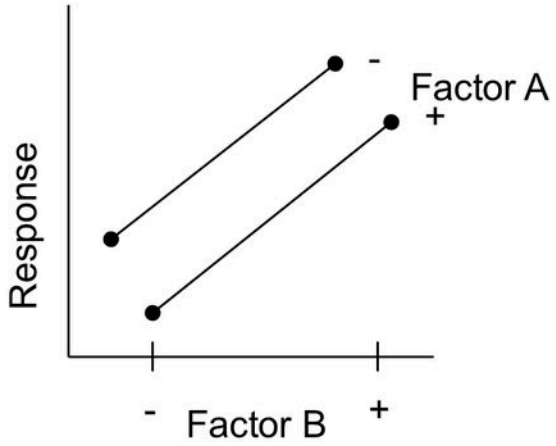
How do truck Size & Tire Type affect Time?

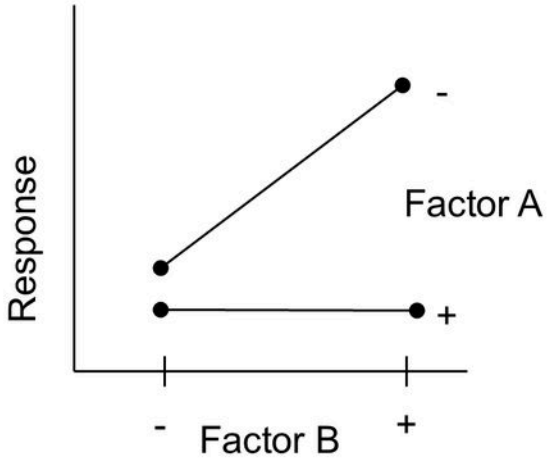
Interaction Effect

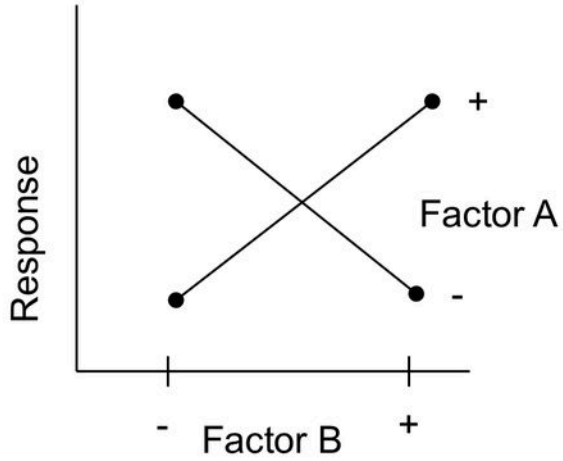
- Differences between 2 or more factor level combinations :
- In our Capital Logistics GB Case Study: On-Time Delivery
- If we are looking at 2 factors: Truck Size & Tire Type.
- Is on-time performance different when:

Interaction Effect

– **Which of the following indicates interactions between factors?**

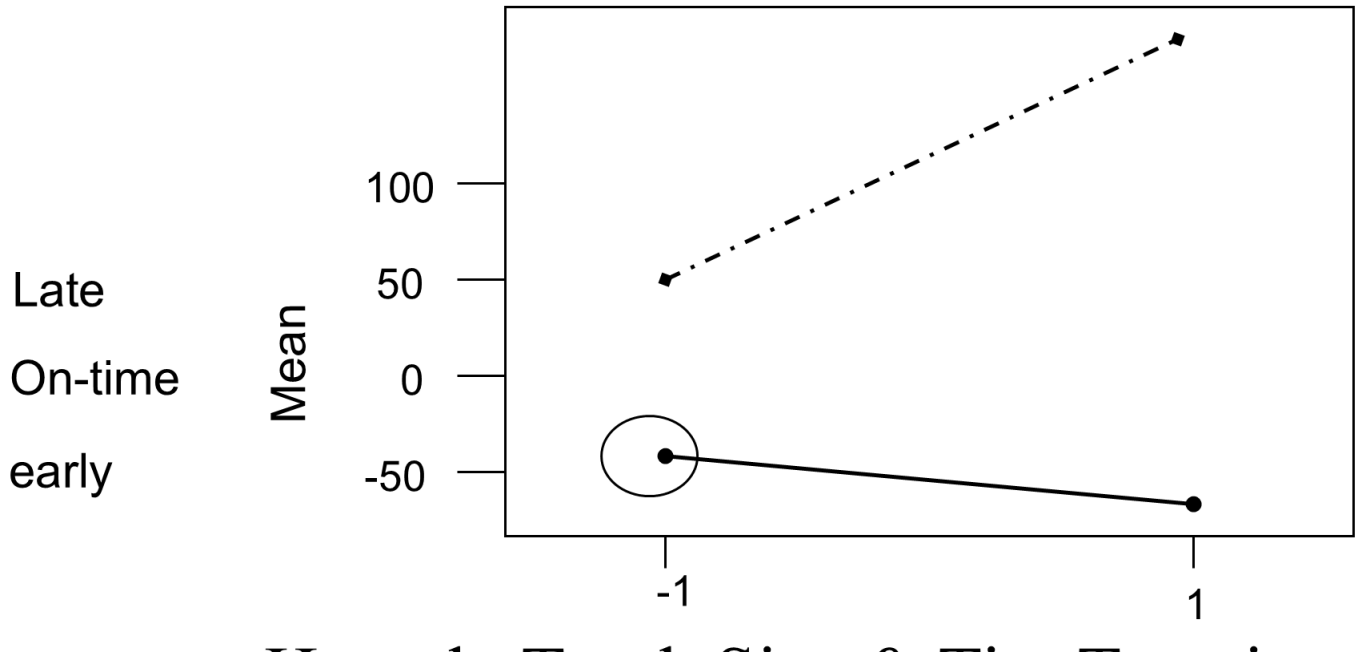






Interaction Effect

– **In our GB Capital Logistics on-time delivery case Study:**



– How do Truck Size & Tire Type interact?
– How do the interactions affect time?

Steps in Analysis: Full Factorial Replicated Designs

- Diagnostics: is data OK?
- Analysis: Make inferences

1. **Plot the raw data**
2. **Plot the residuals**
3. **Examine the factor effects**
4. **Confirm impressions with statistical procedures.**
5. **Summarize conclusions**

factor effects

4. Confirm impressions

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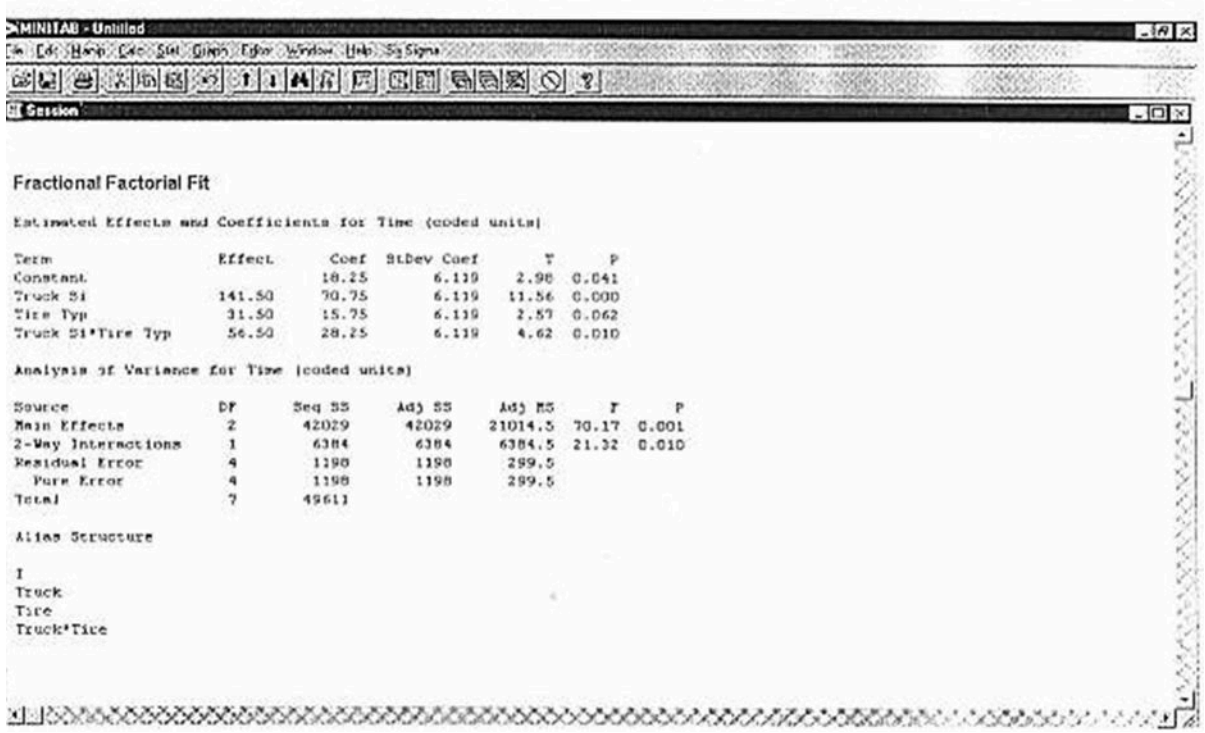
Confirm Impressions

- **The 4th Step in "Analyze data":**
  - **By Now you have a good idea about the influence of the factors in the experiment. You can confirm these impressions statistically by performing a Hypothesis test.**
    - H_o: Factor has no effect on the result.
    - H_a: Factor has an effect on the result
    - $p \geq \dot{\alpha}$:H_o Accept H_o
    - $P < \dot{\alpha}$: Reject H_o

Statistical Analysis Confirms the results of the Graphical Analysis

Confirm Impressions with Minitab

– **In our GB Case Study: What does the Minitab Session Window tell you?**



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Steps in Analysis: Full Factorial Replicated Designs

- Diagnostics: is data OK?
- Analysis: Make inferences

1. **Plot the raw data**
2. **Plot the residuals**
3. **Examine the factor effects**
4. **Confirm impressions with statistical procedures.**
5. **Summarize conclusions**

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Summarize Conclusions

- The 5th Step in "Analyze Data
- List all the conclusions you have made during the analysis
- Interpret the meaning of these results
  - For example, relate them to known physical properties or an expert's personal knowledge of the process.
- Make recommendations
- Formulate and write conclusions in simple language

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Summarize Conclusions

- **For our capital logistics GB case study On-Time Delivery:**
  - **The Green Belt and the project team reviewed the results of the DOE. They saw that there seemed to be a link between on-time performance and the size of the truck unit, as well as an interaction effect between tire type and size of truck on on-time performance.**
  - **They returned to their Fishbone (Cause & Effect) Diagram with greater granularity, and asked why these factors would influence on-time performance. They Validated these reasons through field observations.**

Summarize Conclusions

- **For our capital logistics GB case study On-Time Delivery:**
  - **The team validated the larger size trucks took longer at some locations. In addition to time involved with maneuvering in a cramped environment, on certain routes these trucks had a higher incidence of tire failure (lateral movement caused by tight turns increased the stress on the tires, increasing the number of flat tires)**
  - **By using the data & results from DOE, the team had determined the root cause of the on-time performance problem and were ready to start to brain storm solutions to address / control these "Vital X's".**

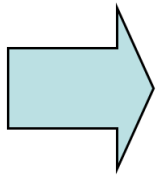
Reducing the size of Experiments Fractional Factorial DOE

Reducing the size of Factorial Experiments

- Many factors potentially impact the quality of any process / product
- The factorial strategy is an efficient approach to experimentation, compares to “one-at-a-time”
- When factors are investigated at 2 levels, the numbers of experimental runs is 2^k (note: the ‘k’ needs to be raised, because it is 2 to the power of k – ‘k’ is the exponent)
- This can result in a large number of runs, even with a relatively small number of factors
- Limited resources allow you time to run only 4 trails in the allotted time.

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Reducing the size of Factorial Experiments



Fractional Factorials

- **Fractional Factorials:**
  - **Provide a middle ground for design of experiment**
  - **Are used to expand on the study of the Vital Few X variables at a low cost**
  - **Can study main effects**
  - **Can study selected (and few in number) interactions**

- Fractional Factorials allow you to reduce the size of an experiment and select certain runs.

Strategy for Choosing Appropriate Design

- The Knowledge line:
  - Current state of process knowledge

	LOW			HIGH
Type of Design	Main effect screening	Fractional Factorials	Full Factorials	Response Surface
Usual # of factors	>5	4-10	1-5	2-3
Purpose Identify	Main effect critical factors Vital few	Some interactions	Relationship among factors	Optimal factor setting
Estimate	Crude direction for improvement	Some interactions	All main effects & interactions	Curvature in response, empirical models
STEP 7	Screening DOE →			
STEP 8			Optimization DOE →	

Design of Experiment

- **Summary**
  - **Provides process performance data under different types of operating parameters**
  - **Establishes cause and effect relationships between a process Y and a possible X's.**
    - **Verify the Vital few X's**
    - **Determine X values for optimal performance.**
    - **Predict future process performance.**
  - **Can be used in both incremental and exponential process improvement studies.**

Caution: Prior to implementing solutions, understand the difference between experimental and actual business / process conditions

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Optimization Design of Experiment Activity (40Minutes)

- **Desired Outcome: Practice how to set-up a DOE**

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Optimization Design of Experiment Activity (40Minutes)

- **Desired Outcome: Practice how to set-up a DOE**

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DOE – Breakout Activity Worksheet

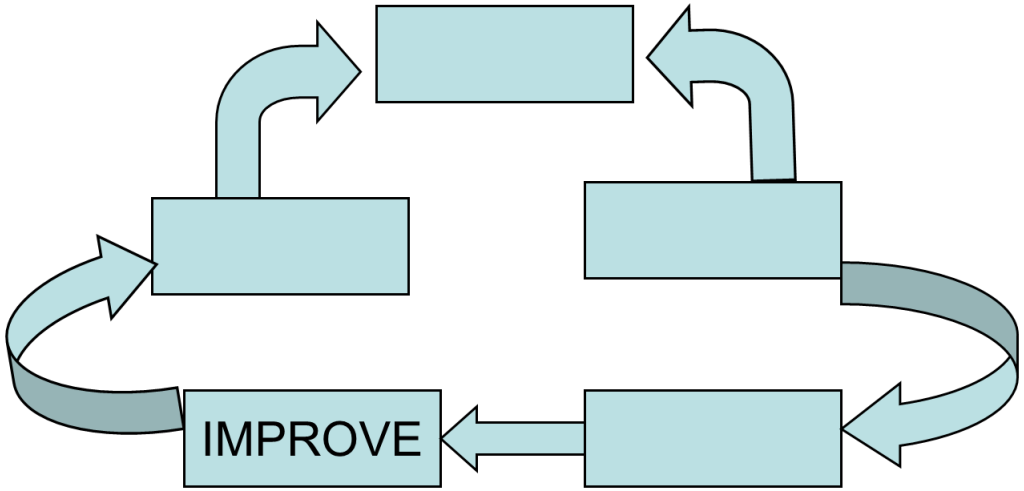
1. **Define the problem** _____
2. **State the Hypothesis** _____
3. **Identify the dependent 9Y) and independent (X) variables**
   - **Dependent Variable (Y) =** _____
   - **Independent Variable (X's) =** _____
   - **=** _____
   - **=** _____
4. **Determine the test levels for each independent variable (X)**

5. Calculate the number of trials to test all combinations of test levels.

DOE – Breakout Activity Worksheet

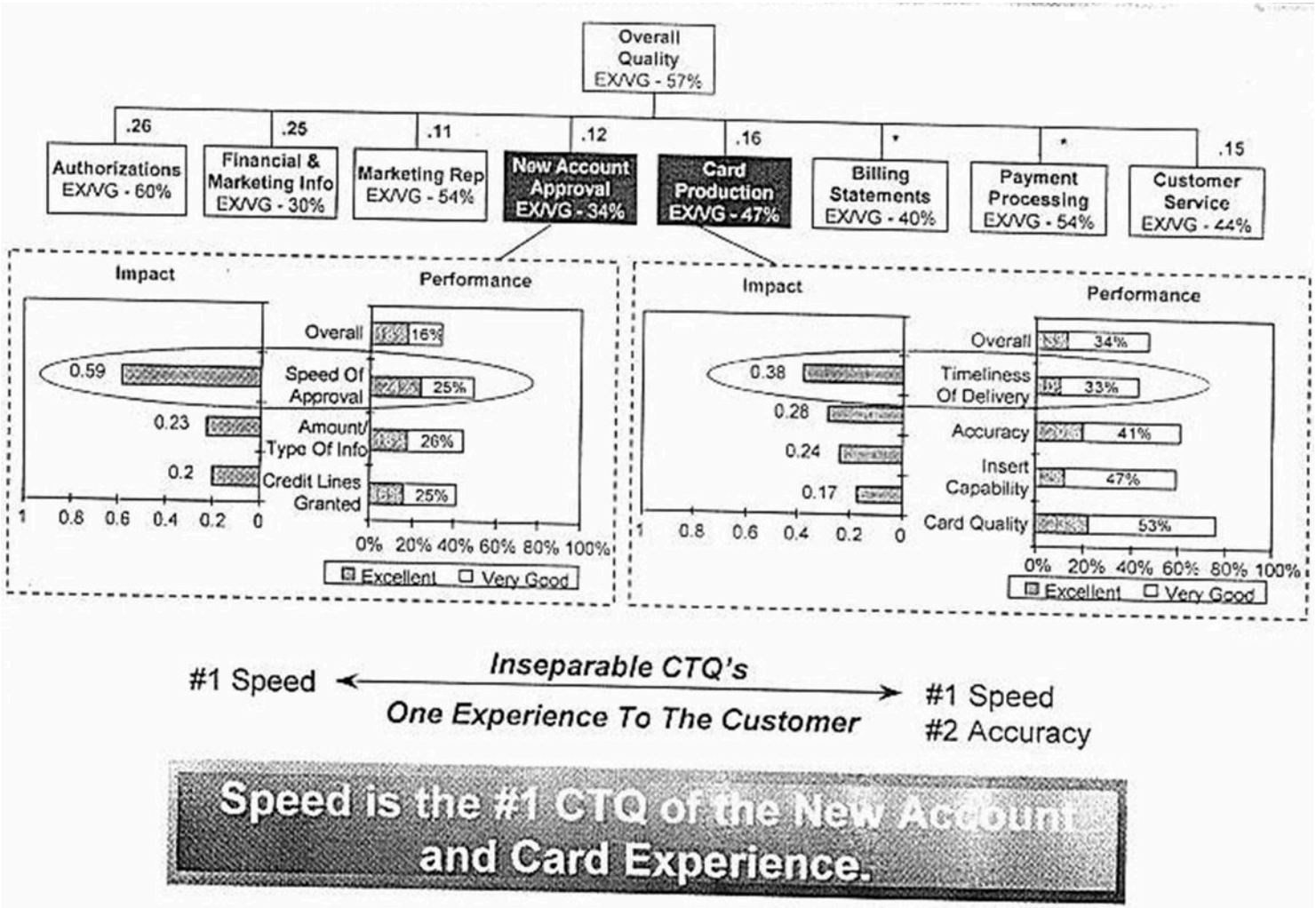
– **Construct the experimental trials table**

Example of designed Experiment: GE Card Service



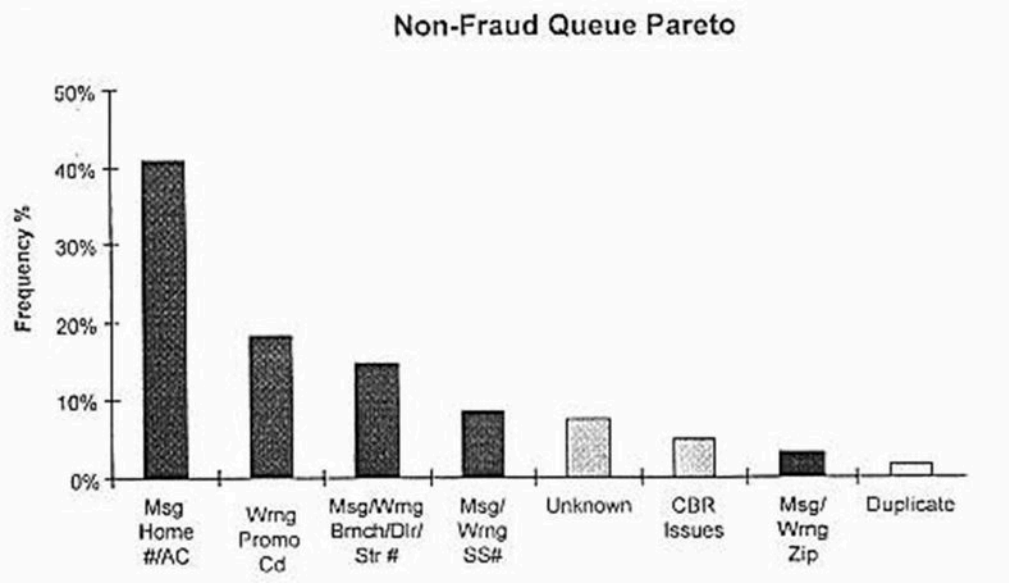
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DOE Example: Defining the CTQ



DOE Example: Causes of Time Consuming Delays

– **79% of queuing for non-fraud! Why?**



– Missing and illegible information is a major cause of delays. What can be done to reduce missing and illegible information?
– Many felt that little could be done to improve the situation for handwritten application.

DOE Example: The Designed Experiment

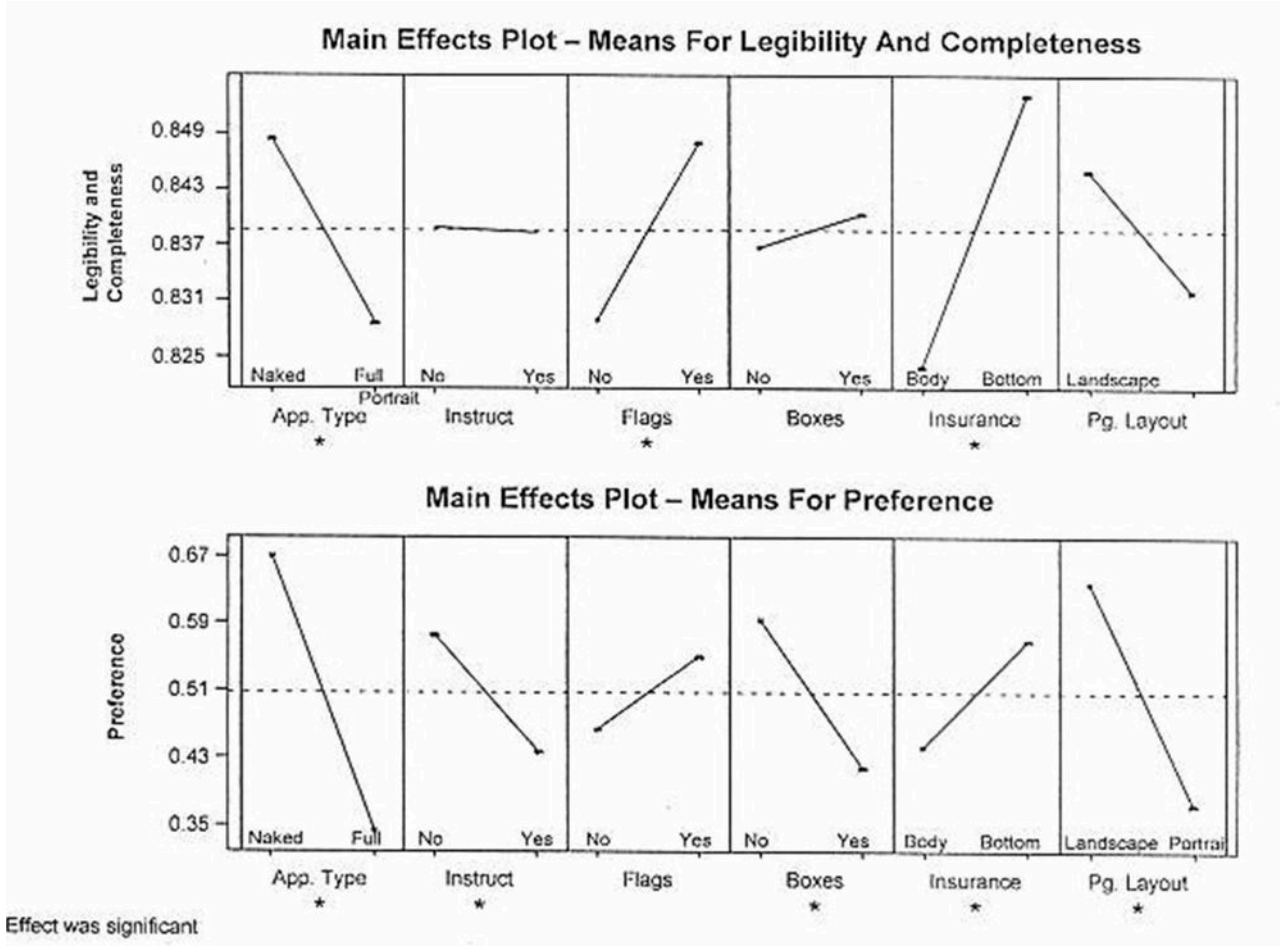
DOE Example: Experimental Complications

- **Complications**
  - The difficulty in tracking many (in this case, 16) different applications through the process.
  - Large amount of person-to-person variation with respect to legibility and completeness, making it difficult to assess the effect of change.
- **Remedies**
  - Test the applications outside the normal process.
  - Have each tester fill out two applications.

DOE Example: Responses

- **Two Key responses were studies**
  - **A combined completeness and legibility score, which was based on the percent of necessary information that was both present and legible.**
  - **A preference score, which told which of the two applications the customer prefered.**

DOE Example: Analysis



Effect was significant

DOE Example: Results

- The data does not show that instructions improve legibility and completeness, but it is felt that instructions should be provided for clarification purpose. Also consideration should be given to improve the effectiveness of the instructions provided on the application.
- Effect not significant.

Design of Experiments: CRD, RBD and Factorial Designs

Design of experiments

Completely Randomized design(CRD)

A product development engineer is interested in maximizing the tensile strength of a new synthetic fiber that is used to make Clothe for shirts. Determine whether varying the cotton content in the synthetic fiber affects the tensile strength of the clothe.

	% of cotton					
		15	20	25	30	35
	1	7	7	15	11	9
	2	12	17	12	18	18
	3	14	18	18	19	19
	4	19	25	22	19	23
	5	7	10	11	15	11

Tensile strength lb/in^2

ANOVA TABLE:

Source	Degrees of freedom	Sum of squares	Mean squares	F_{CAL}	F_{TAB} F(0.05,4,20)
Due to treatments (% of cotton)	4	475.76	118.94	14.76	2.87
Due to error	20	161.20	8.06	------	
Total	24	693.96	-----		

Calculation:

$$\text{Total} = 376$$

$$N = \text{no. of observations}$$

$$\text{Correction factor (CF)} = T^2/N$$
$$= (376)^2/25$$
$$= 5655.04$$

$$\text{SS}_{\text{Treatments}} = \sum C_j^2/5 - \text{CF}$$
$$= (49^2 + 77^2 + 88^2 + 108^2 + 54^2)/5 - 5655.04$$
$$= 475.76$$

$$\text{SS}_{\text{Total}} = \sum\sum X^2 - C.F$$
$$= 636.96$$

TESTING OF HYPOTHESIS

H_O : % Cotton content does not affect tensile strength of the synthetic fiber.

$$i.e.\ \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$$

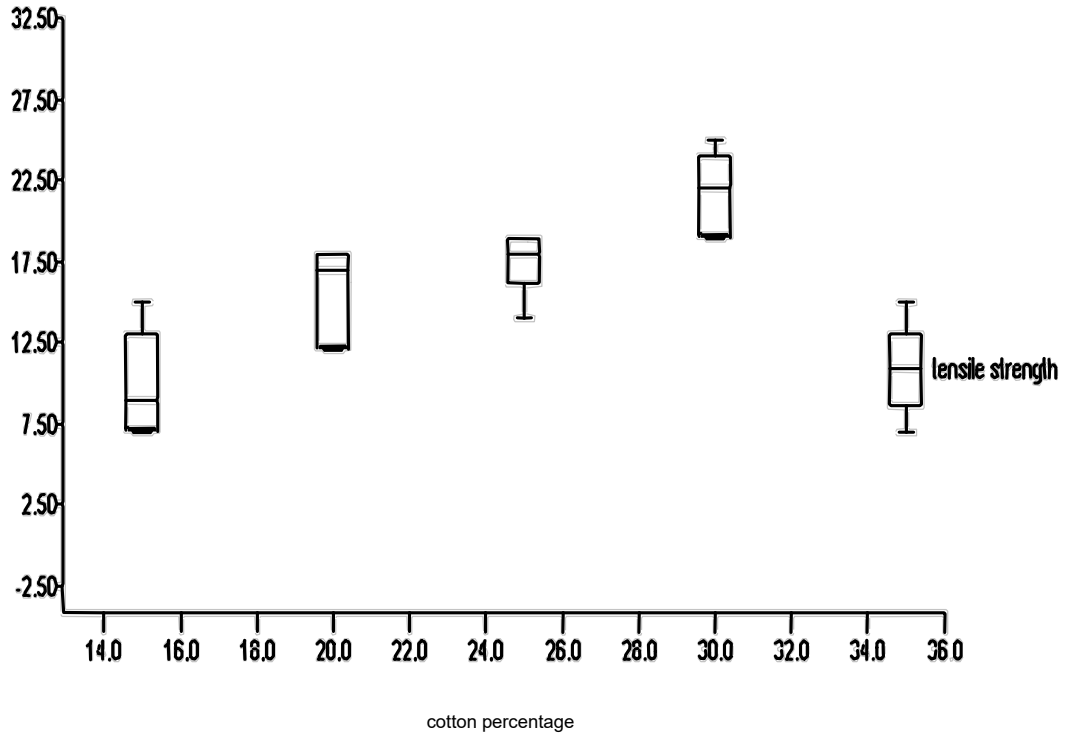
H_I : % of cotton content affect tensile strength of the synthetic fiber.

i.e. $\mu_1 \neq \mu_2 \neq \mu_3 \neq \mu_4 \neq \mu_5$

Since $F_{CAL} > F_{TAB}$ H_O is rejected and may be concluded that averages differ i.e. Percentage of cotton in the synthetic fiber significantly affects the average tensile strength.

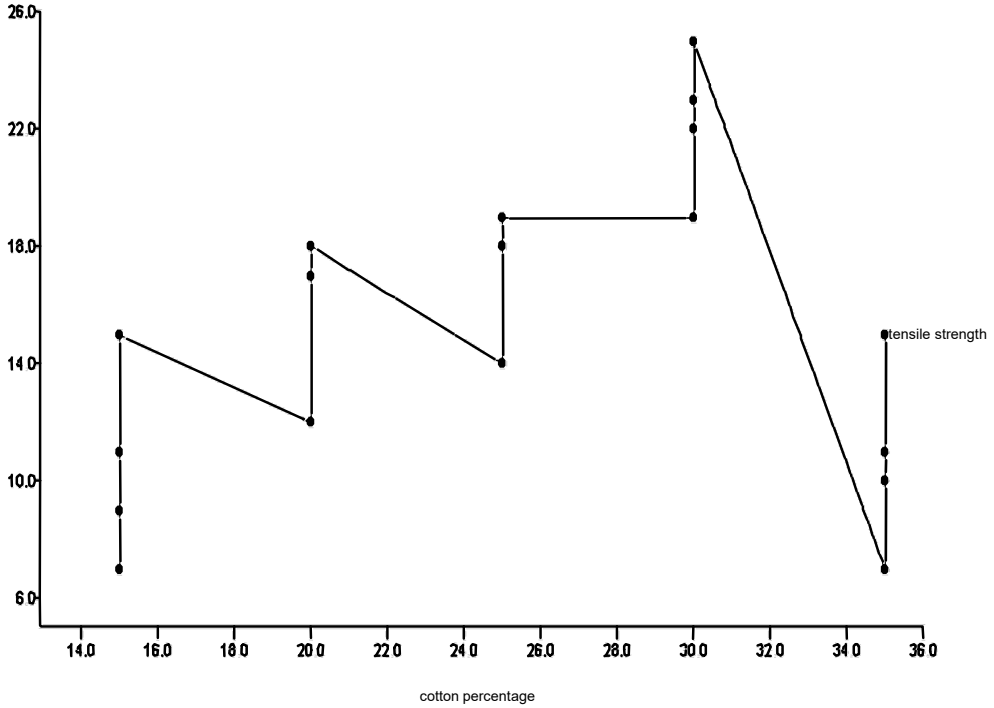
$$
\begin{aligned}
\mathrm{SS}_{\mathrm{Error}} &= \mathrm{SS}_{\mathrm{Total}} - \mathrm{SS}_{\mathrm{Treatments}} \\
&= 636.96 - 475.76 \\
&= 161.20 \\
\mathrm{MS}_{\mathrm{Treatments}} &= \mathrm{SS}_{\mathrm{Treatments}}/\mathrm{df} \\
&= 475.76/4 = 118.94 \\
\mathrm{MS}_{\mathrm{Error}} &= \mathrm{SS}_{\mathrm{Error}}/\mathrm{df} \\
&= 161.20/20 = 8.06 \\
F_{\mathrm{cal}} &= \mathrm{MS}_{\mathrm{Treatments}}/\mathrm{MS}_{\mathrm{Error}} \\
&= 118.94/8.06 = 14.76 \\
F_{\mathrm{tab}} &= F\,(0.05,4,20) \\
&= 2.87
\end{aligned}
$$

Box plot of tensile strength v/s cotton percentage



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Point /line chart for tensile strength v/s cotton percentage



Randomized block design (RBD)

An experimenter is wishes to determine whether or not four different tips produce different readings on a hardness testing machine . The machine operates by pressing the tip into a metal test coupon and from the depth of the resulting depression, the hardness of the coupon can be measured. Four observations are obtained for each tip and tabulated below. The observed response is the Rock well 'C' scale hardness minus 40.

	Type of tip				
		1	2	3	4
	1	9.3	9.4	9.2	9.7
	2	9.4	9.3	9.4	9.6
	3	9.6	9.8	9.5	10.0
	4	10.0	9.9	9.7	10.2

Coupons (blocks)

ANOVA table

Sources of variation	Degrees of freedom	Sum of squares	Mean squares	F_{CAL}	F_{TAB}
Due to treatments	3	0.385	0.128	14.38	3.86
Due to blocks	3	0.825	0.275	30.89	3.86
Due to error	9	0.08	0.009	-----	-----
Total	15	1.29	-----	-----	-----

$$\text{CF} = T^2/N$$
$$= 1542/16$$
$$= 1482.25$$

$$\text{SS}_{\text{TOTAL}} = \sum_{i=1}^{4}\sum_{j=1}^{4} X_{ij}^2 - \text{C. F}$$
$$= 1483.54 - 1482.25$$
$$= 1.29$$

$$\text{SS}_{\text{TIPS}} = \sum_{j=1}^{4} C_j^2 \,/4 - \text{C. F}$$
$$= 1482.635 - 1482.25$$
$$= 0.385$$

$$SS_{COUPONS} = \sum_{j=1}^{4} R_j^2 /4 - C.F$$
$$= 1483.075 - 1482.24$$
$$= 0.825$$

$$F_{CAL(TIPS)} = MS_{treatments}/MS_{error}$$
$$= 0.128/0.0089$$
$$= 14.38$$

$$F_{CAL(coupons)} = MS_{coupons}/MS_{error}$$
$$= 0.275/0.0089$$
$$= 30.89$$

$$F_{\text{TAB}} = F(0.05,3,9)$$
$$= 3.86$$

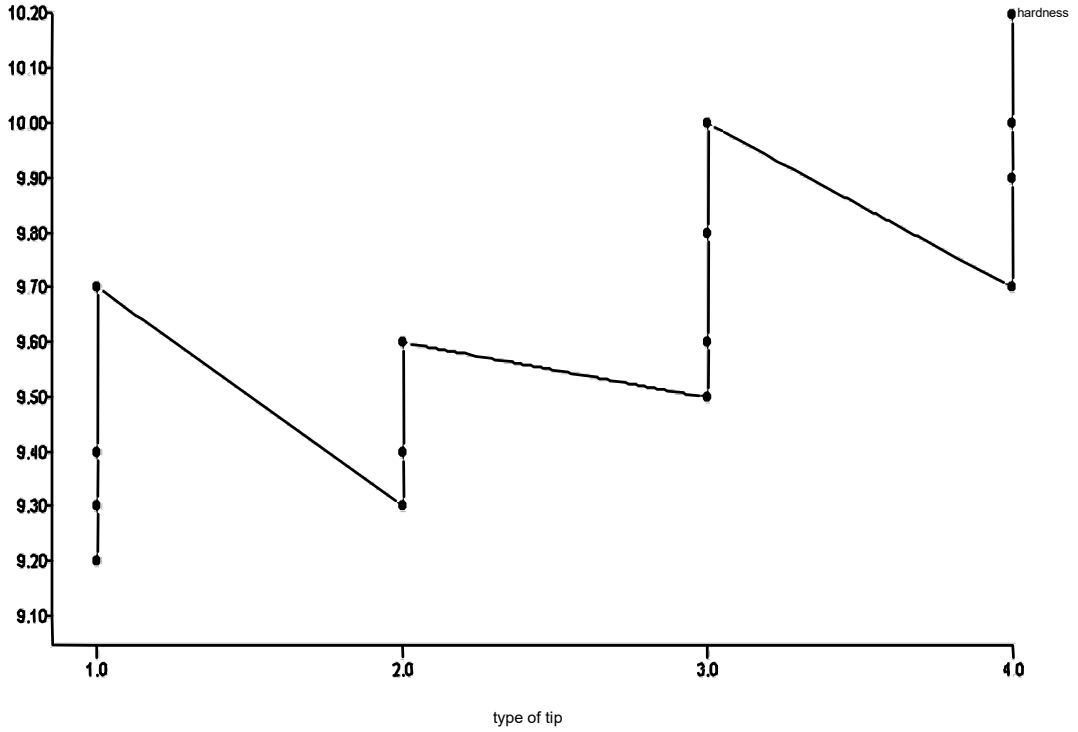
TESTING OF HYPOTHESIS :

$$H_O = \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$$
$$H_I = \mu_1 \neq \mu_2 \neq \mu_3 \neq \mu_4 \neq \mu_5$$

Since F CAL > FTAB HO is rejected. It may be concluded that type of tip affects the mean hardness reading.Also the blocks(coupons) differ significantly since the mean square for blocks is larger relative to error.

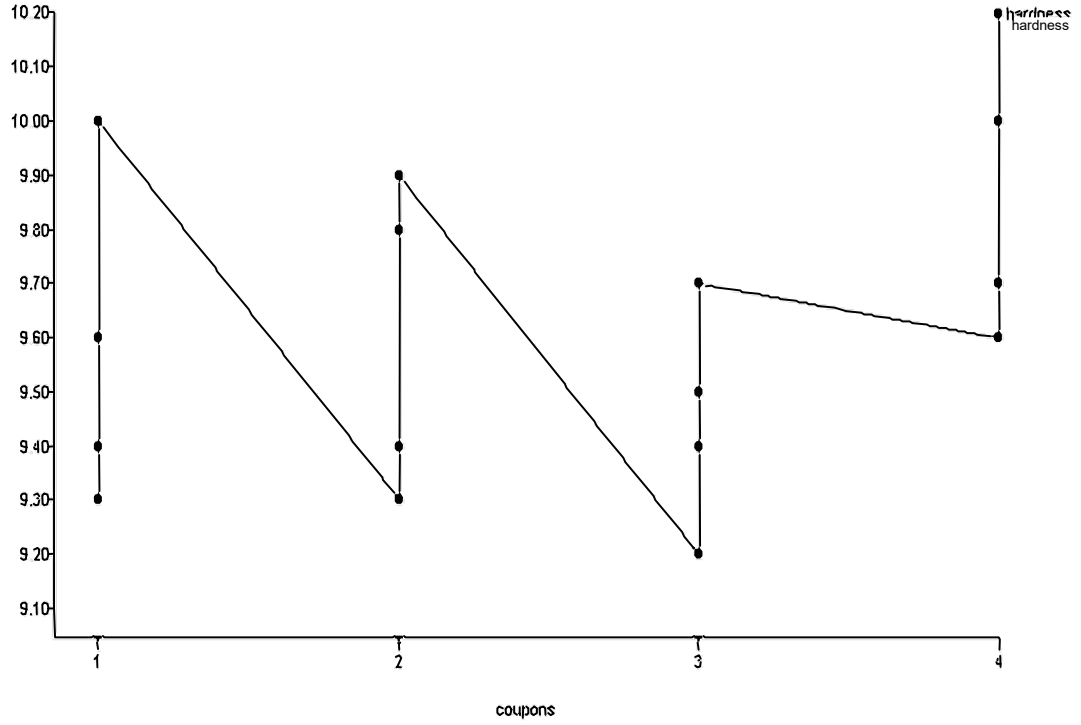
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Point /line chart for type of tip v/s hardness



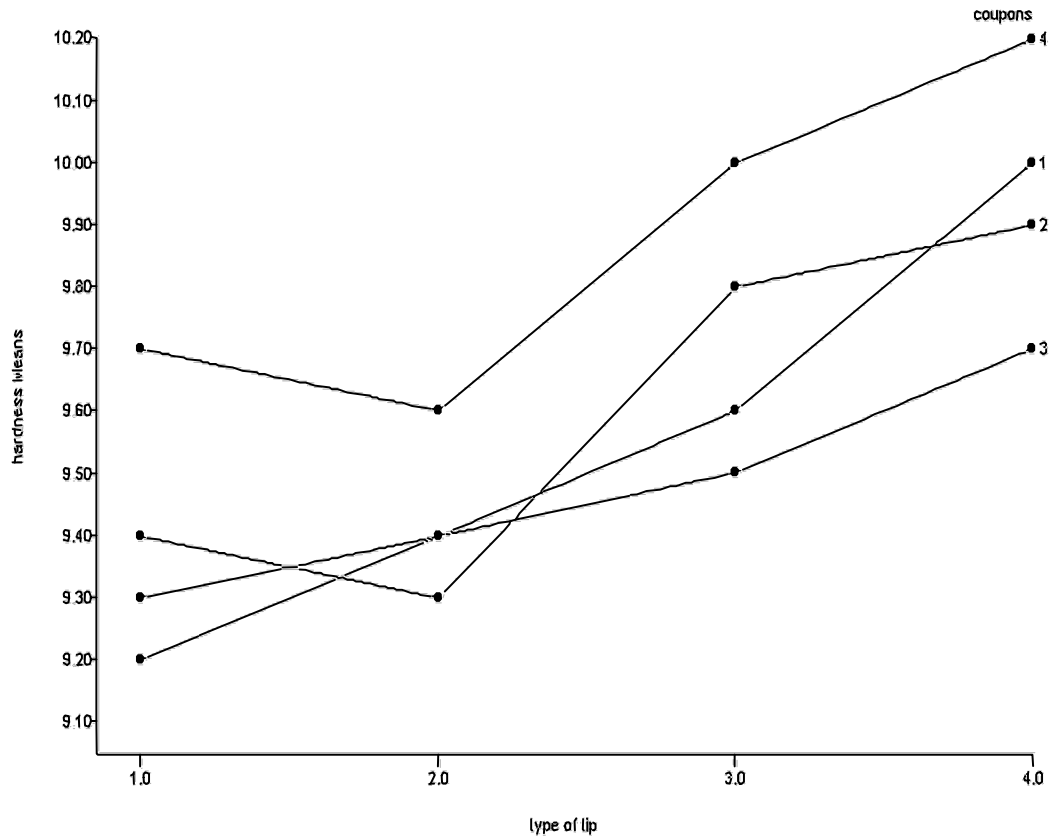
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Point /line chart for coupon v/s hardness



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Interaction chart of hardness v/s coupons



2^2 factorial design

2^2FACTORIAL DESIGN

An experiment was carried out to investigate into the effect of the concentration of the reactant (A) and the amount of the catalyst (B) on the reaction time of a chemical process. Two levels of each factor are chosen and three replicates of a 2^2 factorial design are run. The results are shown below.

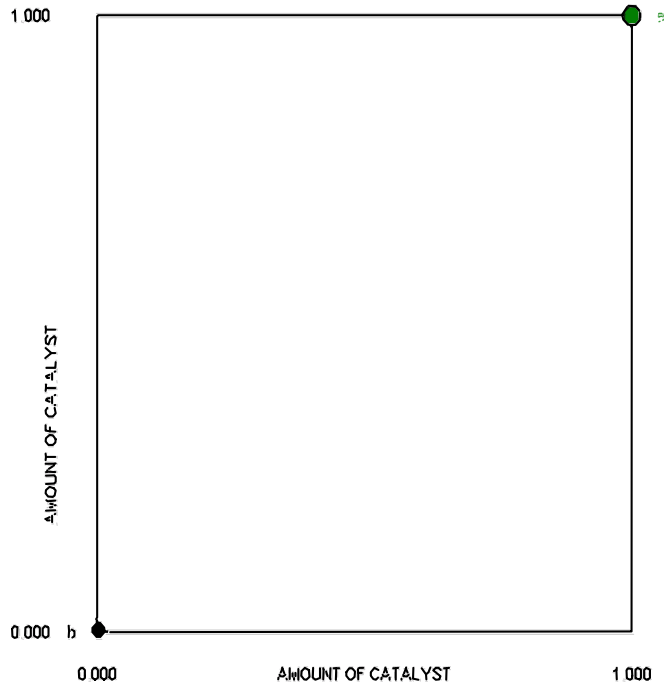
Which effects appear to be large? Use the analysis of variance to appear to be large? Use the analysis of variance to confirm your conclusions. Test at 1% significance level.

		Replicate			
		I	II	III	TOTAL
TREATMENT	1	28	25	27	80
	A	36	32	32	100
	B	18	19	23	60
	AB	31	30	29	90

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Graphical representation of the experiment

a(1,0,1,0) =	29.000	36.000	27.000
	30.000	b(0,0,0,0) = 18.000	28.000
	31.000	19.000	
	32.000	23.000	
	32.000	25.000	



ANOVA table

Source of variation	DOF	SS	MS	F_{CAL}	F_{TAB} F(0.01,1,8)
A	1	208.33	208.33	53.15	11.26
B	1	75	75	19.13	11.26
AB	1	8.33	8.33	2.125	11.26
ERROR	8	31.34	3.92	-----	-----
TOTAL	11	323	-----	-----	-----

$$
\begin{aligned}
\mathrm{SS}_A &= (-1 + a - b + \mathrm{ab})^2 / 2^n * r \\
&= (-80 - 100 + 60 + 90)^2 / \, 2^2 * 3 \\
&= 208.33 \\
\mathrm{SS}_B &= (-1 - a + b + \mathrm{ab})^2 / 2^n * r \\
&= (-80 - 100 + 60 + 90)^2 / 2^2 * 3 \\
&= 75
\end{aligned}
$$

$$SS_{AB} = (+1 - a - b + ab)^2/2^n r$$
$$= (80 - 100 - 60 + 90)^2/2^2 \cdot 3$$
$$= 8.33$$

$$\text{Correction factor} = T^2/N$$
$$= 330^2/12$$
$$= 9075$$

$$SS_{TOTAL} = \sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{3} X_{ijk}^2 - T^2/N$$
$$= 9398 - 9075 = 323.0$$

$SS_{ERROR} = SS_{TOTAL} - SS_A - SS_B - SS_{AB}$

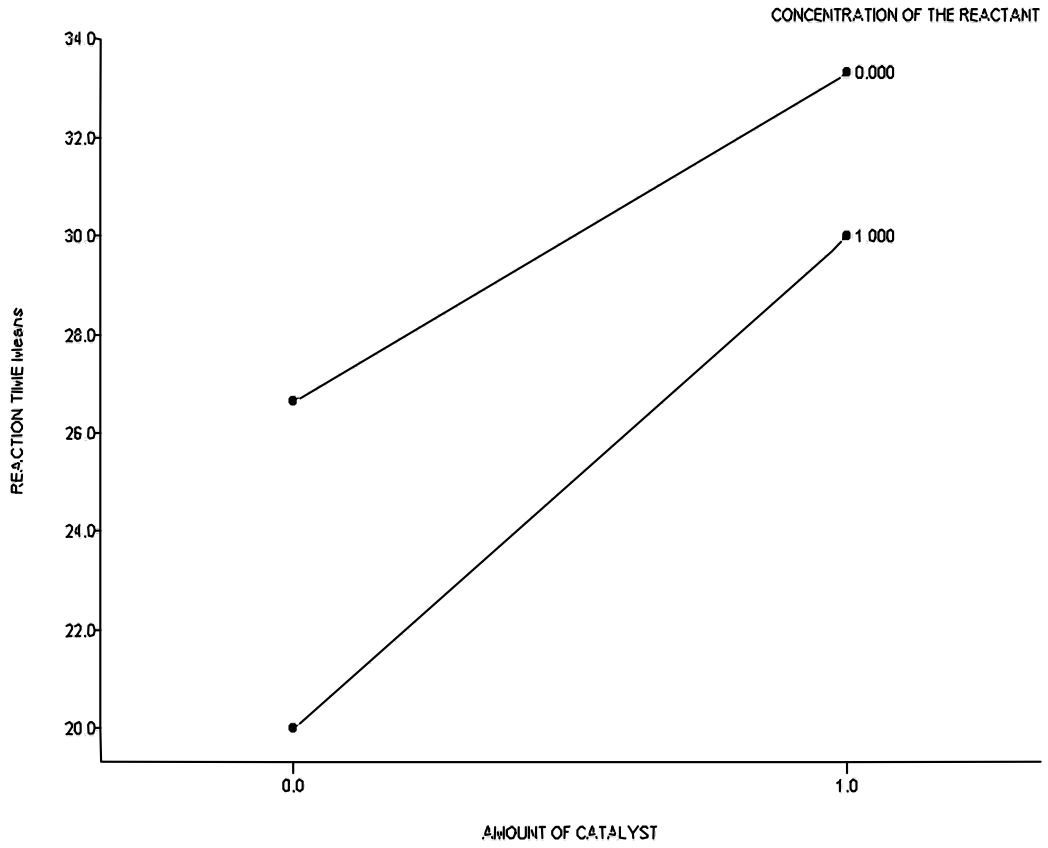
$$= 323 - 208.33 - 75 - 8.33$$
$$= 31.34$$

Final interpretation :

Both the main factors are significant . Interaction effect is statistically not significant.

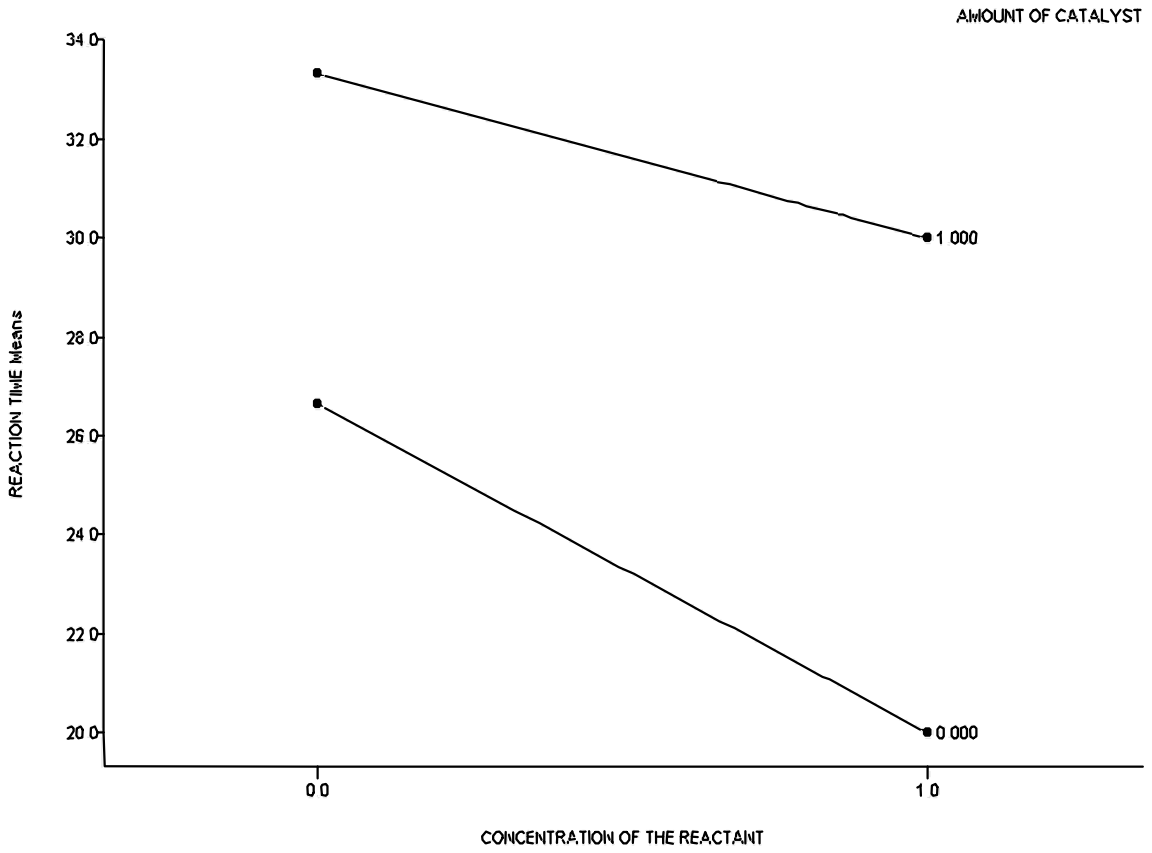
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Interaction chart of reaction time v/s concentration of reactant



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Interaction chart of reaction time v/s amount of catalyst



2^3 FACTORIAL DESIGN

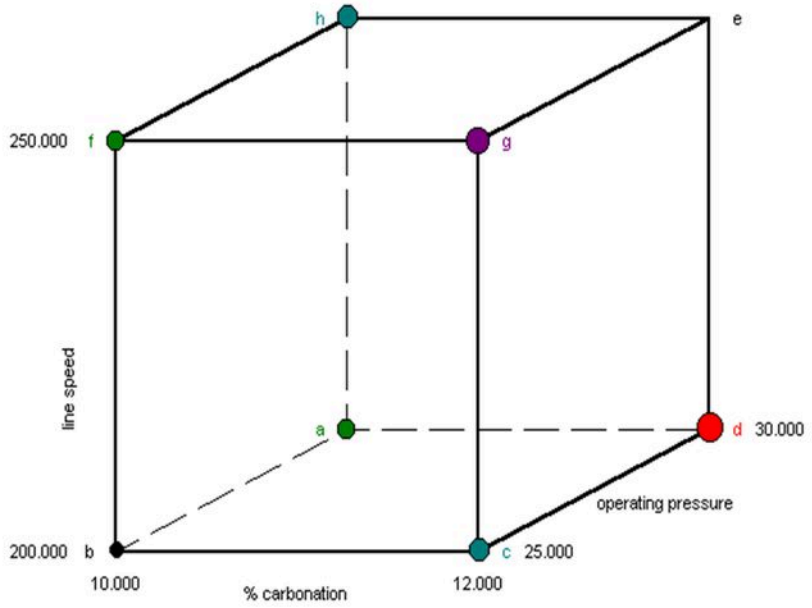
A soft drink bottler is interested in obtaining more uniform fill heights in the bottles produced by his manufacturing process. The process engineer can control three variables during the filling process. The percent carbonation (A), the operating pressure in the filler (B), and the bottles produced per minute (C), He decides to run two replicates of a factorial design in these three factors, with all 24 runs taken in random order. The response variable observed is the average deviation from the target fill height. The results of the experiment are shown below. Use the analysis of variance to analyze the results.

		Operating pressure (B)			
		25 PSI		30 PSI	
		C=200	C=250	C=200	C=250
% CARBO-NATION	10	-3 -1	-1 0	-1 0	1 1
	12	0 1	2 1	2 3	6 5

C=line speed in bpm

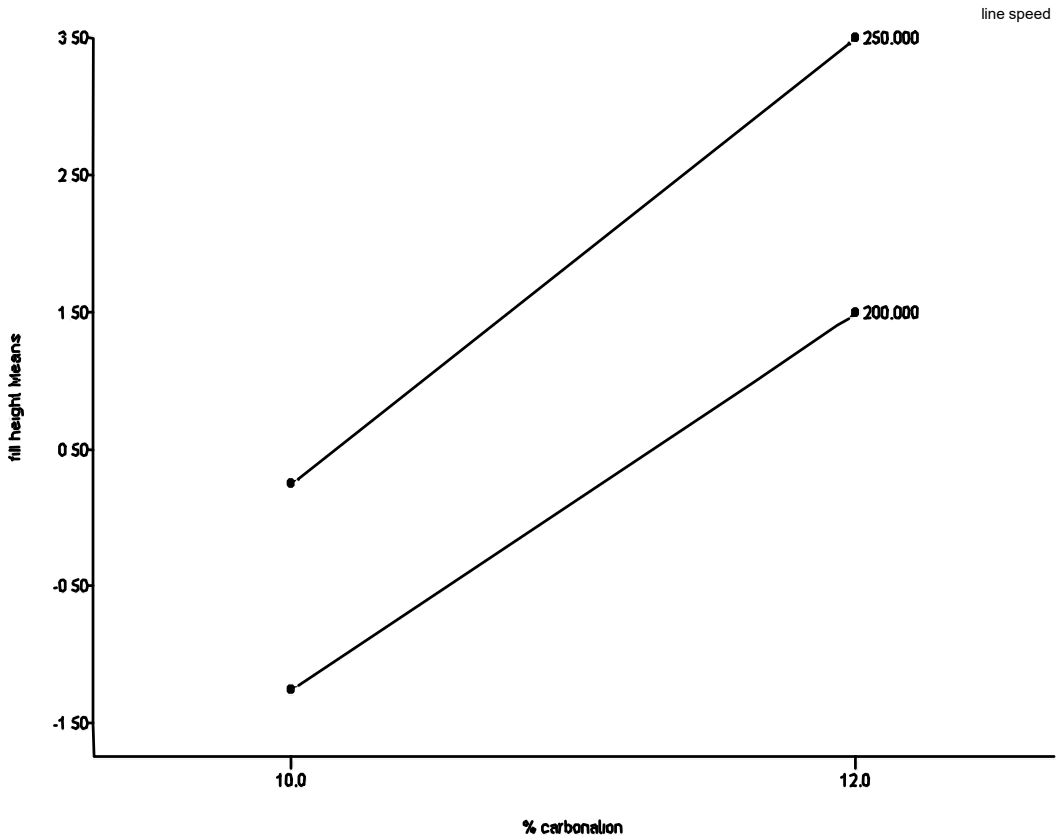
Graphical representation of 23 design

a(10.0,200.0,30.0) = -1.000	0.000
0.000	g(12.0,250.0,25.0) = 1.000
b(10.0,200.0,25.0) = -3.000	2.000
-1.000	h(10.0,250.0,30.0) = 1.000
c(12.0,200.0,25.0) = 0.000	1.000
1.000	
d(12.0,200.0,30.0) = 2.000	
3.000	
e(12.0,250.0,30.0) = 5.000	
6.000	
f(10.0,250.0,25.0) = -1.000	



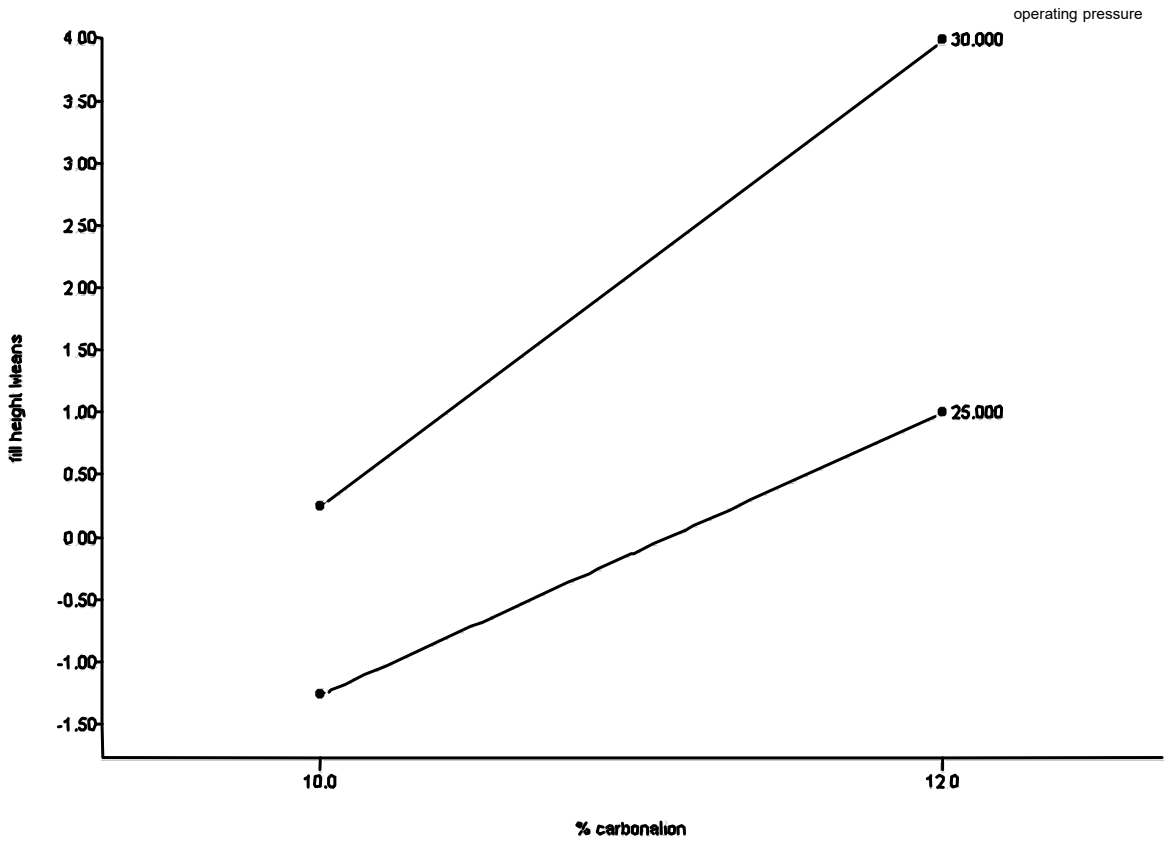
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Interaction plot of line speed v/s fill height



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Interaction plot of operating pressure v/s fill height



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Yate's method for 2^3 design

Treat ment	response	I	II	III	Effect	Main effect	Sum of squares
1	-4	-3	1	16	1	---	---
a	1	4	15	24	A	3.0	36.0
b	-1	2	11	18	B	2.25	20.25
ab	5	13	13	6	AB	0.75	2.25
c	-1	5	7	14	C	1.75	12.25
ac	3	6	11	2	AC	0.25	0.25
bc	2	4	1	4	BC	0.5	1.00
abc	11	9	5	4	ABC	0.5	1.00

ANOVA table

TREATMENT	DOF	MS	F_{CAL}	F_{TAB}
A	1	36.00	57.6 *	3.46
B	1	20.25	32.40 *	3.46
C	1	12.25	19.60 *	3.46
AB	1	2.25	3.6	3.46
AC	1	0.25	0.4	3.46
BC	1	1.0	1.6	3.46
ABC	1	1.0	1.6	3.46
ERROR	8	0.625	---	---
TOTAL	15		---	---

INFERENCE :

At 10% significance level FTAB < FCAL for A,B,C and AB. It may be concluded that

- the percentage carbonation ,
- the operating pressure in the filler and
- the bottles produced per minute

affect the fill heights in the bottles.

There is mild interaction between carbonation and pressure.

2^4 factorial design

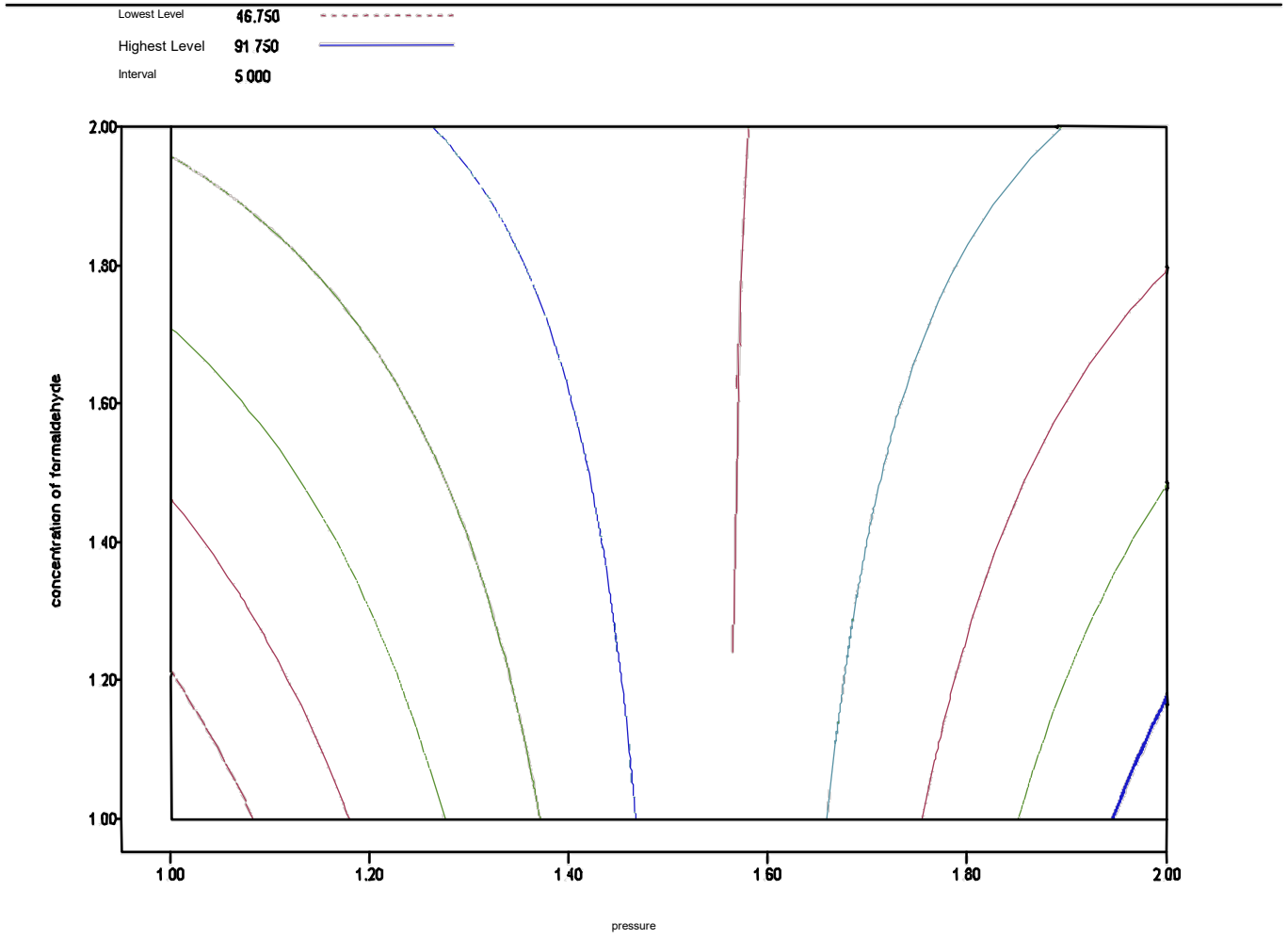
A chemical product is produced in a pressure vessel. A factorial experiment is carried out in the pilot plant to study the factors thought to influence the filtration rate of this product. The four factors are temperature(A),pressure(B), concentration of formaldehyde(C) ,and stirring rate (D). Each factor is at two levels, and the data obtained from a single replicate of 2^4experiment are shown in the table below. The 16 runs are made in random order. The process engineer is interested in maximizing the filtration rate. Analyze the data.

Treatment combination	Filtration rate
1	45
A	71
B	48
AB	65
C	68
AC	60
BC	80
ABC	65

Treatment combination	Filtration rate
D	43
AD	100
BD	45
ABD	104
C_D	75
ACD	86
BCD	70
ABCD	96

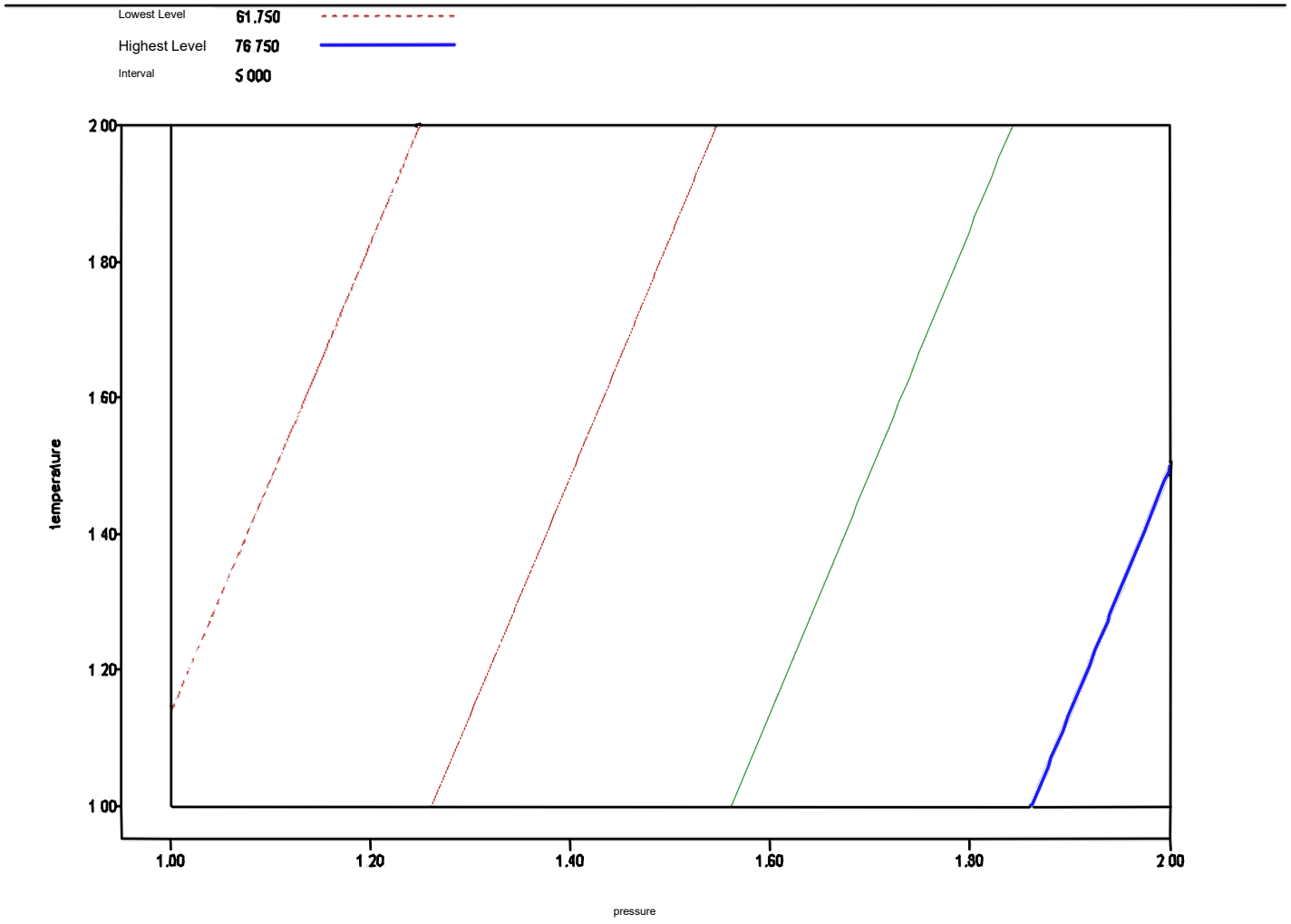
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Contour plot of pressure v/s concentration of formaldehyde



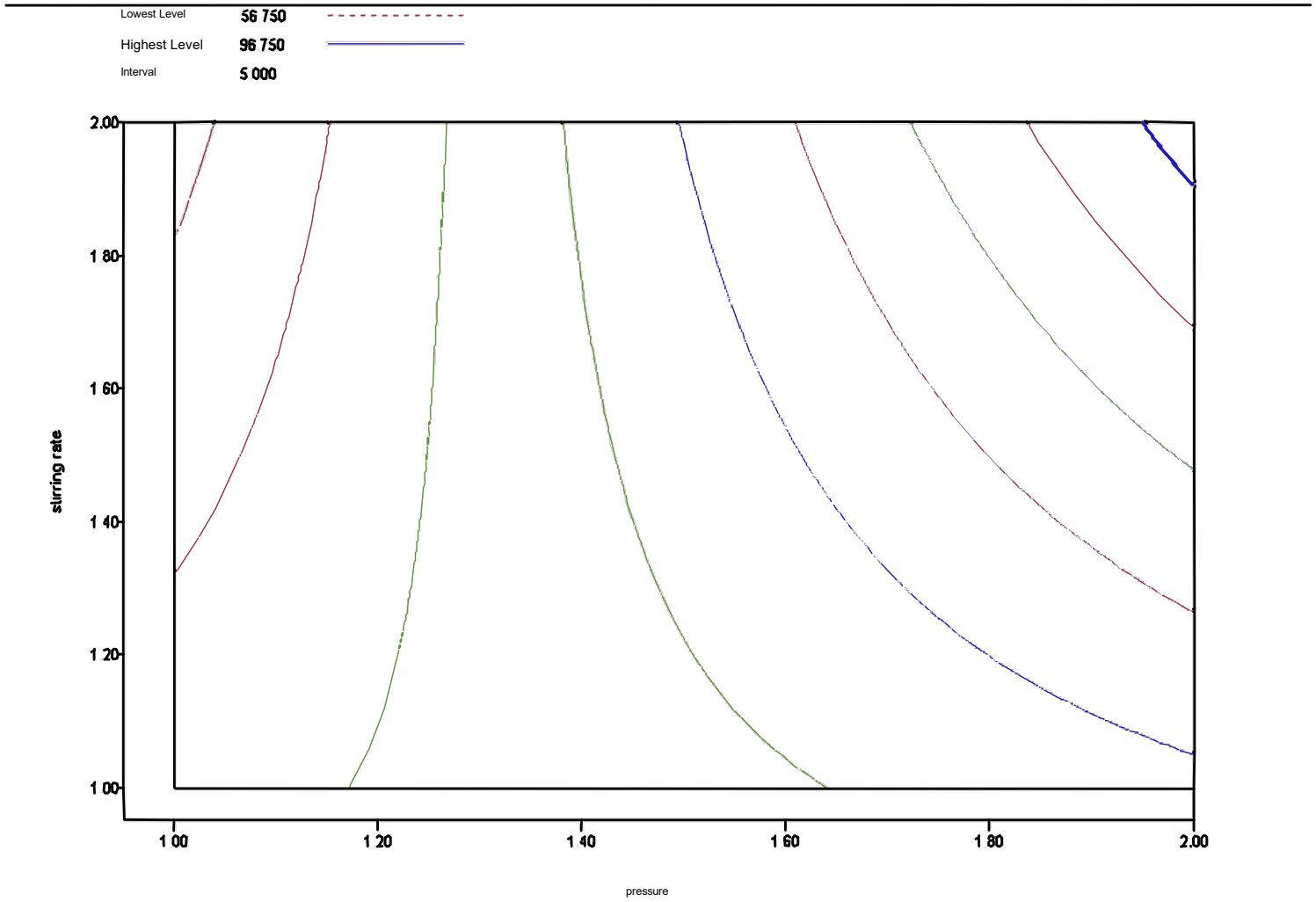
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Contour plot of temperature v/s pressure



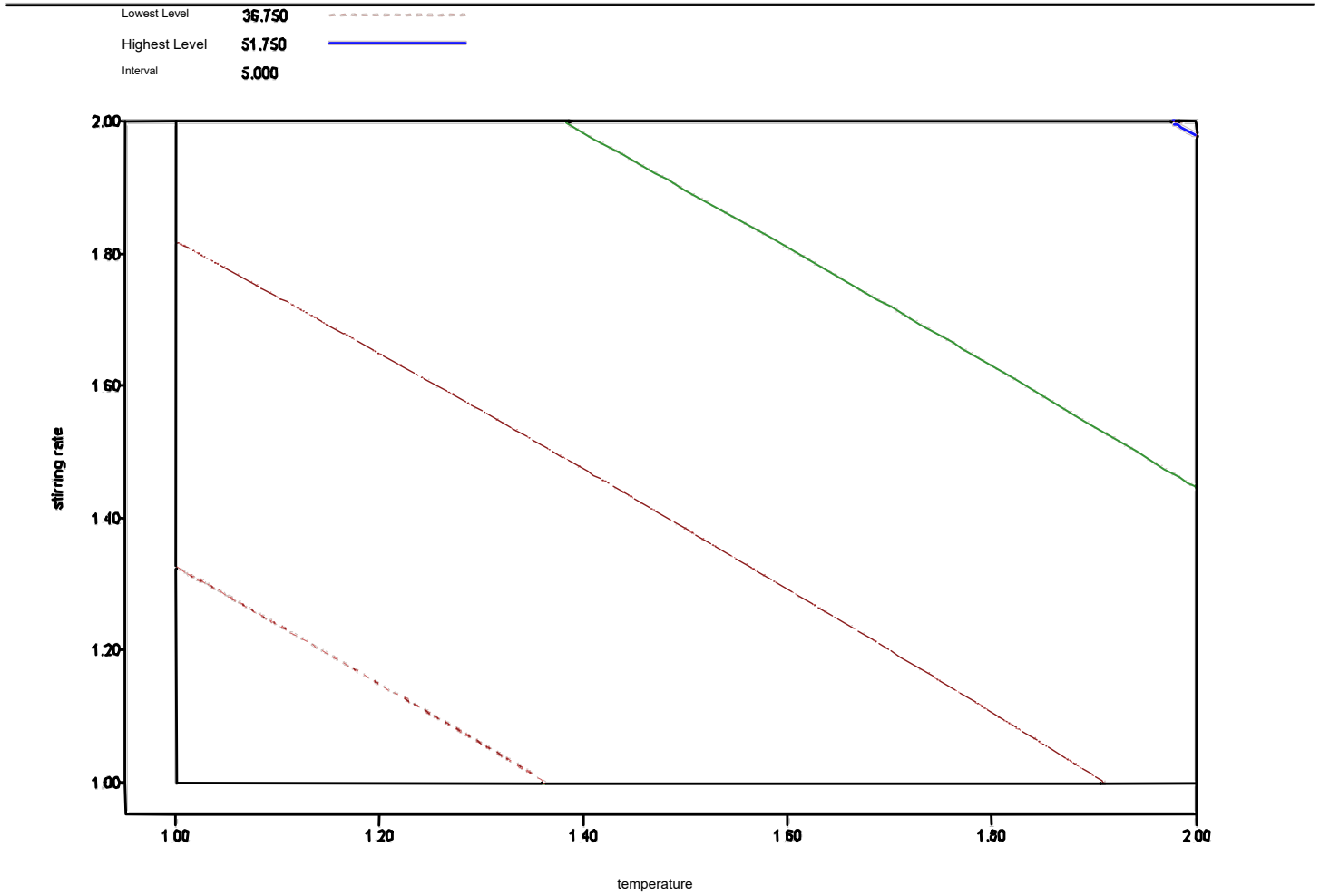
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Contour plot of pressure v/s stirring rate



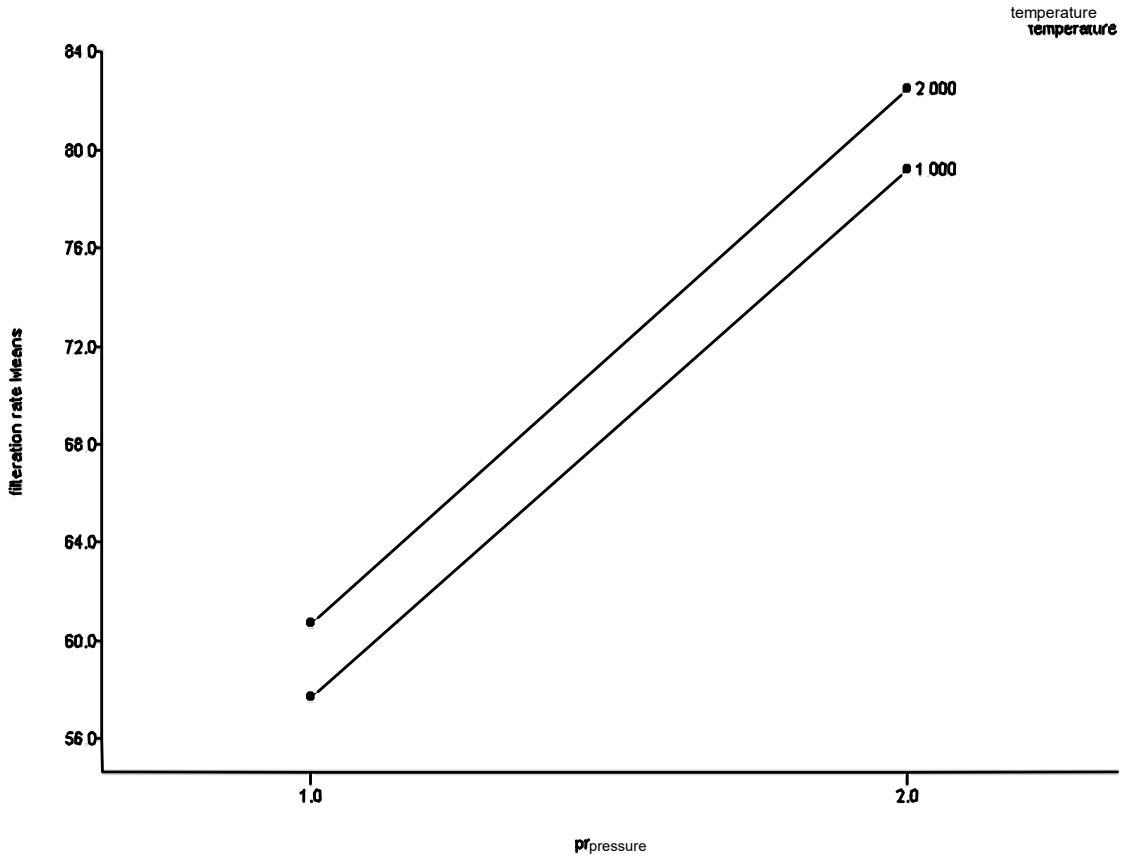
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Contour plot of temperature v/s stirring rate



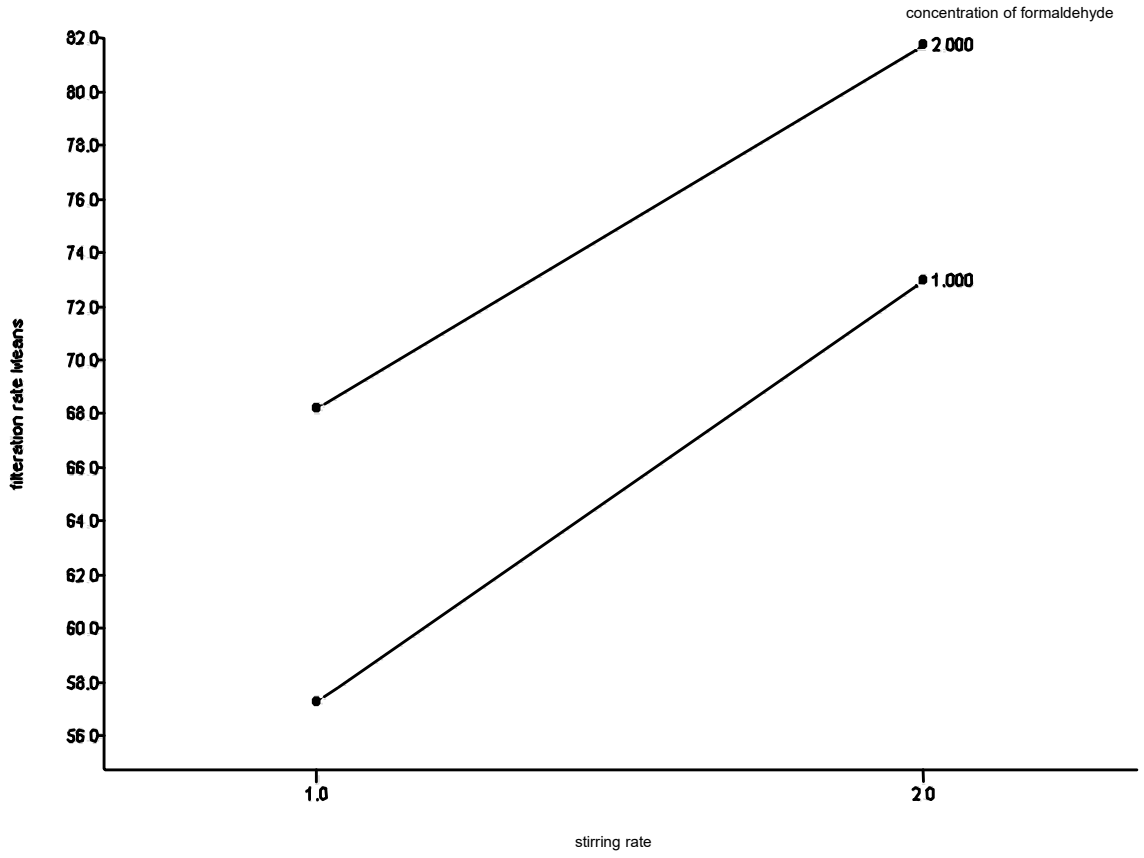
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Interaction plot of filtration rate v/s pressure



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Interaction plot of concentration of filtration rate v/s stirring rate



COMPUTATION TABLE

EFFEC	RESP	I	II	III	IV	ME	SS
1	45	116	229	502	1121	-	---
A	71	113	273	619	173	20.38	1870.56
B	48	128	292	20	25	3.13	39.06
AB	65	145	327	153	11	-1.13	0.06
C	68	143	43	14	79	9,88	390.06
AC	60	149	-23	11	-145	-19.38	1311.06
BC	80	161	116	-16	19	2.3	22.56
ABC	65	166	37	17	15	0.63	14.06
D	43	26	-3	44	117	14.63	855.56
AD	100	17	17	35	133	17.88	1105.56
BD	45	-8	6	-66	-3	-0.38	0.56
ABD	104	-15	5	-79	33	5.38	68.06
C_D	75	57	-9	20	-9	-1.13	5.06
ACD	86	59	-17	-1	-13	-0.38	10.56
BCD	70	11	2	10	-21	-2.63	27.56
ABCD	96	26	15	13	11	2.63	7.56

3^2 FACTORIAL DESIGN

An engineer is designing a battery for use in a device that will be subjected to some extreme variations in temperature, which may affect battery life. The only design parameter that he can select at this point is the plate material for the battery. The engineer decides to test all three plate materials at three temperature levels –15°F, 70°F, and 125°F. four batteries are tested at each combination of plate material and temperature, by running a 3^2 experiment. The observed battery life data are given in the following table.

		Temperature (A)		
		15	70	125
Material type (M)	1	130 155 74 180	34 40 80 75	20 70 82 58
	2	150 188 159 126	136 122 106 115	25 70 58 45
	3	138 110 168 160	174 120 150 139	96 104 82 60

POLYNOMIAL EFFECTS OF TEMPERATURE

TEMPERATURE	TREATMENT TOTALS	ORTHOGONAL CONTRAST COEFFICIENT	
		LINEAR	QUADRATIC
15	1738	-1	+1
70	1291	0	-2
125	770	+1	+1

$$T_L = -968 \; T_Q = -74$$

$$SS_{TL} = 39042.67 \qquad SS_{TQ} = 76.05$$

$$SS_T = SS_{TL} + SS_{TQ} = 39118$$

$$CF = 4009000.03$$
$$SS_M = 4939005/12 - C.F.$$
$$= 10683.72$$

Polynomial effects of M-T interaction:

Temp	Totals for material type			Orthogonal contrast coefft.	
	1	2	3	linear	quadratic
15	539	623	576	-1	+1
70	229	479	583	0	-2
125	230	198	342	+1	+1

$M_I T_L = -309$
$M_1 T_Q = 311$

$M_2 T_L = -425$
$M_2 T_Q = -137$

$M_3 T_L = -234$
$M_3 T_Q = -248$

$$
\begin{aligned}
SS_{MXTL} &= ((-309)^2 + (-425)^2 + (-234)^2)/(4 * 2) - 968^2/(12 * 2) \\
&= 2315.08 \\
SS_{MXTQ} &= ((311)^2 + (-137)^2 + (-248)^2)/4 * 6 - (-74)^2/(12 * 6) \\
&= 7298.70 \\
SS_{MXT} &= SS_{MXTL} + SS_{MXTQ} \\
&= 2315.08 + 7298.70 \\
&= 9613.78
\end{aligned}
$$

Source of var	DOF	SS	MS	F_{CAL}	F_{TAB}
Mat type M	2	10683.72	5431.86	7.91 A	5.49
Temp (T)	2	39118.72	19559.36	28.97 A	5.49
TL	(1)	(39042.67)	39042.67	57.82 A	5.49
TQ	(1)	(76.05)	76.05	0.12	---
Interaction MXT	4	9613.78	2403.44	3.56 B	2.73
MXTL	(2)	(2315.78)	1157.54	1.71	---
MXTQ	(2)	(7298.70)	3649.35	5.41 B	2.73
Error	27	18230.75	672.21	---	---
Total	35	77646.92	---	---	---

A- significant at 1%

B – significant at 5%

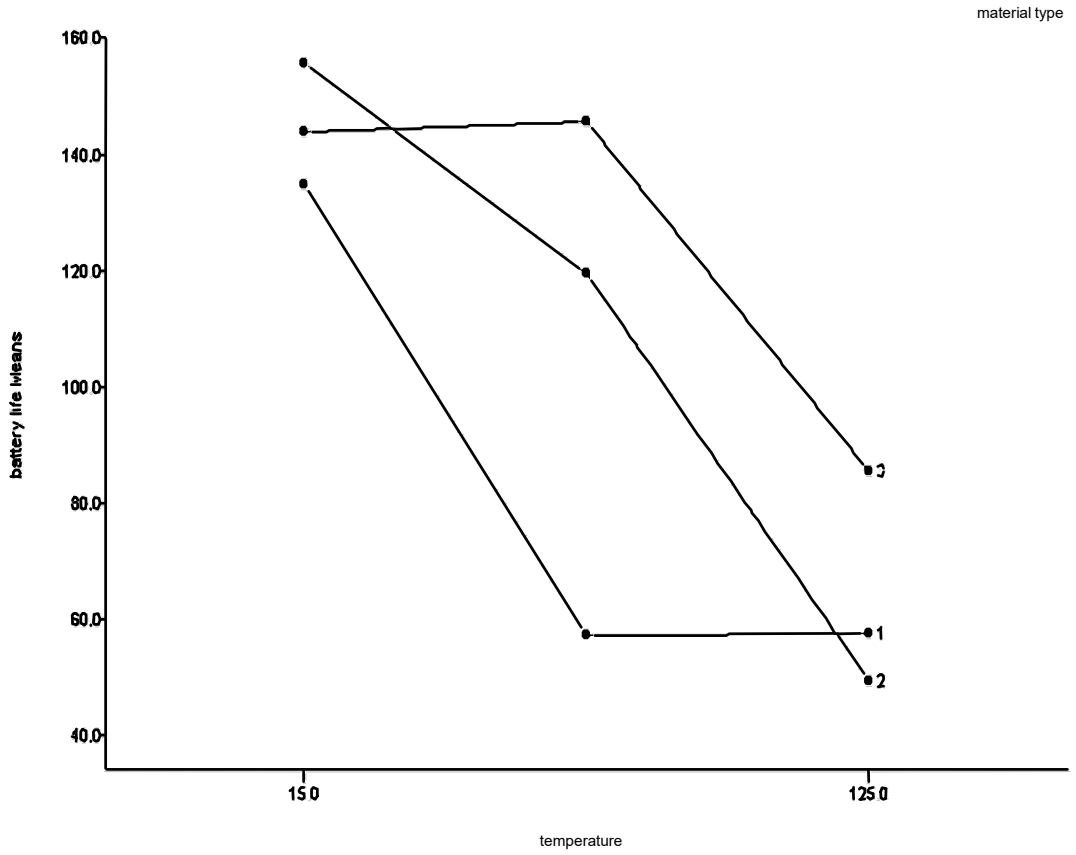
Effect of temperature is is significant.

linear effect of temperature is evident and quadratic temperature effect is not significant.

Interaction effect is significant due to result of the MXT_Q component.

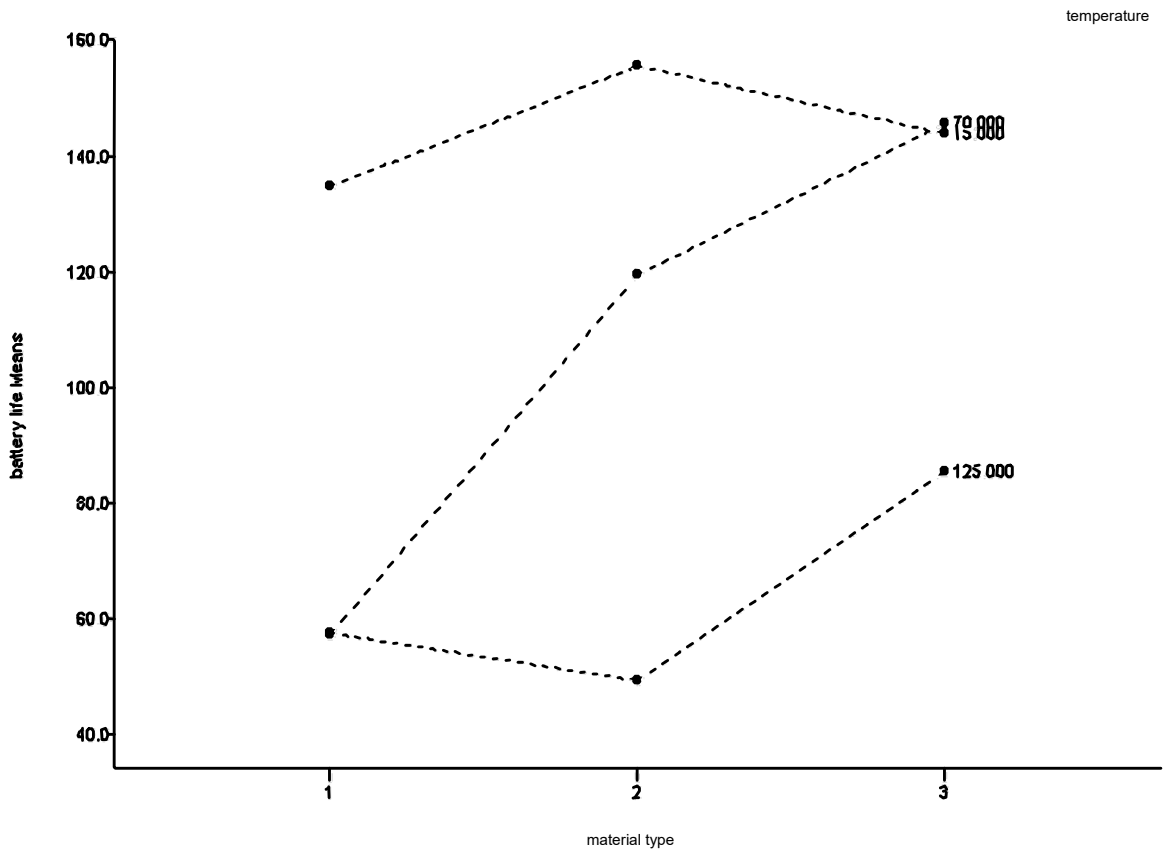
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INTERACTION PLOT OF BATTERY LIFE V/S MATERIAL TYPE



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INTERACTION PLOT OF BATTERY LIFE V/S TEMPERATURE



Orthogonal array Technique

In a study to examine process control parameter for joining 'BUS-BAR' of an electrical system, the following factors and levels were identified for investigation with respect to efficiency of joining.

Factor code	Factor description	levels	
		1	2
A	Torque	25 ftlb	40 ftlb
B	Compound	nil	Silicon grease
C	Surface finish	As is	Ground
D	Joint overlap	8mm	16 mm

Develop an OA layout to evaluate the main effects and interactions AxB and CxD.

Solution :

1. Degrees of freedom required.

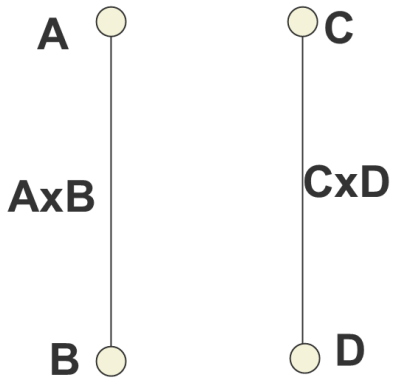
$$\text{TDF} = \text{main effects (4)} + \text{interaction effects (2)}$$
$$= 6$$

$$2.\ \text{MNE} = \text{TDF} + 1$$
$$= 6 + 1 = 7$$

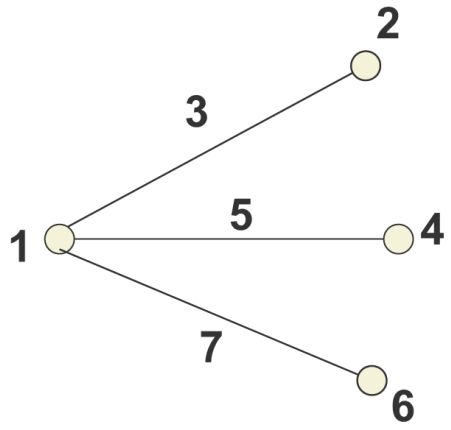
3 Choice of series 2^n- 2^3 is chosen.

OA layout $L_8 2^7$ – trial layout

4. Required linear graph:

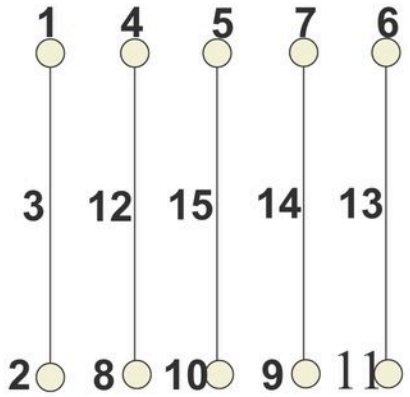


5. Standard linear graph (SLG) :

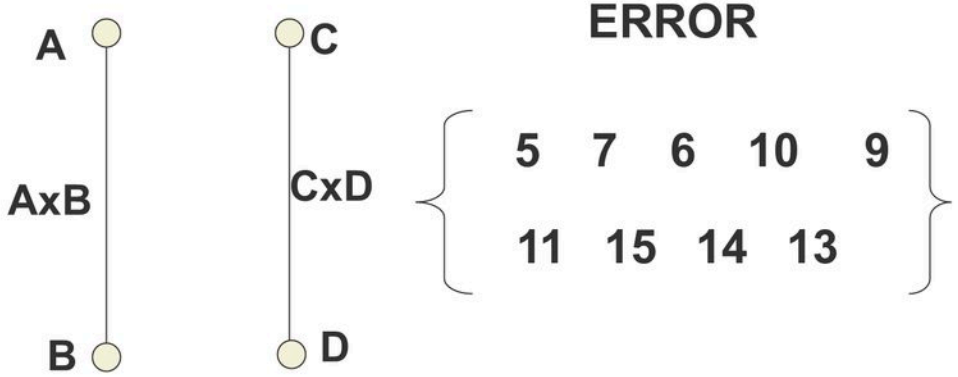


It is not possible to modify the standard linear graphs to meet the required linear graph. Choosing a layout at one level higher - $L_{16}2^{15}$

SLG :



MLG :



7. OA layout :

Expt no.	A (1)	B (2)	C (4)	D (8)	AxB (3)	CxD(12)	E- 5,7,6,10,9,11, 15,14,13
1	1	1	1	1	1	1	
2	1	1	1	2	1	2	
3	1	1	2	1	1	2	
4	1	1	2	2	1	1	
5	1	2	1	1	2	1	
6	1	2	1	2	2	2	
7	1	2	2	1	2	2	
8	1	2	2	2	2	1	
9	2	1	1	1	2	1	
10	2	1	1	2	2	2	
11	2	1	2	1	2	2	
12	2	1	2	2	2	1	
13	2	2	1	1	1	1	
14	2	2	1	2	1	2	
15	2	2	2	1	1	2	
16	2	2	2	2	1	1	

PHYSICAL LAYOUT

Expt no.	A	B	C	D	Response
1	25	Nil	As is	8	8
2	25	Nil	As is	16	6
3	25	Nil	Ground	8	6
4	25	Nil	Ground	16	2
5	25	SG	As is	8	4
6	25	SG	As is	16	4
7	25	SG	Ground	8	3
8	25	SG	Ground	16	6
9	40	Nil	As is	8	5
10	40	Nil	As is	16	1
11	40	Nil	Ground	8	1
12	40	Nil	Ground	16	2
13	40	SG	As is	8	7
14	40	SG	As is	16	10
15	40	SG	Ground	8	10
16	40	SG	Ground	16	9

ANOVA table

Source of variance	DOF	SS	MS	F_{CAL}	F_{TAB}
Due to A	1	2.25	2.25	3.682	5.12
B	1	30.25	30.25	49.59	5.12
C	1	2.25	2.25	3.689	5.12
D	1	6.25	6.25	10.246	5.12
AxB	1	90.25	90.25	147.95	5.12
CxD	1	0.25	0.25	0.410	5.12
Error	9	5.5	0.61	---	---
Total	15	137	---	---	---

Problem 2:

In an investigation on the stability of power factor of capacitor it was decided to examine the following factors.develop a OA design and physical layout.

Sl. No.	Factor	levels	
1	Leakage of current of foil (A)	A1: ≤ 2 m	A2: ≥2 m
2	Power factor of foil (B)	B1 : ≤ 3%	B2 : 3.5%
3	Capacitance value of foil(C)	C1 : 8-10 µm	C2: 11-13 µm
4	Soaking time during impregnation (D)	D1: 2 hrs	D2: 4 hrs
5	Loading pattern(E)	E1 : 70%	E2 : 100%

Interactions required: AxB, BxC

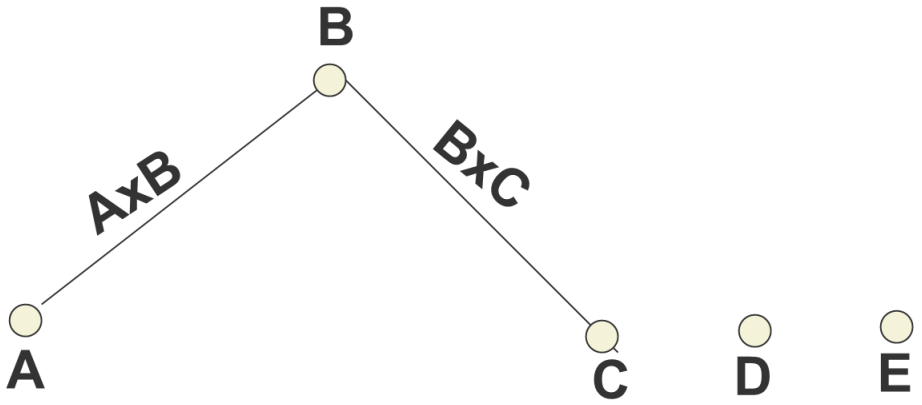
STEP 1 : degrees of freedom required

$$= A(1) + B(1) + C(1) + D(1) + E(1) + \text{AxB}(1) + \text{BxC}(1)$$
$$= 7 \text{ df}$$

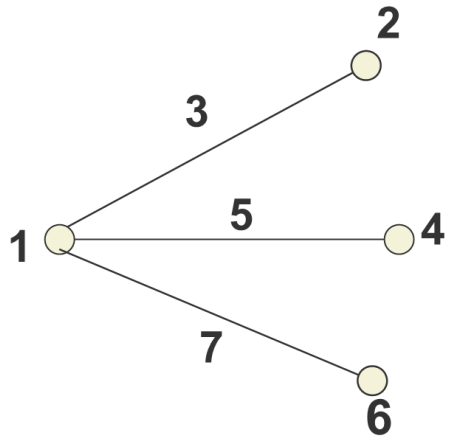
STEP 2: Minimum no. of experiment (MNE)

$$\text{MNE} = 7 + 1 = 8$$

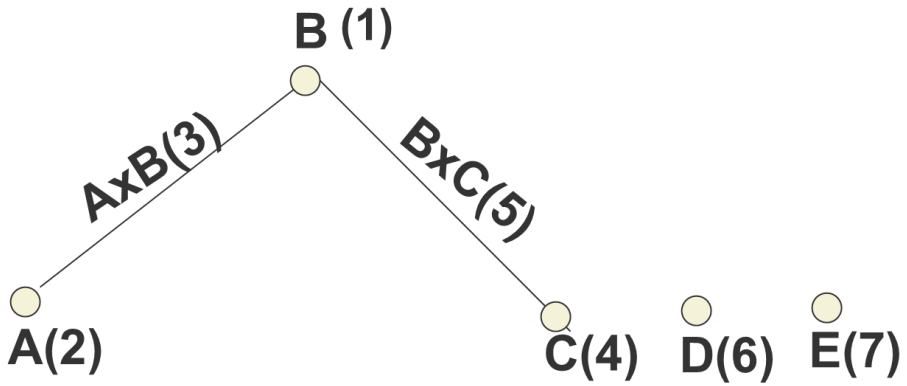
STEP3 : Required linear graph is as follows :



STEP 4 : Standard linear graph (SLG)



STEP 5 : Modified linear graph (MLG)



STEP- 6 : OA layout :

Expt . no.	B (1)	A (2)	C (4)	D (6)	E (7)	AxB (3)	BxC (5)
1	1	1	1	1	1	1	1
2	1	1	2	2	2	1	2
3	1	2	1	2	2	2	1
4	1	2	2	1	1	2	2
5	2	1	1	1	2	2	2
6	2	1	2	2	1	2	1
7	2	2	1	2	1	1	2
8	2	2	2	1	2	1	1

STEP 7: PHYSICAL LAYOUT:

TRIAL NO.	LEVEL OF A B C D E				
1	>2 mA	3.0%	8-10 µf	2 hrs	70%
2	>2 mA	3.0%	11-13 µf	4 hrs	100%
3	>2 mA	3.0%	8-10 µf	4 hrs	100%
4	>2 mA	3.0%	11-13 µf	2 hrs	70%
5	<2 mA	3.5%	8-10 µf	2 hrs	100%
6	<2 mA	3.5%	11-13 µf	4 hrs	70%
7	<2 mA	3.5%	8-10 µf	4 hrs	70%
8	<2 mA	3.5%	11-13 µf	2 hrs	100%

Problem 3

In an investigation on getting required color in a wrist watch dial plating, following factors were to be studied.

Sl. No.	Factor	Levels
1	A: Temperature	40°C, 47.5°C, 55°C
2	B: Voltage	3.5v,4.5v,2.5v
3	C: Time of immersion	30s, 40s, 50s
4	D: Connection of batch	30%, 35%, 40%

Give a layout of design to estimate all main effects and Interactions AB, AC, BC

STEP 1: Total degrees of freedom (TDF)

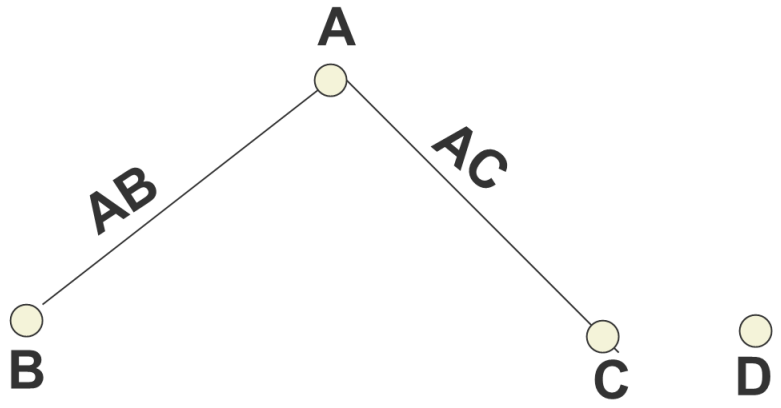
$$= A(2) + B(2) + C(2) + D(2) = \mathrm{AB}(4) + \mathrm{AC}(4) + \mathrm{BC}(4)$$
$$= 20$$

STEP 2: Minimum no. of experiments(MNE)

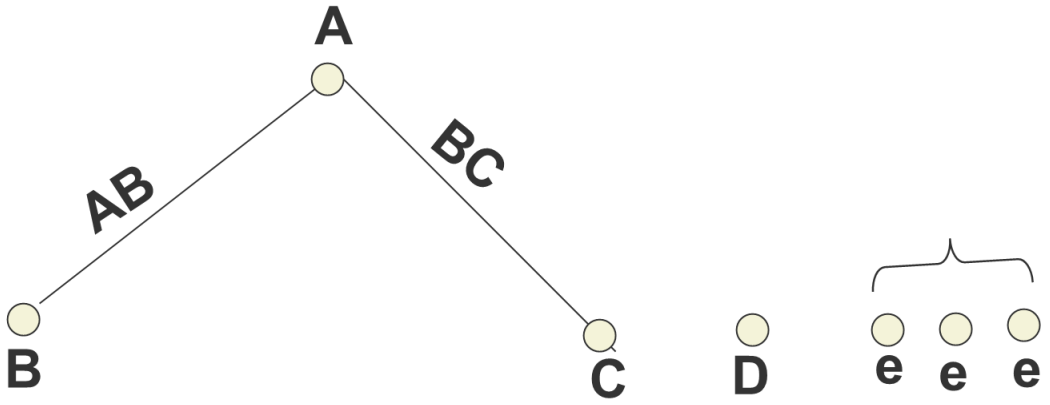
$$= \mathrm{TDF} + 1$$
$$= 21$$

STEP 3: Trial layout $= L_{27}(3^{13})$

STEP 4: Required linear graph (RLG)



STEP 5: Modified linear graph:



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STEP 7: OA LAYOUT

Expt no.	A 1	B 2	C 5	D 9	AB (3,4)	BC (8,11)	AC (6,7)	E(10,12,13)
1	1	1	1	1				
2	1	1	2	2				
3	1	1	3	3				
4	1	2	1	2				
5	1	2	2	3				
6	1	2	3	1				
7	1	3	1	3				
8	1	3	2	1				
9	1	3	3	2				
10	2	1	1	2				
11	2	1	2	3				
12	2	1	3	1				
13	2	2	1	3				
14	2	2	2	1				

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STEP 7: OA LAYOUT (continued)

Expt no.	A(°C) 1	B 2	C 5	D 9	AB (3,4)	BC (8,11)	AC (6,7)	E(10,12,13)
15	2	3	3	2				
16	2	1	1	1				
17	2	1	2	2				
18	2	1	3	3				
19	3	2	1	3				
20	3	2	2	1				
21	3	2	3	2				
22	3	3	1	1				
23	3	3	2	2				
24	3	3	3	3				
25	3	1	1	2				
26	3	1	2	3				
27	3	1	3	1				

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STEP 7: PHYSICAL LAYOUT

Expt no.	A(°C) 1	B(V) 2	C(S) 5	D(%) 9
1	40	3.5	30	30
2	40	3.5	40	35
3	40	3.5	50	40
4	40	4.5	30	35
5	40	4.5	40	40
6	40	4.5	50	30
7	40	2.5	30	40
8	40	2.5	40	30
9	40	2.5	50	35
10	47.5	3.5	30	35
11	47.5	3.5	40	40
12	47.5	3.5	50	30
13	47.5	4.5	30	40
14	47.5	4.5	40	30

STEP 7: PHYSICAL LAYOUT (continued)

Expt no.	A(°C)	B(V)	C(S)	D(%)
15	47.5	4.5	50	35
16	47.5	2.5	30	30
17	47.5	2.5	40	35
18	47.5	2.5	50	40
19	55	3.5	30	40
20	55	3.5	40	30
21	55	3.5	50	35
22	55	4.5	30	30
23	55	4.5	40	35
24	55	4.5	50	40
25	55	2.5	30	35
26	55	2.5	40	40
27	55	2.5	50	30

ANALYSIS METHOD OF MEASURABLE DATA

Problem:

In an investigation to find the best condition of molar ratio n (at least 2.25) for the production of sodium silicate, the following factors with their levels and interactions were chosen for experiment.

Interactions : AxB, AxD, AxE, BxD and DxE

Sl no.	Factor	Factor code	levels	
1	Reaction temp	A	190+ 3	180+ 3
2	Reaction time	B	4	5
3	Rate of temp increase	C	30 – 35	15 – 200
4	Agitation	D	100	150
5	Silica sand type	E	‘Gen’ no.5	‘Gen’No.6
6	Relative excess of silica sand	F	20	10
7	Active carbon	G	‘Edcol’T3	Edcol’3A
8	Quantity of active carbon added	H	0.13	0.06
9	Evaculated water temperature	I	80+ 5	40+ 5

Step 1: The total degrees of freedom (TDF)

$$\text{TDF} = A(1) + B(1) + C(1) + D(1) + E(1) + F(1) + G(1) + H(1) + I(1)$$

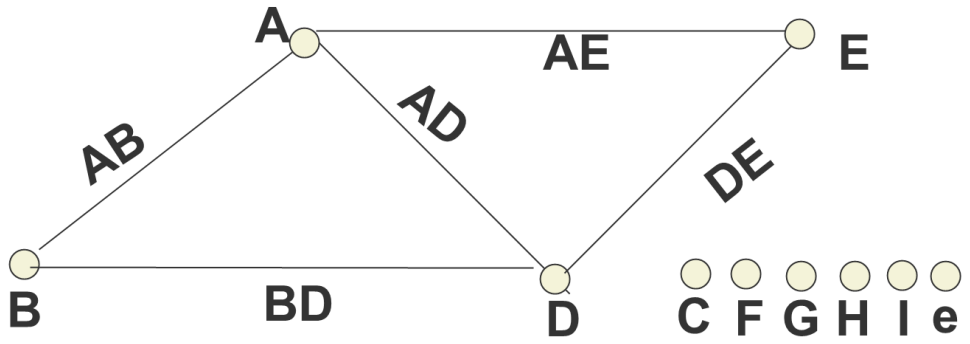
+AB(1)+AD(1)+AE(1)+BD(1)+DE(1)

$$= 13$$

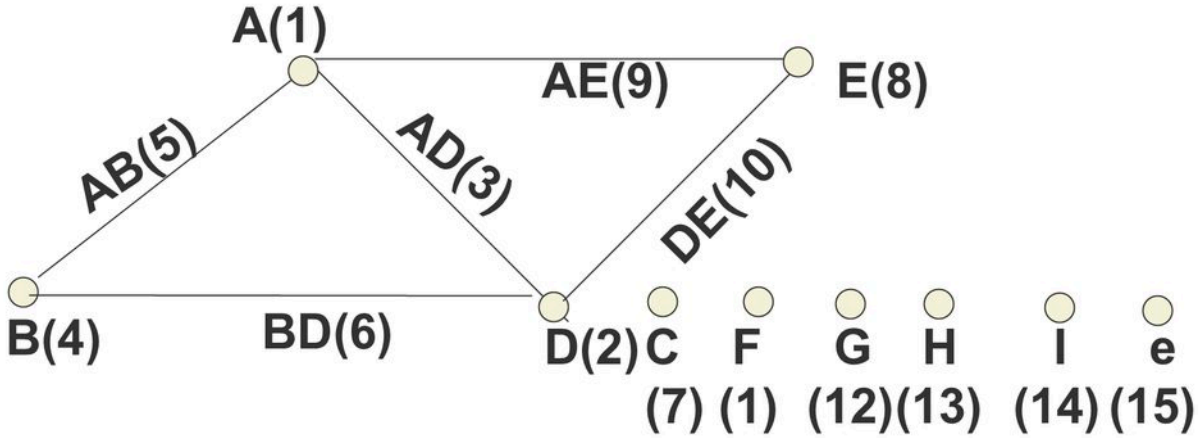
Step 2: Minimum no. of experiments (MNE):

$$\begin{aligned} \text{MNE} &= \text{TDF} + 1 \\ &= 13 + 1 \\ &= 14 \end{aligned}$$

Step 3: The required linear graph (RLG):



The assignment of the column was made by referring to the interaction table of $L_{16}[2^{15}]$. The assignment are as follows.



The design layout is given in the following table along with the data on molar ratio. Two measurements were taken for molar ratio.

The design layout with the data

Sl no	A 1	D2	AXD 3	B4	AXB 5	BXD 6	C7	E8	AXE 9	DXE 10	F 11	G 12	H 13	I 14	e	Molar ratio	
																1	2
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2.11	2.12
2	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2.16	2.17
3	1	1	1	2	2	2	2	1	1	1	1	2	2	2	2	2.21	2.20
4	1	1	1	2	2	2	2	2	2	2	2	1	1	1	1	2.16	2.18
5	1	2	2	1	1	2	2	1	1	2	2	1	1	2	2	2.07	2.09
6	1	2	2	1	1	2	2	2	2	1	1	2	2	1	1	2.30	2.32
7	1	2	2	2	2	1	1	1	1	2	2	2	2	1	1	2.16	2.14

Sl no	A	D	AxD	B	AxB	BxD	C	E	AxE	DXE	F	G	H	I	e	Molar ratio	
																1	2
8	1	2	2	2	2	1	1	2	2	1	1	1	1	2	2	2.17	2.30
9	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1.87	1.90
10	2	1	2	1	2	1	2	2	1	2	1	2	1	2	1	2.10	2.12
11	2	1	2	2	1	2	1	1	2	1	2	2	1	2	1	1.97	1.98
12	2	1	2	2	1	2	1	2	1	2	1	1	2	1	2	2.23	2.23
13	2	2	1	1	2	2	1	1	2	2	1	1	2	2	1	1.97	1.96
14	2	2	1	1	2	2	1	2	1	1	2	2	1	1	2	2.03	2.08
15	2	2	1	2	1	1	2	1	2	2	1	2	1	1	2	2.02	2.05
16	2	2	1	2	1	1	2	2	1	1	2	1	2	2	1	2.12	2.14

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Analysis of data

Total response table for main factors and for the columns of interactions/errors.

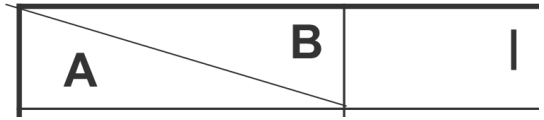
Sl.no.	Factor/ code	Factor/ interaction	Levels		Total
			1	2	
1	A	Reaction temperature	34.96	32.77	67.73
2	B	Reaction time	33.37	34.36	67.73
3	C	Rate of temperature increase	33.85	33.88	67.73
4	D	Agitation	33.71	34.02	67.73
5	E	Silica sand type	32.82	34.91	67.73
6	F	Relative excess of silica sand	34.51	33.22	67.73
7	G	Active carbon type	33.72	34.01	67.73

Sl.no.	Factor/ code	Factor/ interaction	Levels		Total
			1	2	
8	H	Quality of active carbon additive	33.65	34.08	67.73
9	I	Evacuated water temperature	33.90	33.83	67.73
10	AXB		34.083	32.65	67.73
11	AXD		33.68	34.05	67.73
12	AXE		34.15	33.58	67.73
13	BXD		33.75	33.98	67.73
14	DXE		33.92	33.81	67.73
15	e		33.85	33.88	67.73

Total response table for interactions

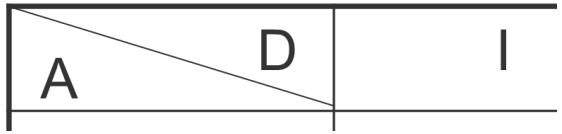
.

For interaction AxB



	I	II	III
1	17.34	17.62	34.96
2	16.03	16.74	32.77
Total	33.37	34.36	67.73

For interaction AxD



	I	II	III
1	17.31	17.65	34.96
2	16.40	16.37	32.77
Total	33.71	34.02	67.73

For interaction BxD

B \ D	I	II	III
1	16.55	16.82	33.37
2	17.16	17.20	34.36
Total	33.71	34.02	67.73

For interaction DxE

D \ E	I	II	III
1	16.36	17.35	33.71
2	16.46	17.56	34.02
Total	32.82	34.91	67.73

For interaction AxE

A \ E	I	II	III
1	17.1	17.86	34.96
2	15.72	17.05	32.77
Total	32.82	34.91	67.73

Sum of squares calculation

1. Correction factor $(C.F) = (67.73)^2/32 = 143.355$

2. S_A =sum of squares due to factor A

$$= ((34.96)^2 + (32.77)^{2)}/16 - \text{CF}$$

$$= 0.149878125$$

It is to be noted that to find out the sum of squares of a factor having two levels, the following formula can be used $(A_1 - A_2)^2/n$ where A_i=sum of the observations when the factor A is at level 'i' and n=total no. of observations, I=1,2.

This is possible since

$$2(A_1^2+A_2^2)/n - (A_1+A_2)^2/n$$

$$= 2A_1^2 + 2A_2^2 - A_1^2 - A_2^2 - 2A)/n \; - \; (A_1 + A_2)^2/n$$

$$= \; (A_1^2 + \; A_2^2 \; - \; 2A_1A_2)/n$$

$$= \; (A_1 \; + \; A_2)^2/n$$

$$
\begin{aligned}
S_B &= 0.030628125 \\
S_C &= 0.000028125 \\
S_D &= 0.003003125 \\
S_E &= 0.136503125 \\
S_F &= 0.052003125 \\
S_G &= 0.005778125 \\
S_H &= 0.000153125 \\
S_I &= 0.000153125 \\
S_{\mathrm{AXB}} &= 0.005778125
\end{aligned}
$$

$$S_{AXD} = 0.004278125$$
$$S_{AXE} = 0.10153125$$
$$S_{BXD} = 0.004278125$$
$$S_{DXE} = 0.00378125$$

S_{E1} =Sum of squares due to column 15 which is free in the layout of the design = 0.000028125

S_{T1} =Sum of squares due to '16' treatments(which are the combination of different levels of the factors A,B,…G,H,I)

$$= ((4.23)^2 + (4.33)^2 + \ldots. + (4.26)^2)/2 - C.F.$$
$$= 0.402871875$$

S_{T2} = Total sum of squares

$$= (2.11)^2 + (2.12)^2 + (2.16)^2 + \ldots (2.12)^2 + (2.14)^2 - C.F$$
$$= 143.7617 - C.F$$
$$= 0.40692188$$

S_{e2} = Error sum of squares due to replication

$$= S_{T2} - S_{T1}$$
$$= 0.004050005$$

ANOVA TABLE:

Source	D.f.	S.S	M.S	F CAL	F T_{AB}	P%
A	1	0.149878125	0.149878125	646.37	4.35	36.77
B	1	0.030628125	0.030628125	132.09	4.49	7.47
C	1	0.000028125	0.000028125	646.37		---
D	1	0.003003125	0.003003125	12.95		0.68
E	1	0.136503125	0.136503125	588.69		33.49
F	1	0.052003125	0.052003125	224.27		12.72
G	1	0.002628125	0.002628125	11.33		0.59
H	1	0.005778125	0.005778125	24.92		1.36
I	1	0.002628125	0.000153125			--
AXB	1	0.005778125	0.005778125	24.92		1.36
AXD	1	0.004278125	0.004278125	18.45		0.99
AXE	1	0.010153125	0.010153125	43.79		2.44

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ANOVA TABLE (contd..):

Source	D.f.	S.S	M.S	F CAL	F T_{AB}	P%
BXD	1	0.001653125	0.001653125	7.13		0.35
DXE	1	0.000378125	0.000378125			-
e1	1	0.000028125	0.000028125			-
ST1	15	0.402871875				-
e2	16	0.004050005				-
ST2	31	0.40692188				-
e1	17	0.00407813	0.00023989			-
e11	19	0.00425938	0.000224177			-
e111	20	0.004637505	0.000231875			1.78

The average response table for significant main factors

Sl. no.	Factor code	Level	
		I	I
1	A	2.185	2.048125
2	B	2.085625	2.1475
3	D	2.106875	2.12625
4	E	2.05125	2.181875
5	F	2.156875	2.07625
6	G	2.1075	2.125625
7	H	2.103125	2.13

AVERAGE RESPONSE TABLES FOR SIGNIFICANT INTERACTIONS

For interaction AXB

A \ B	1	2
1	2.1675	2.2025
2	2.00375	2.0925

For interaction AXD

A \ D	1	2
1	2.16375	2.20625
2	2.05	2.04625

For interaction AXE

A \ E	1	2
1	2.1375	2.2325
2	1.965	2.13125

For interaction BXD

B \ D	1	2
1	2.06875	2.1025
2	2.145	2.15

ANALYSIS METHOD FOR 0-1 DATA

WHEN THE NUMBER OF REPLICATIONS IS CONSTANT

PROBLEM:

In an investigation of minimizing wave soldering defects namely “no. of joints having blow hole” in PCB, the following factors and levels and interactions were considered.

Sl. No.	Factor	Factor code	Levels		
			1	2	3
1	Sold temperature	A	210	215	330
2	Conveyor speed	B	2.50	3.25	4
3	Preheat setting	C	2.50	3.25	4
4	Drier setting	D	2.5	3.25	4.0

Interactions required are AXB, AXC, BXC

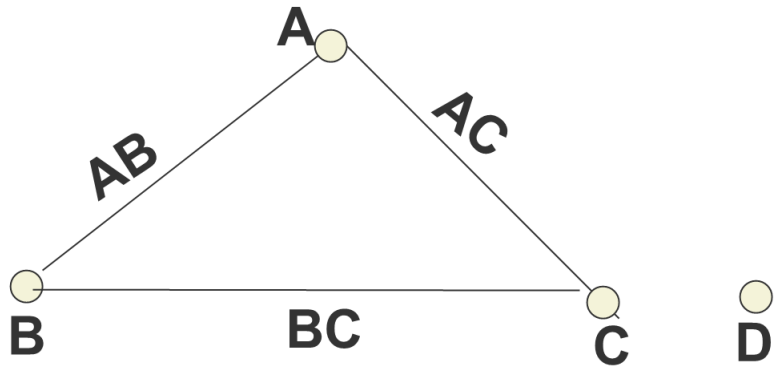
1. Total degrees of freedom required

$$= \text{ Total of } (d.f. \text{ of main effects } + \ d.f \text{ of interactions})$$
$$= \ 2 + 2 + 2 + 2 + 4 + 4 + 4$$
$$= 20$$

2. Minimum number of experiments (MNE) required.

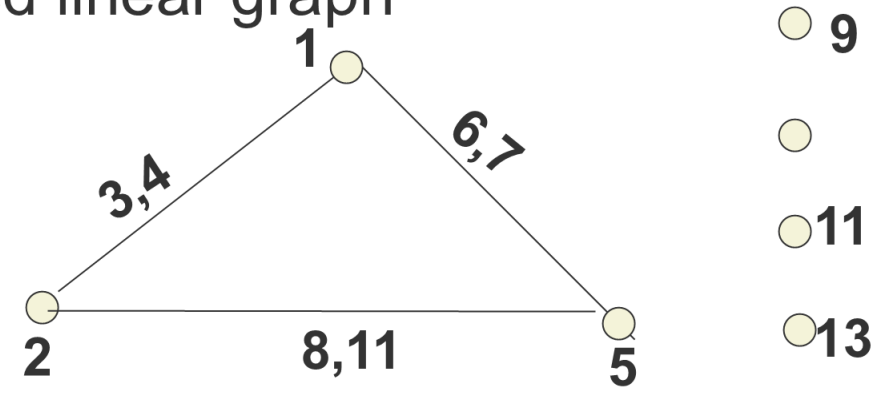
$$\text{MNE } = \ 20 + 1 \ = 21$$

3. Required Linear Graph (RLG)



Since each factor is at 3 levels, $L_{27}3^{13}$ was chosen

4. Modified linear graph



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Master plan of the experiment with response

Sl.no.	Solder Temp (A)	Conveyor speed (B)	Preheat setting (C)	Drier setting (D)	defects “blow holes”			Total
					1	2	3	
1	210	2.5	2.5	2.5	55	42	38	135
2	210	2.5	3.25	3.25	41	36	28	105
3	210	2.5	4	4	54	48	43	145
4	210	3.25	2.5	3.25	44	43	29	116
5	210	3.25	3.25	4	33	25	50	108
6	210	3.25	4	2.5	36	40	45	121
7	210	4	2.5	4	50	46	45	141
8	210	4	3.25	2.5	35	25	34	94
9	210	4	4	3.25	31	32	27	90
10	215	2.5	2.5	3.25	34	26	40	100

Sl.no.	Solder Temp (A)	Conveyor speed (B)	Preheat setting (C)	Drier setting (D)	defects “blow holes”			Total
					1	2	3	
11	215	2.5	2.5	3.25	47	34	25	106
12	215	2.5	3.25	3.25	23	22	18	63
13	215	3.25	4	4	45	63	32	140
14	215	3.25	2.5	2.5	39	35	36	110
15	215	3.25	3.25	3.25	22	24	24	70
16	215	4	4	2.5	43	40	45	128
17	215	4	2.5	3.25	48	28	33	109
18	215	4	3.25	4	24	23	27	74
19	220	2.5	4	4	45	41	40	126
20	220	2.5	2.5	2.5	29	24	20	73

Sl.no.	Solder Temp (A)	Conveyor speed (B)	Preheat setting (C)	Drier setting (D)	defects “blow holes”			Total
					1	2	3	
21	220	2.5	4	3.25	47	34	25	106
22	220	3.25	2.5	3.25	23	22	18	63
23	220	3.25	3.25	4	45	63	32	140
24	220	3.25	4	2.5	39	35	36	110
25	220	4	2.5	3.25	22	24	24	70
26	220	4	3.25	2.5	43	40	45	128
27	220	4	4	3.25	48	28	33	109


Thank You!